

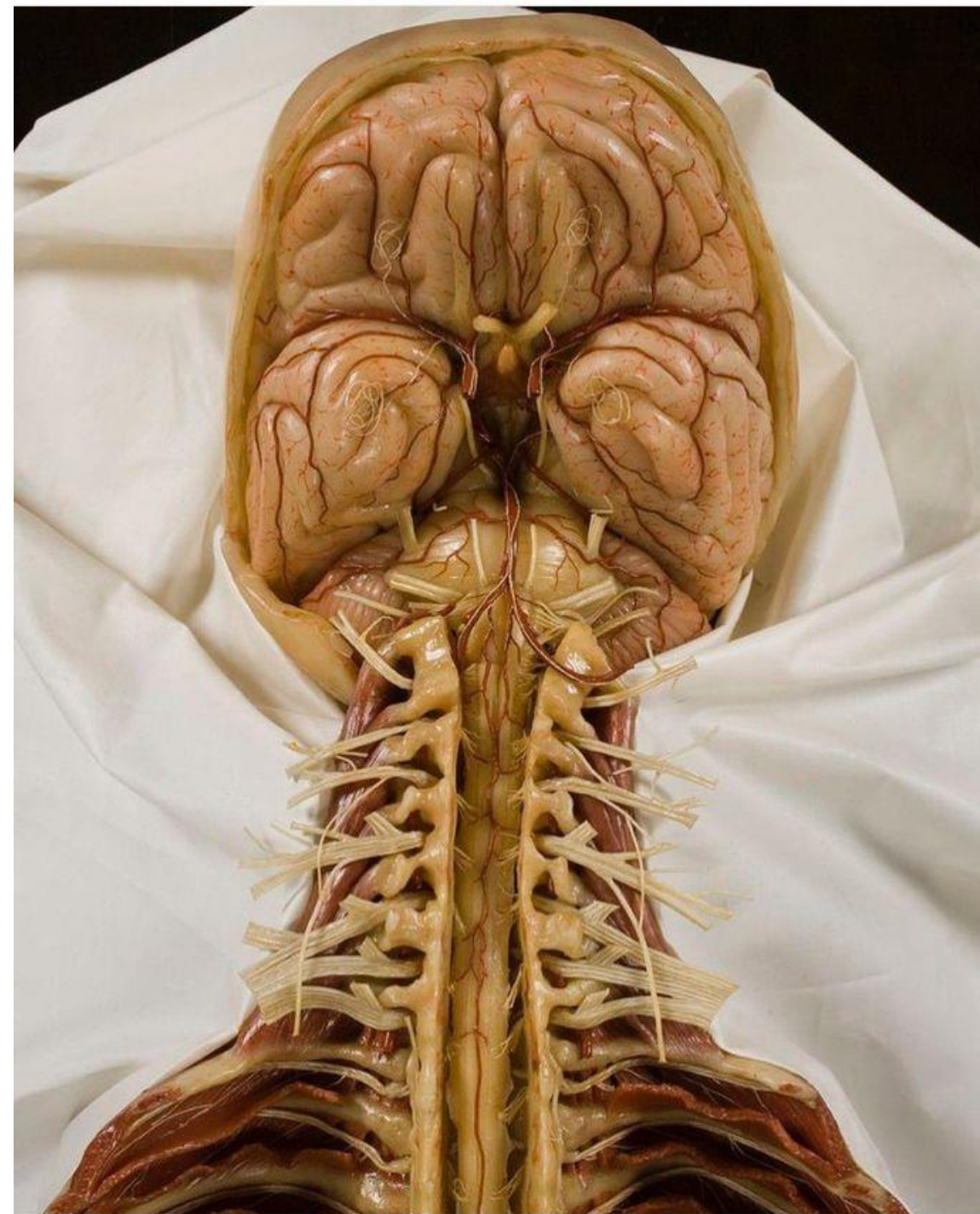
Nervous System

Nervous System Components

- The nervous system enables the body to react to continuous changes in its internal and external environments.
- It also controls and integrates the various activities of the body, such as circulation and respiration

Nervous System

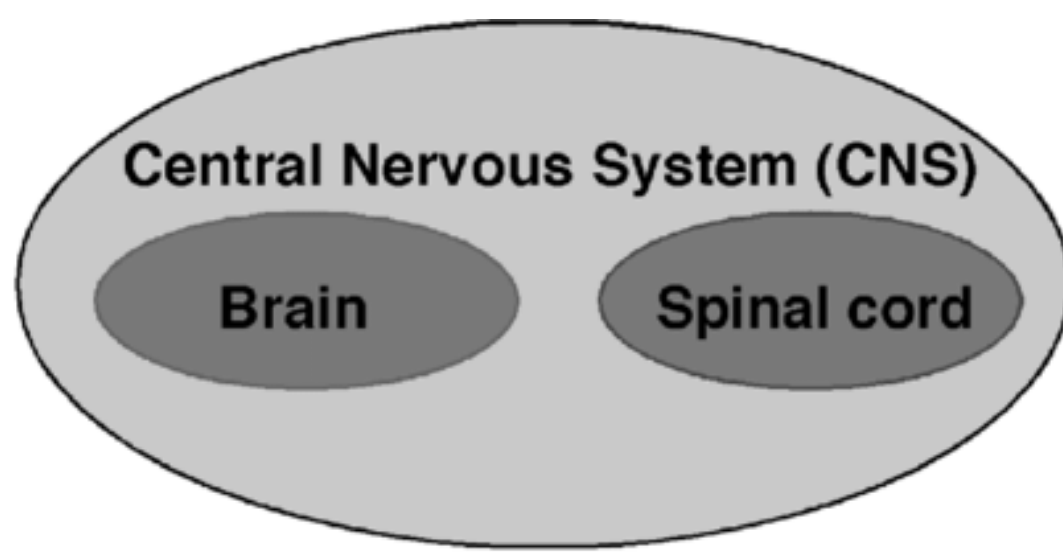
- The nervous system is divided into:
 - Central nervous system
 - Consist of
 - brain and
 - spinal cord
 - Peripheral nervous system
 - Consist of
 - Cranial nerves
 - Spinal nerves
 - Ganglia



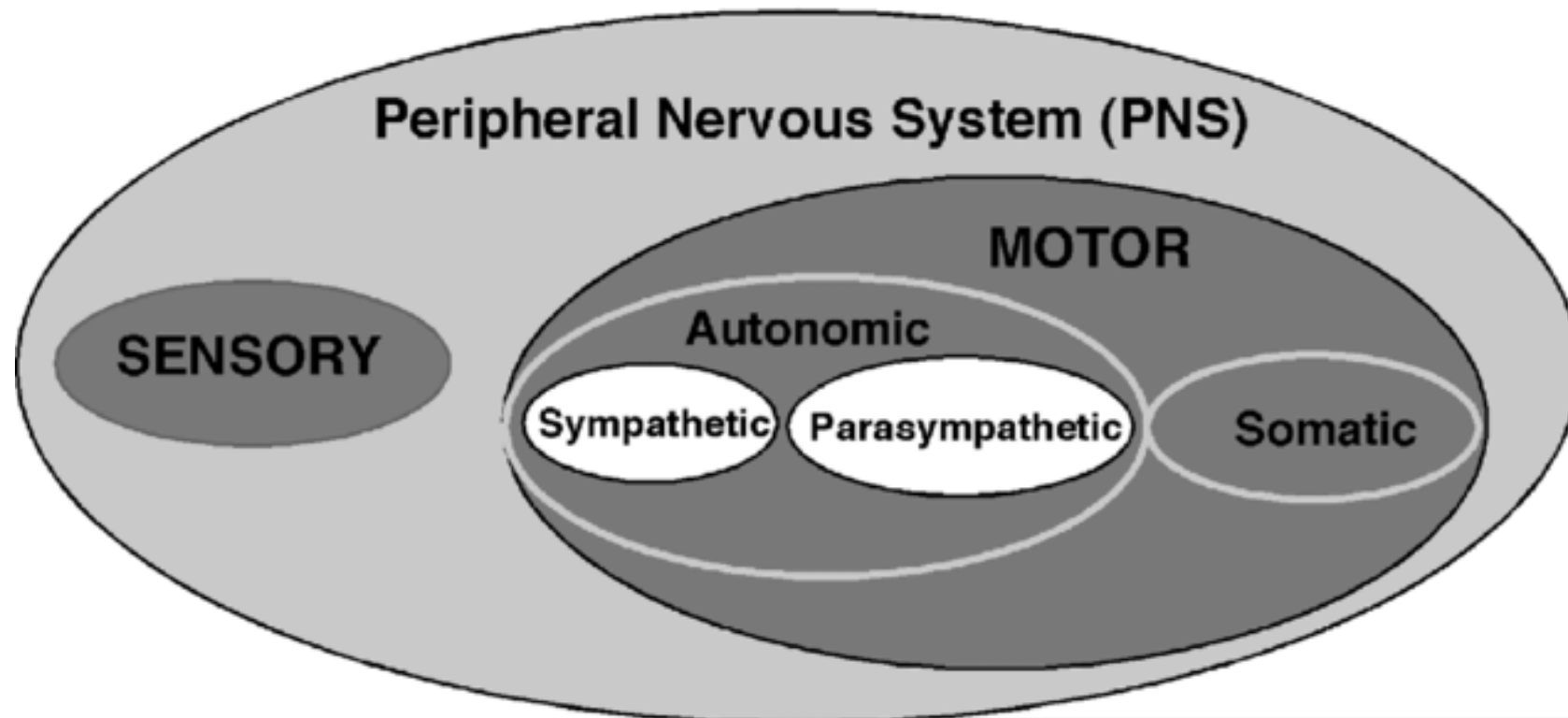
Nervous System

- The central and peripheral parts have
 - Somatic
 - Innervation of skeletal muscles
 - Transmission of sensory information
 - Autonomic
 - Control of:
 - cardiac muscles
 - smooth muscles
 - glands

- For descriptive purposes, the nervous system is divided:
- Structurally
 - *central nervous system* (CNS) : the brain and spinal cord,
 - *peripheral nervous system* (PNS): cranial and spinal nerves and their associated *ganglia*).
- Functionally
 - *somatic nervous system* (SNS): innervation of skeletal muscle (efferent pathway) and transmission of sensory information (afferent pathway)
 - *autonomic nervous system* (ANS): control of cardiac muscle, smooth muscle and glands (have afferent and efferent pathways)

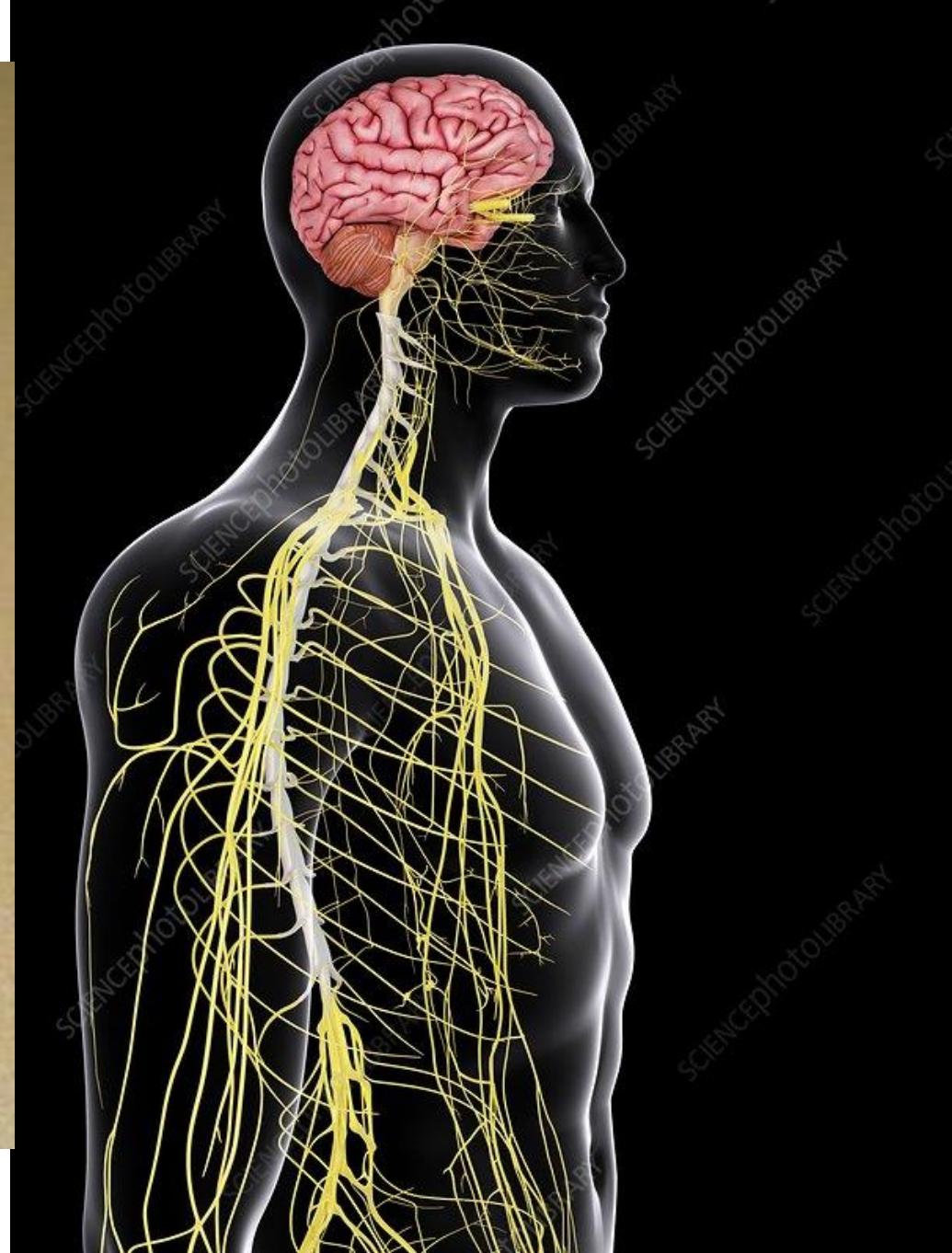
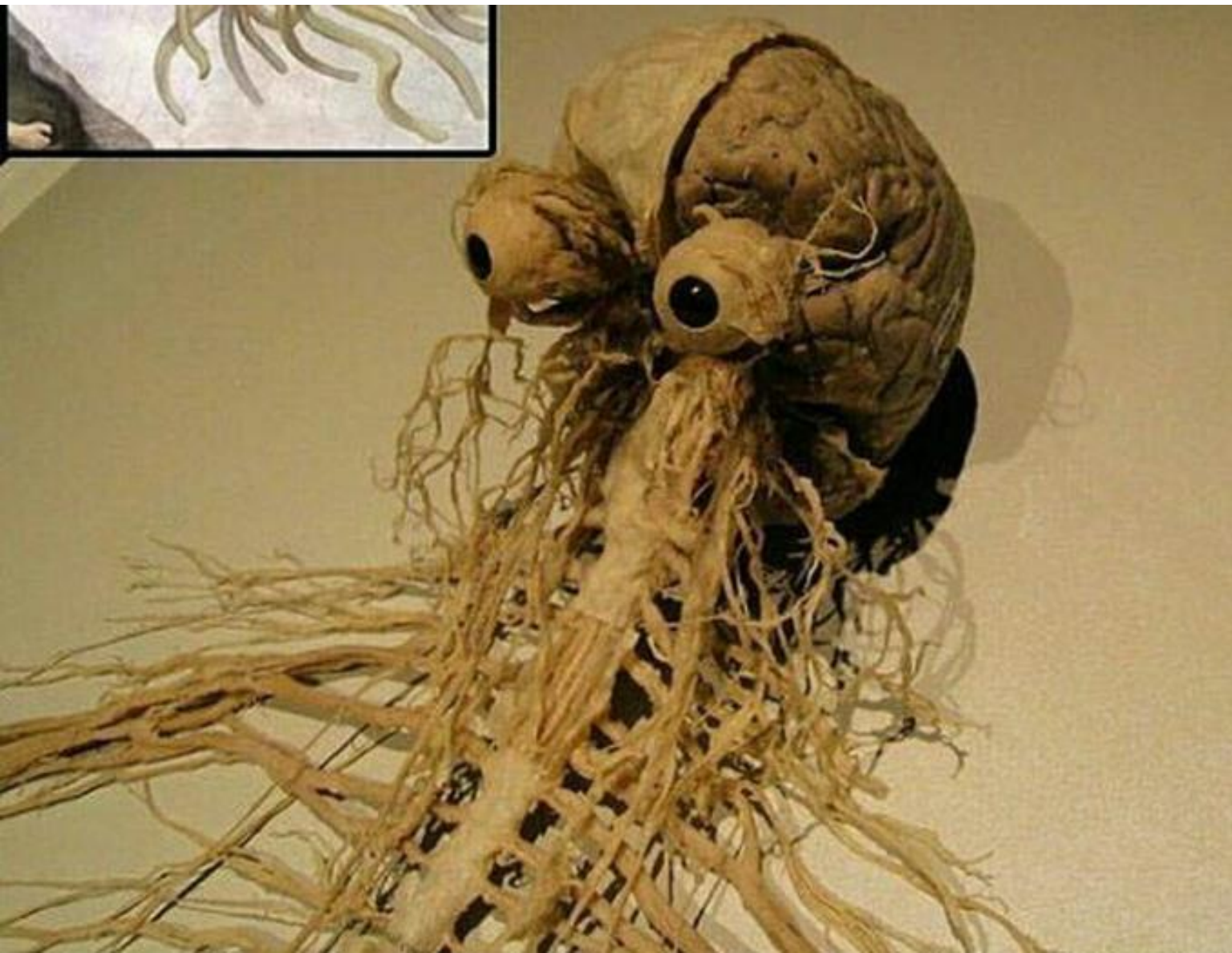


CNS
Integrates inputs
and commands
response



PNS
Sensory:
Afferent fibers
send signals from
receptors to CNS

Motor:
Efferent fibers
send signals from
CNS to effector organs



Central Nervous System

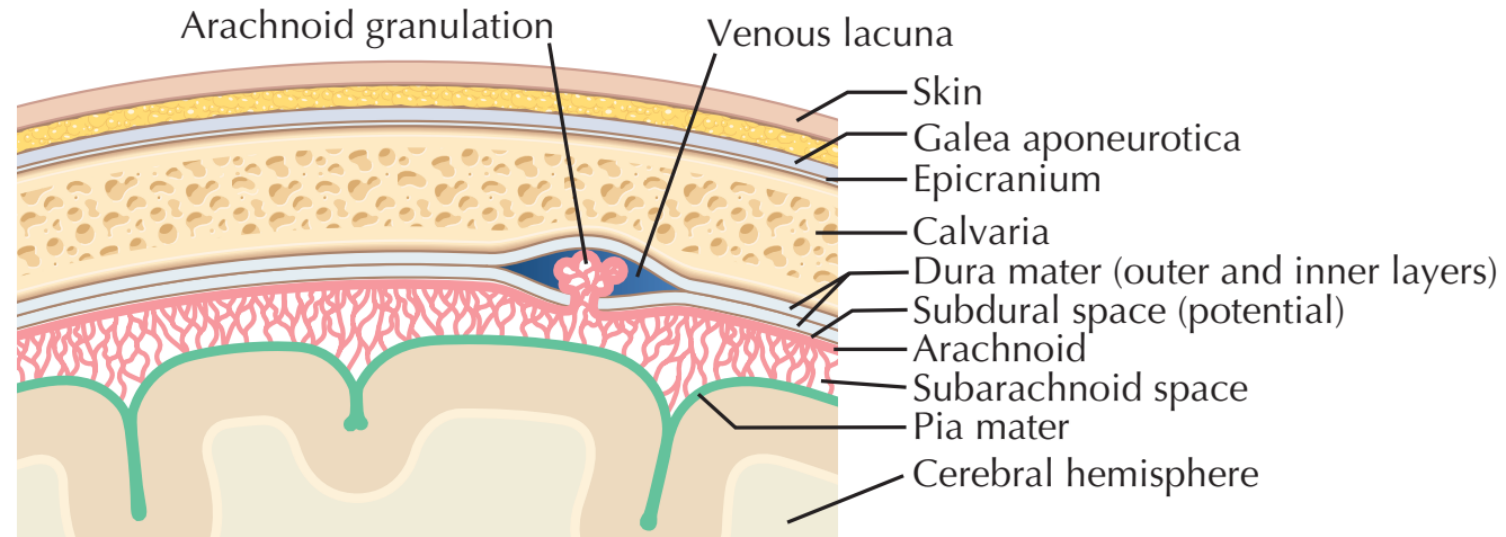
Learning Objectives

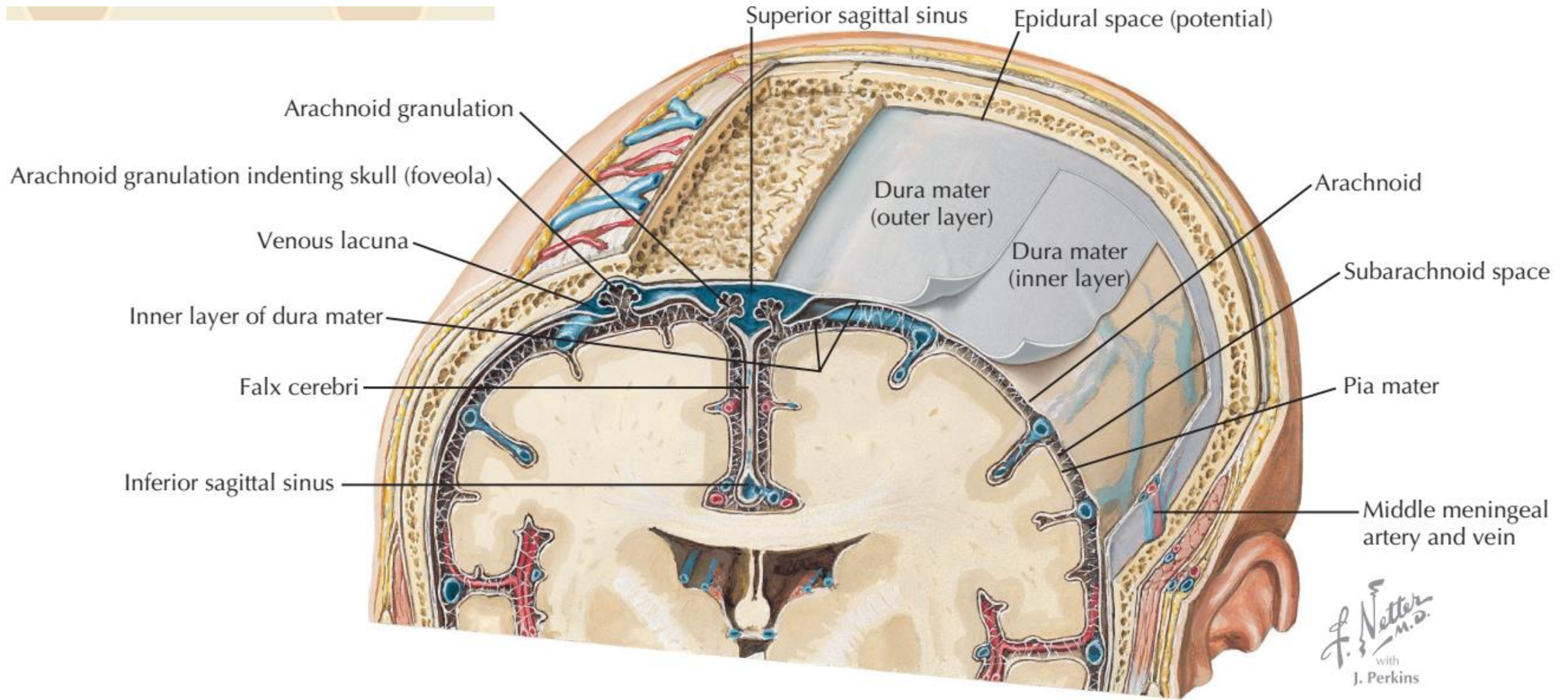
At the end of the lecture students should be able to:

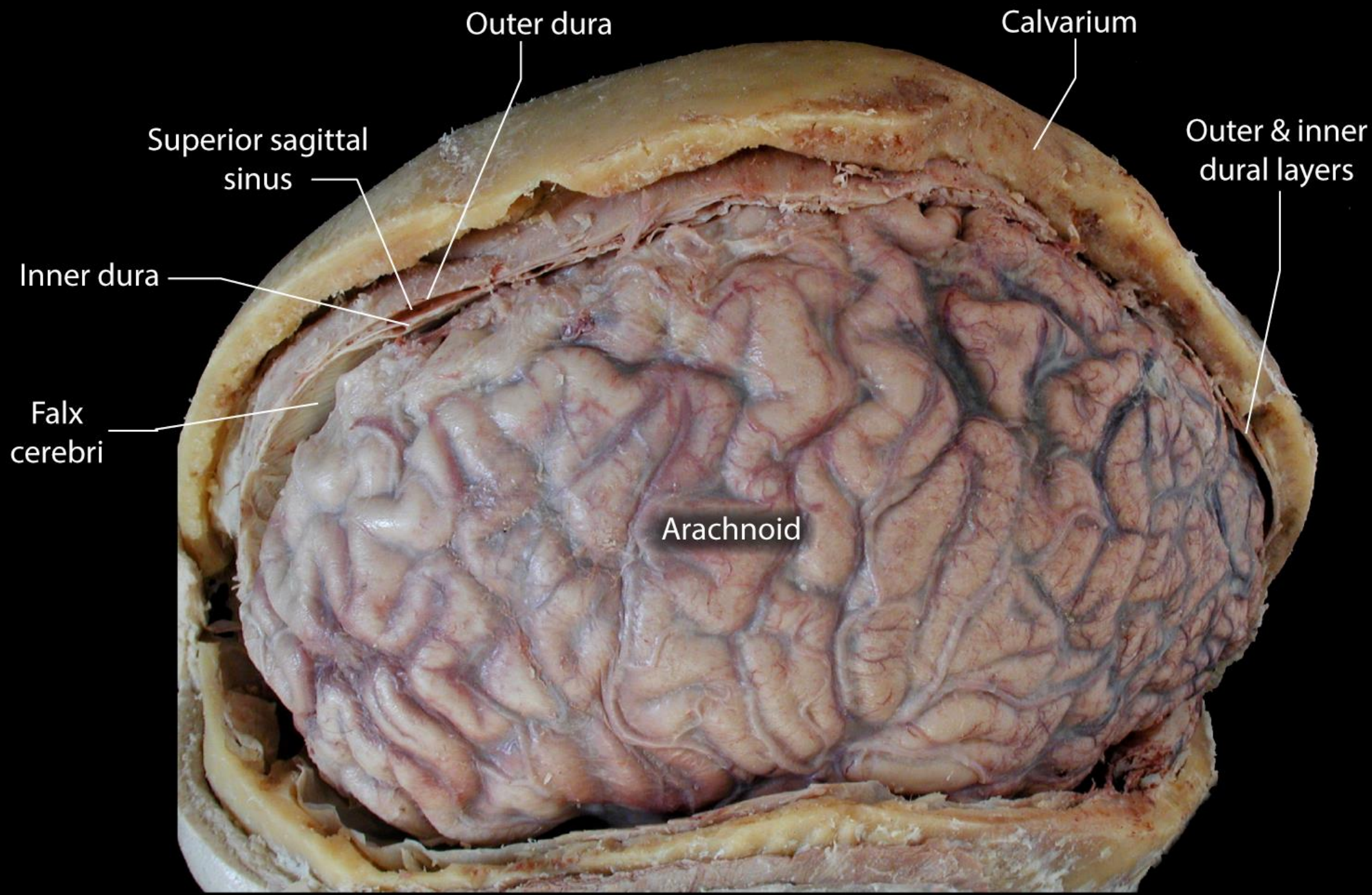
- State the parts of the brain
- Describe the gyri and sulci of the cerebrum
- Understand the functional cortical areas (Brodmann numbers)
- State the parts and function of the basal nuclei
- Describe the blood supply to the cerebrum

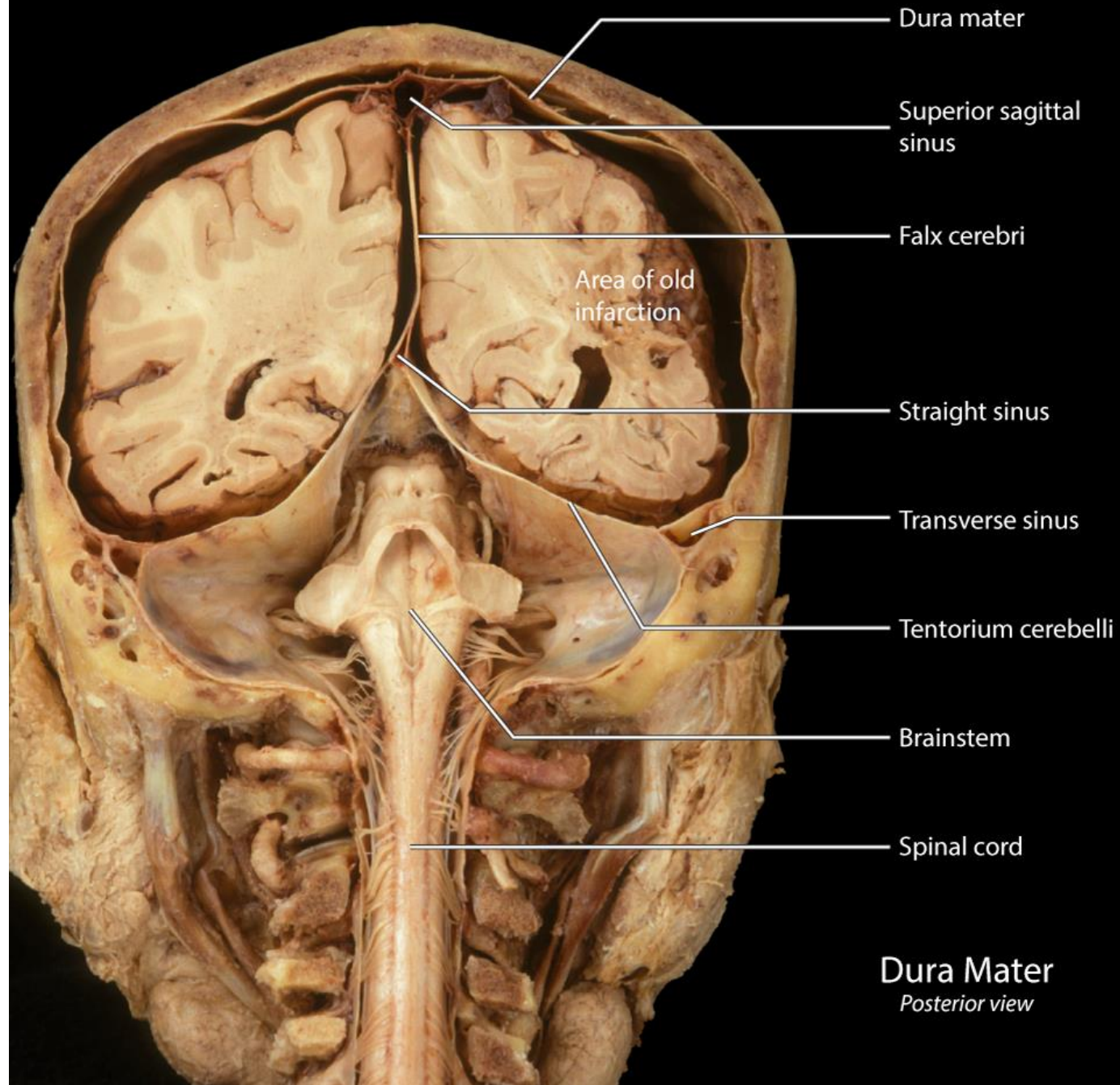
Brain Meninges

- The brain covering (meninges) comprises 3 connective tissue membranes
 - Dura mater (pachymeninx)
 - Leptomeninges comprise
 - Arachnoid mater
 - Pia mater
- Provide protection and support

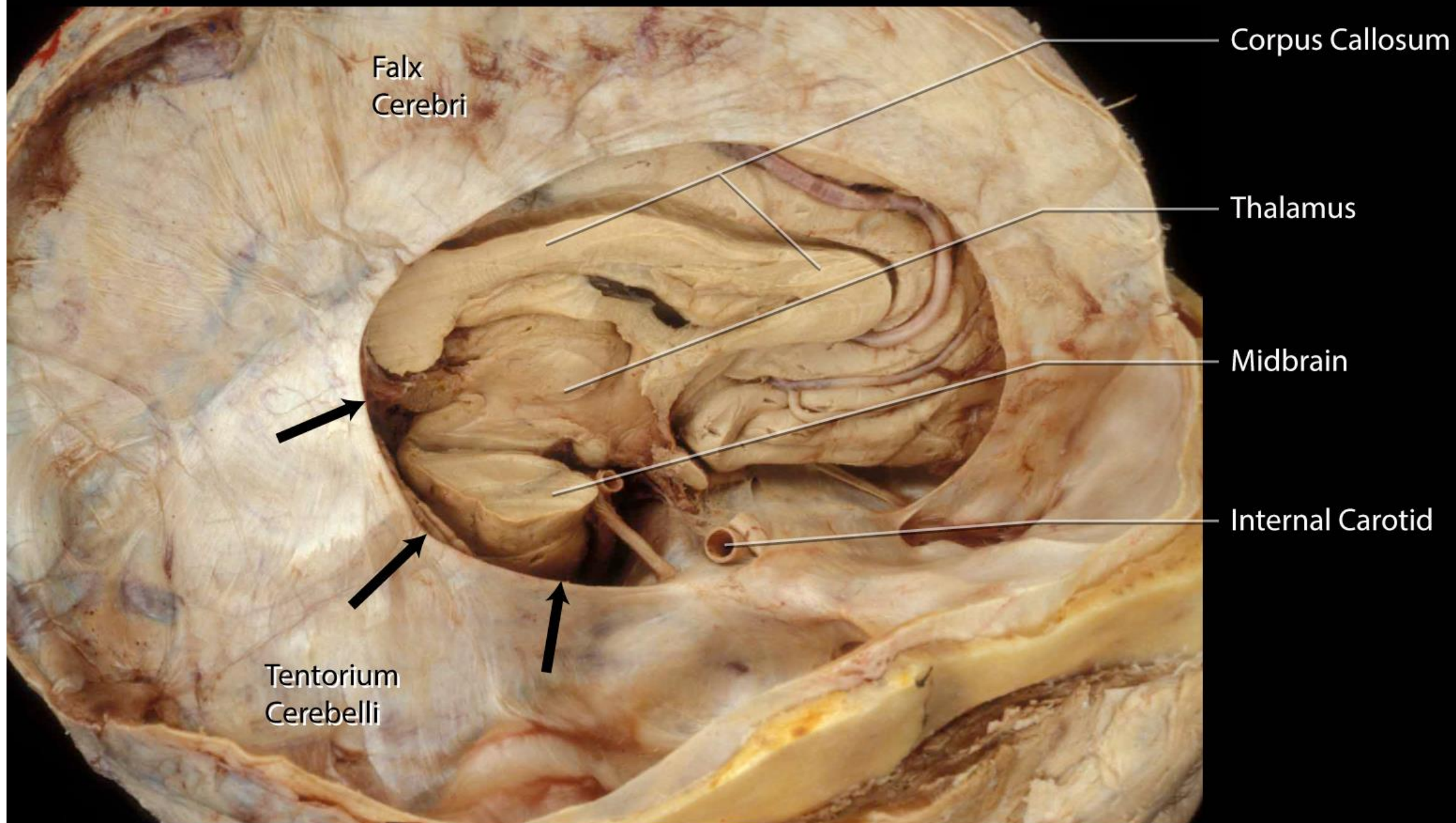






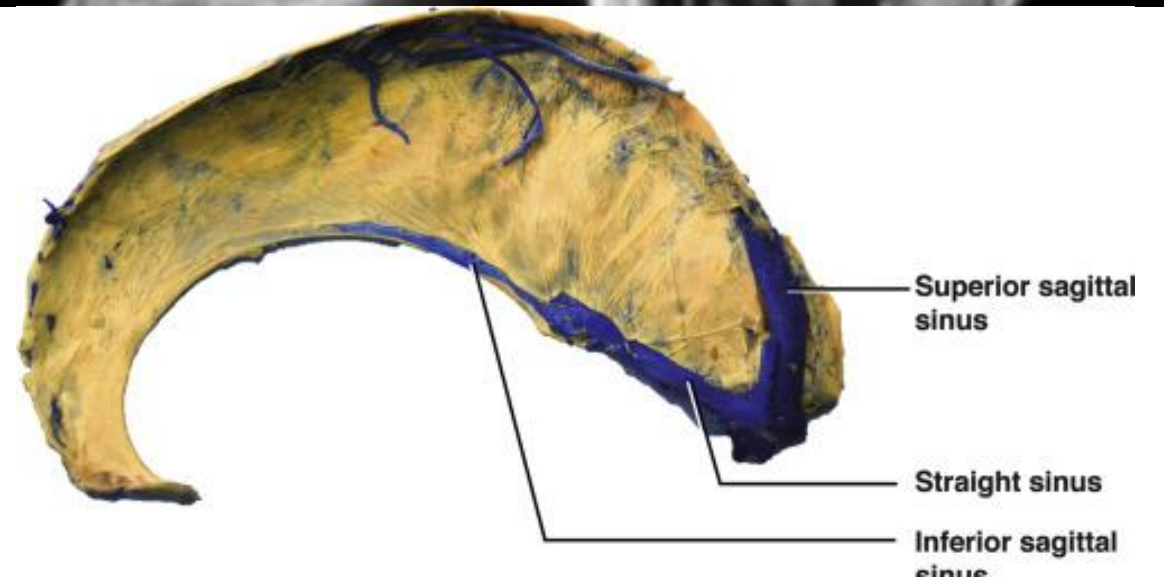
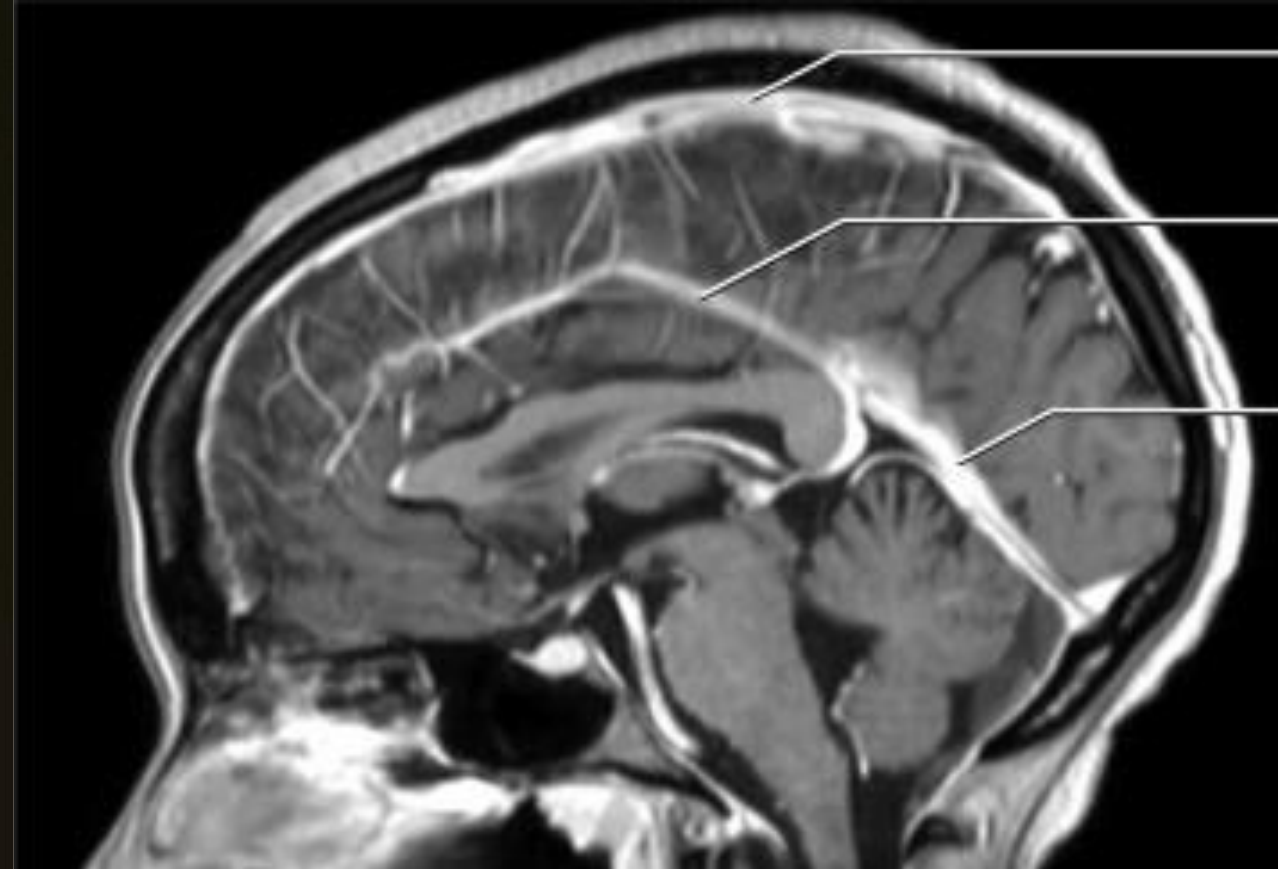
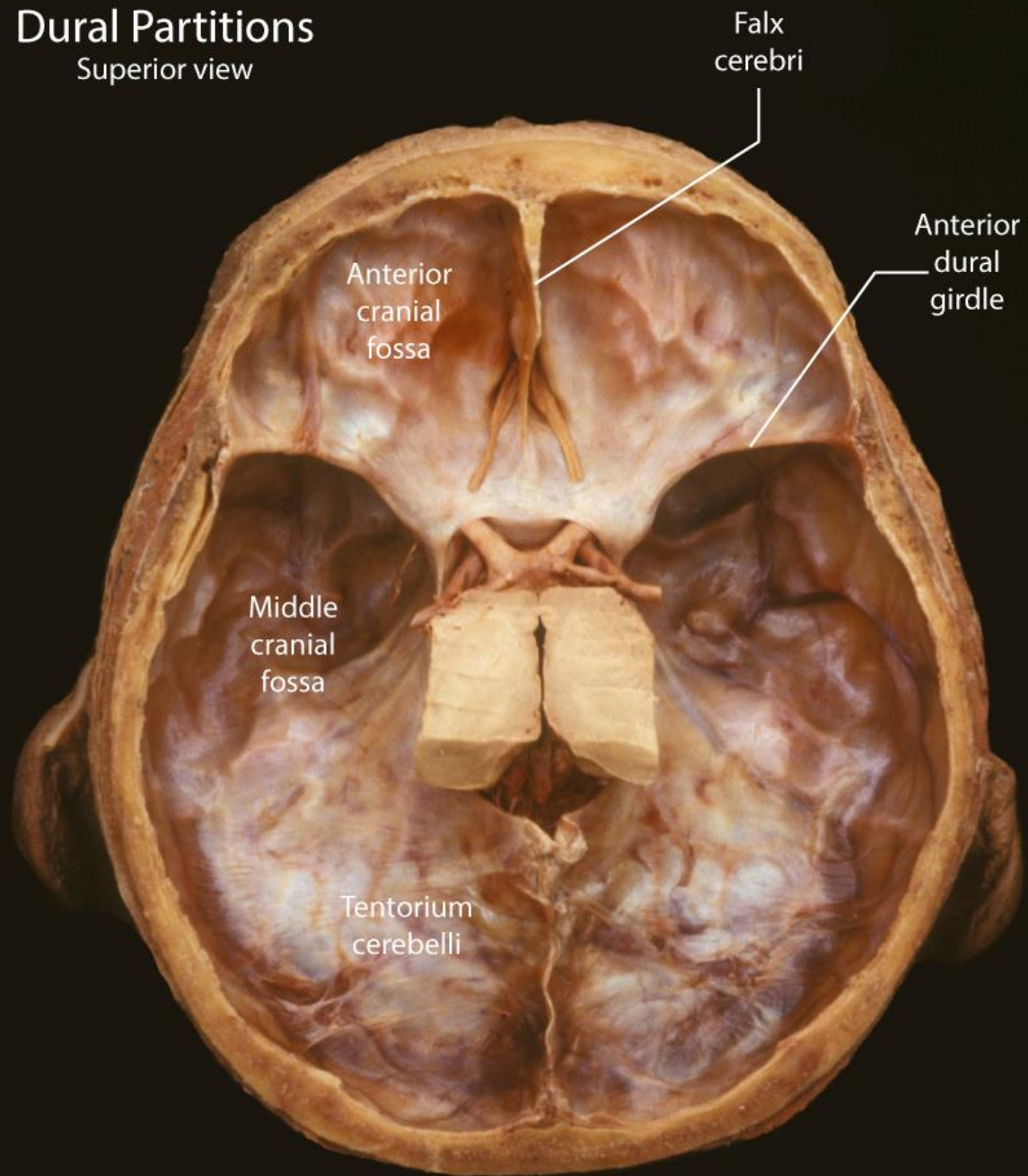


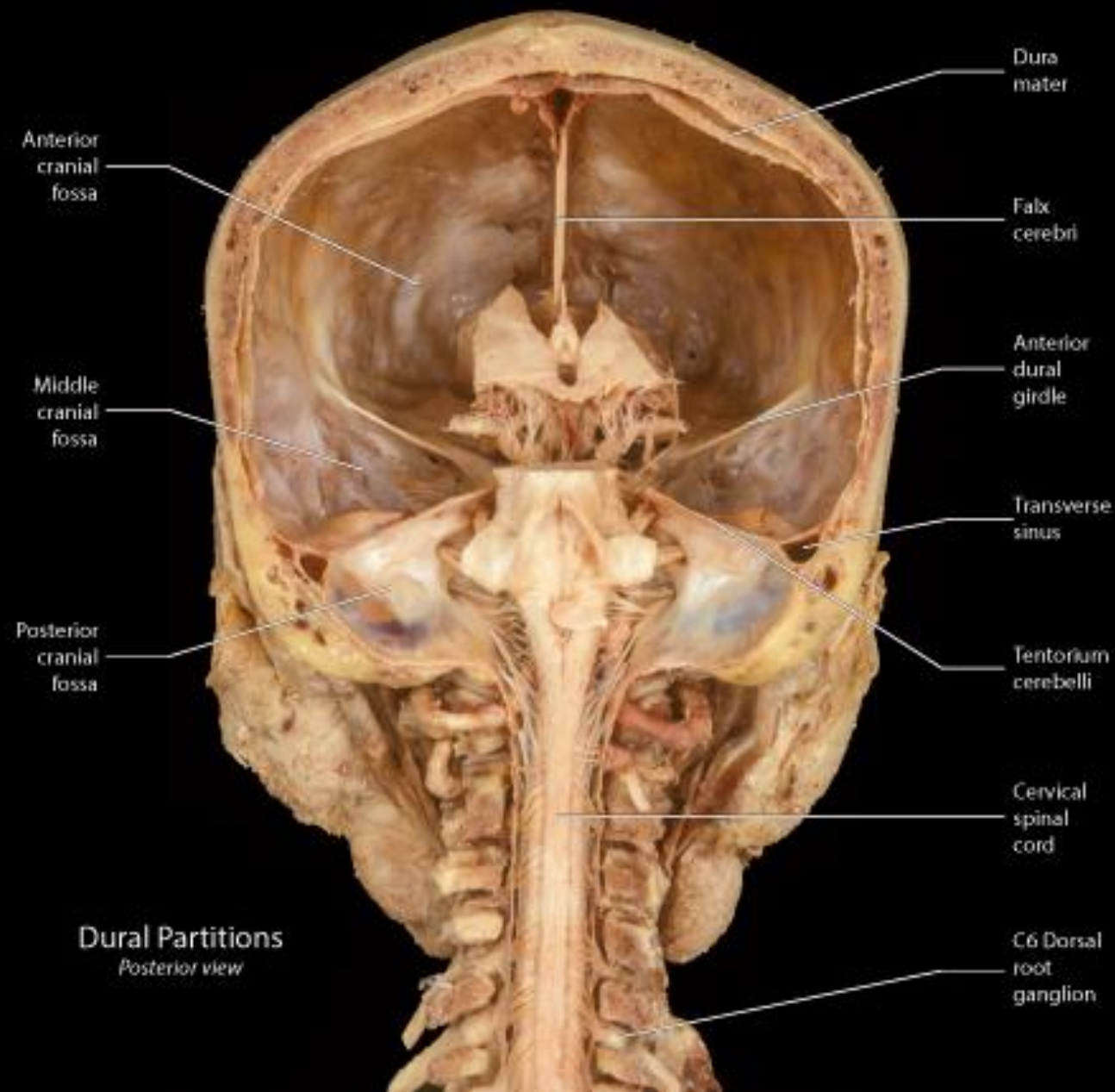
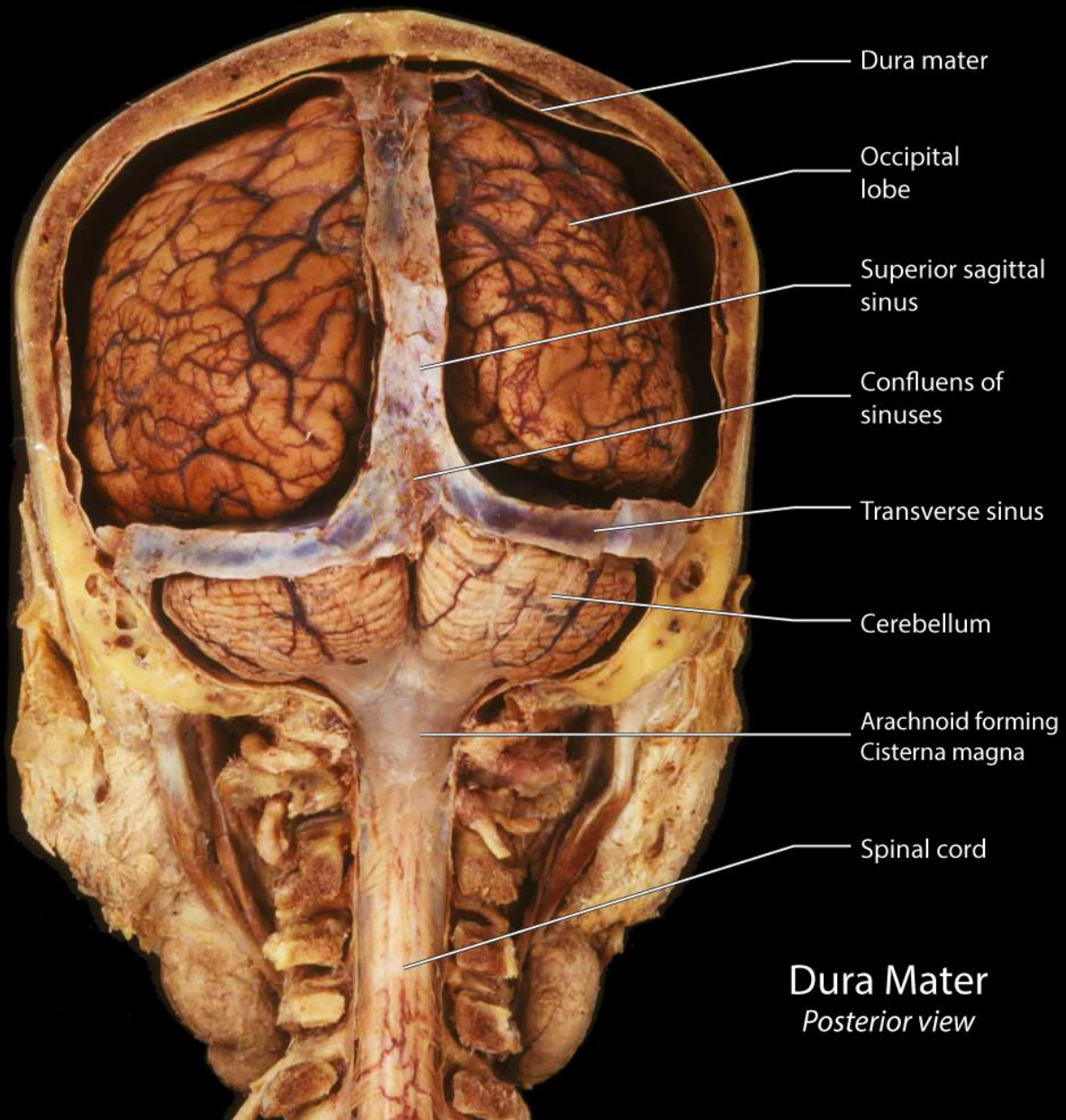
Dura Mater
Posterior view

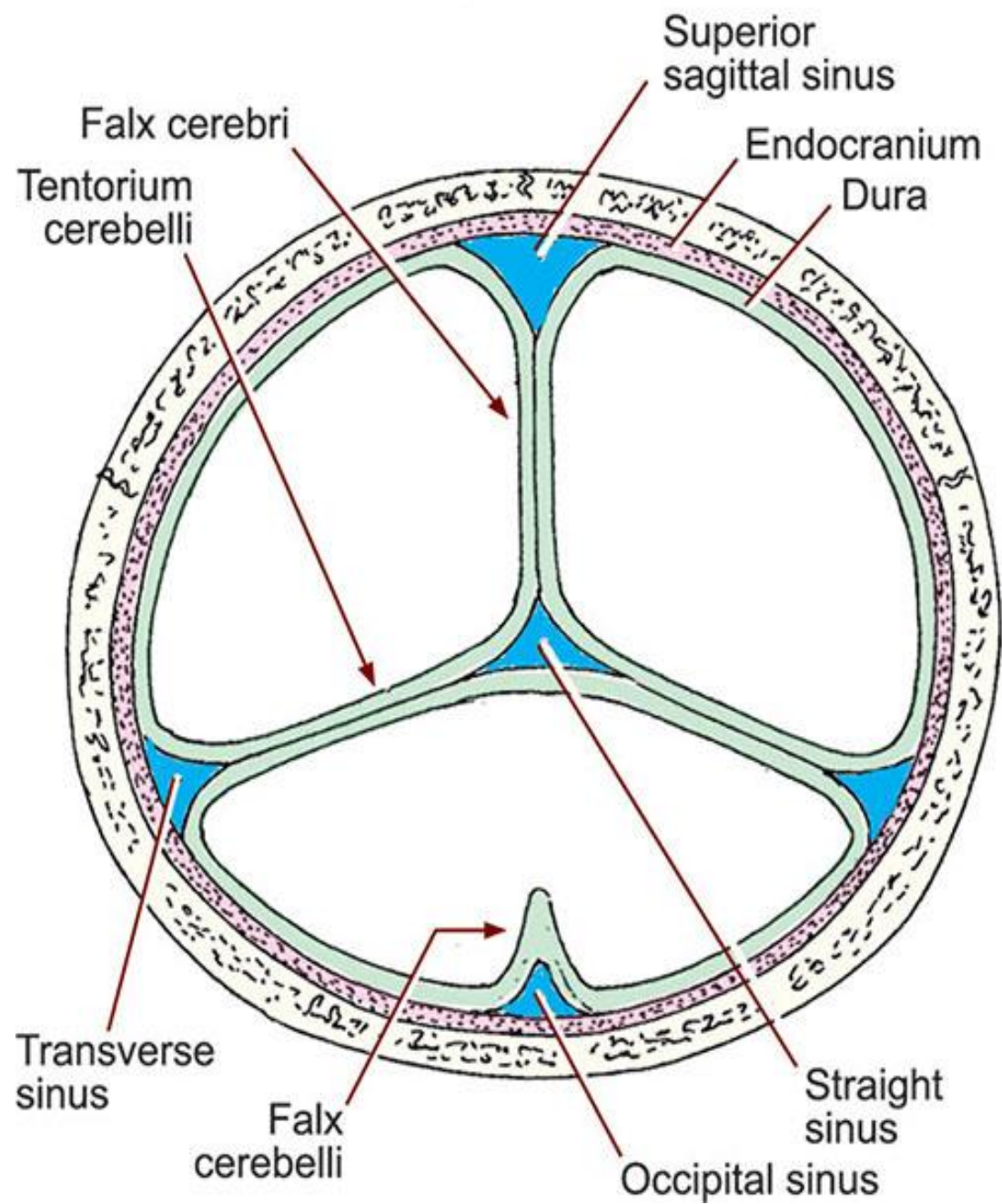


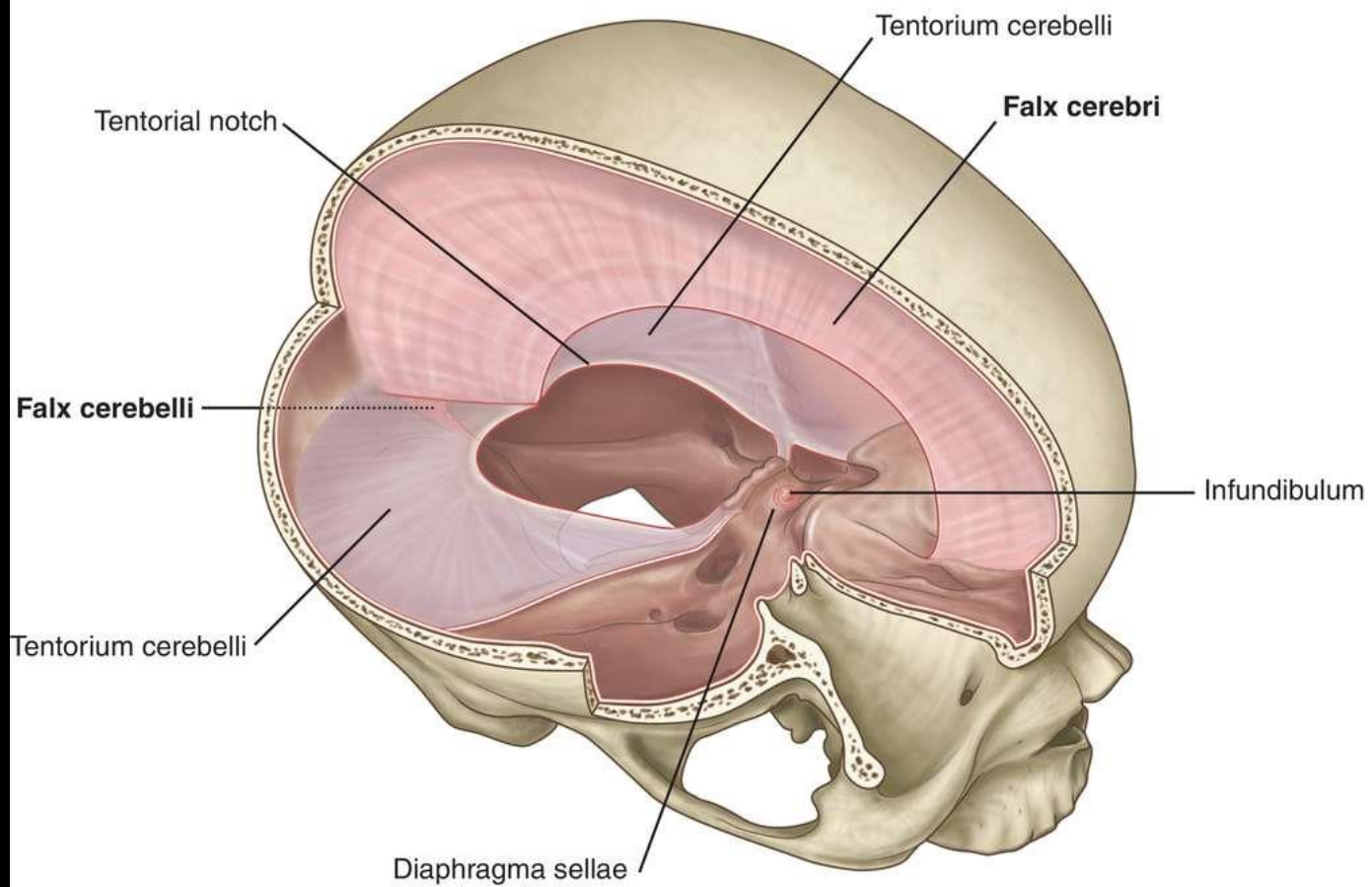
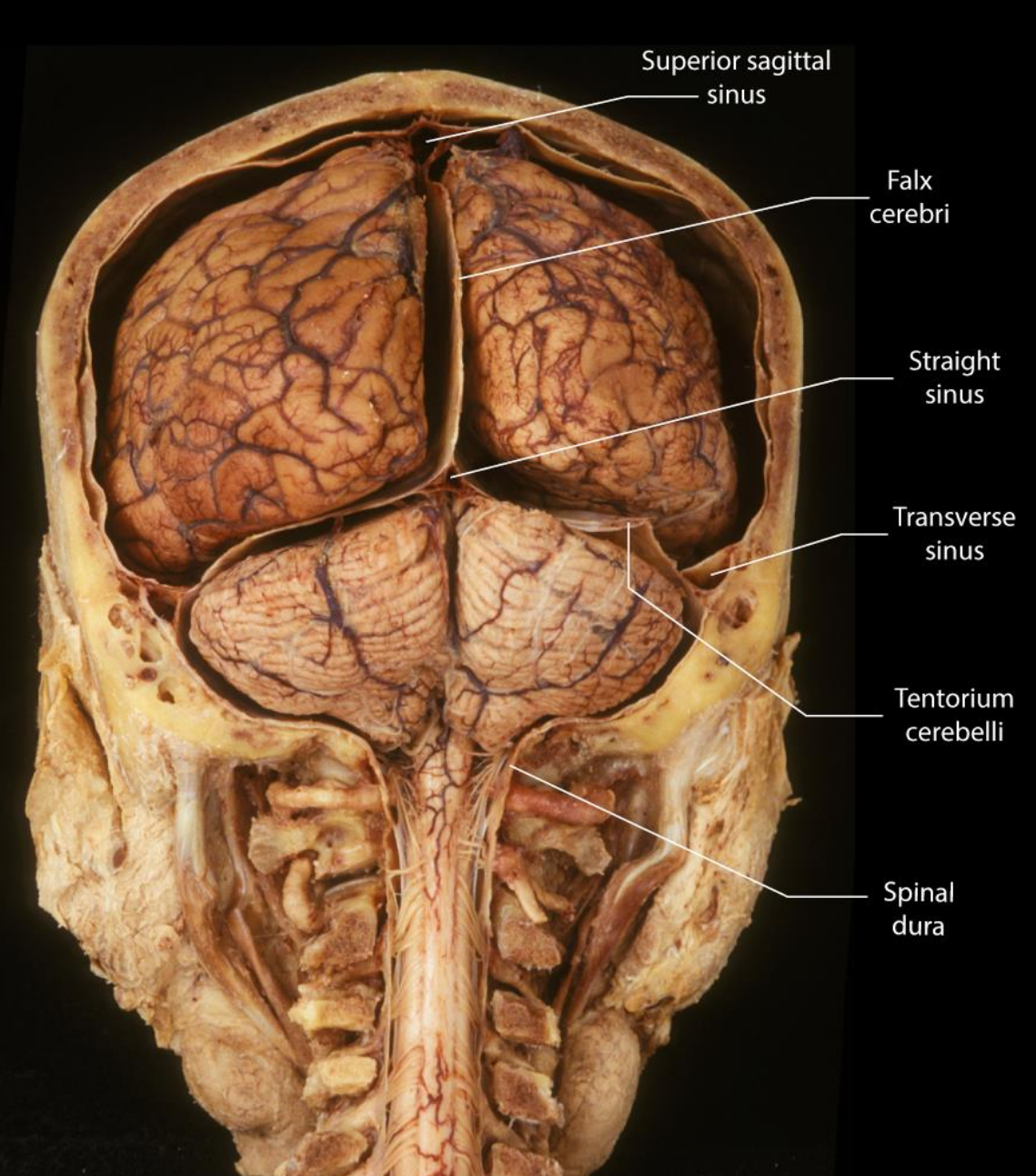
Dural Partitions

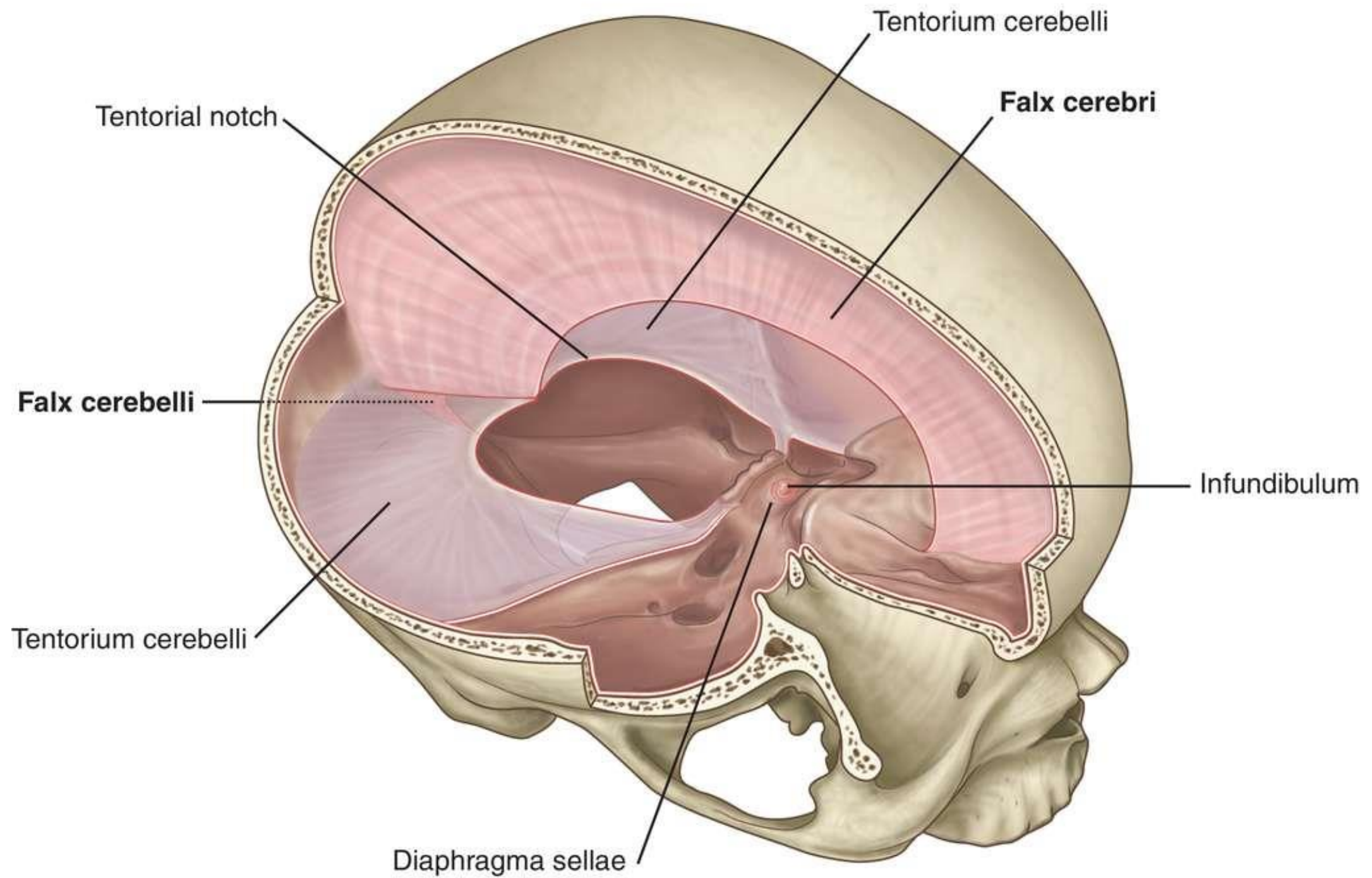
Superior view





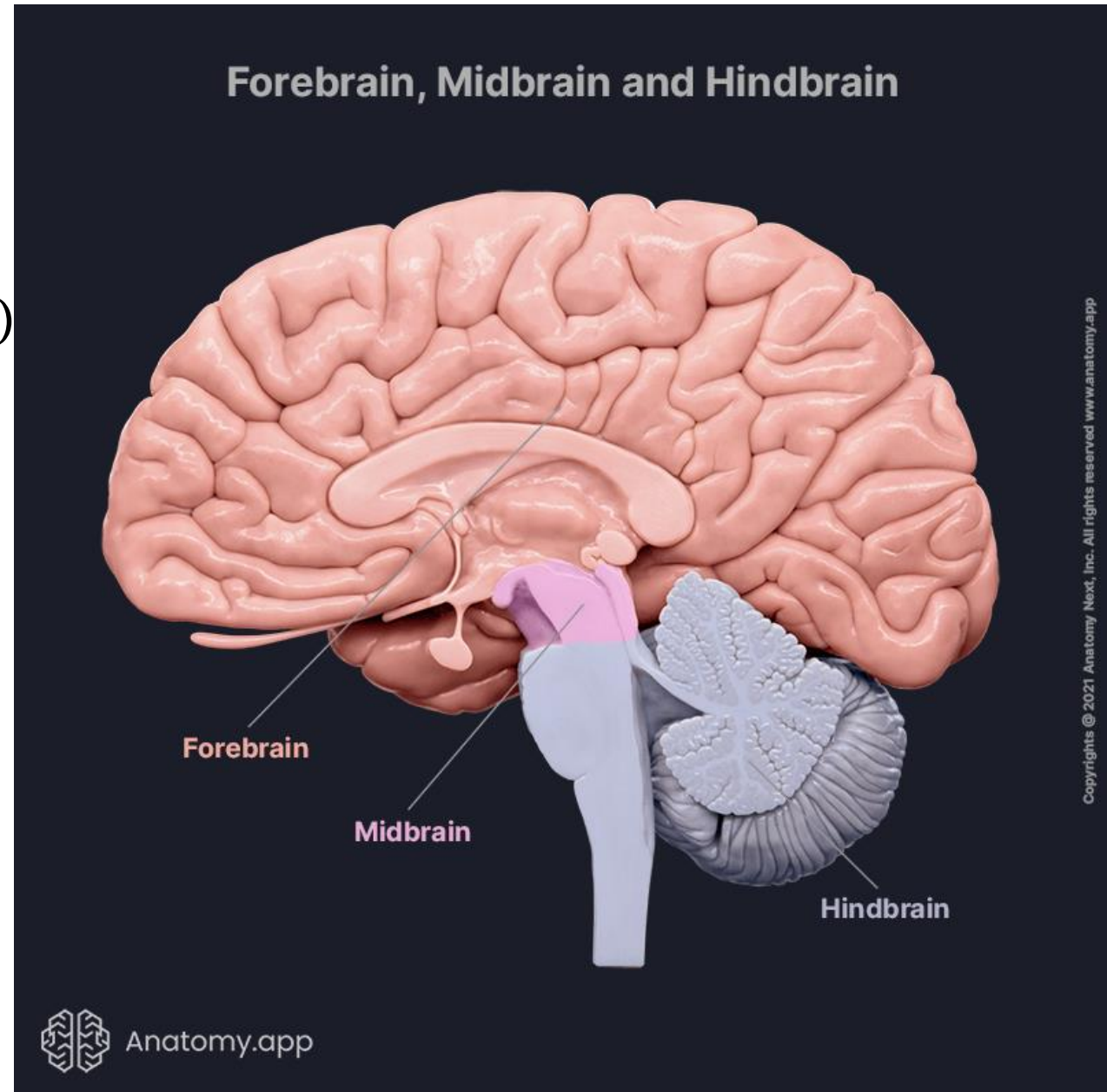






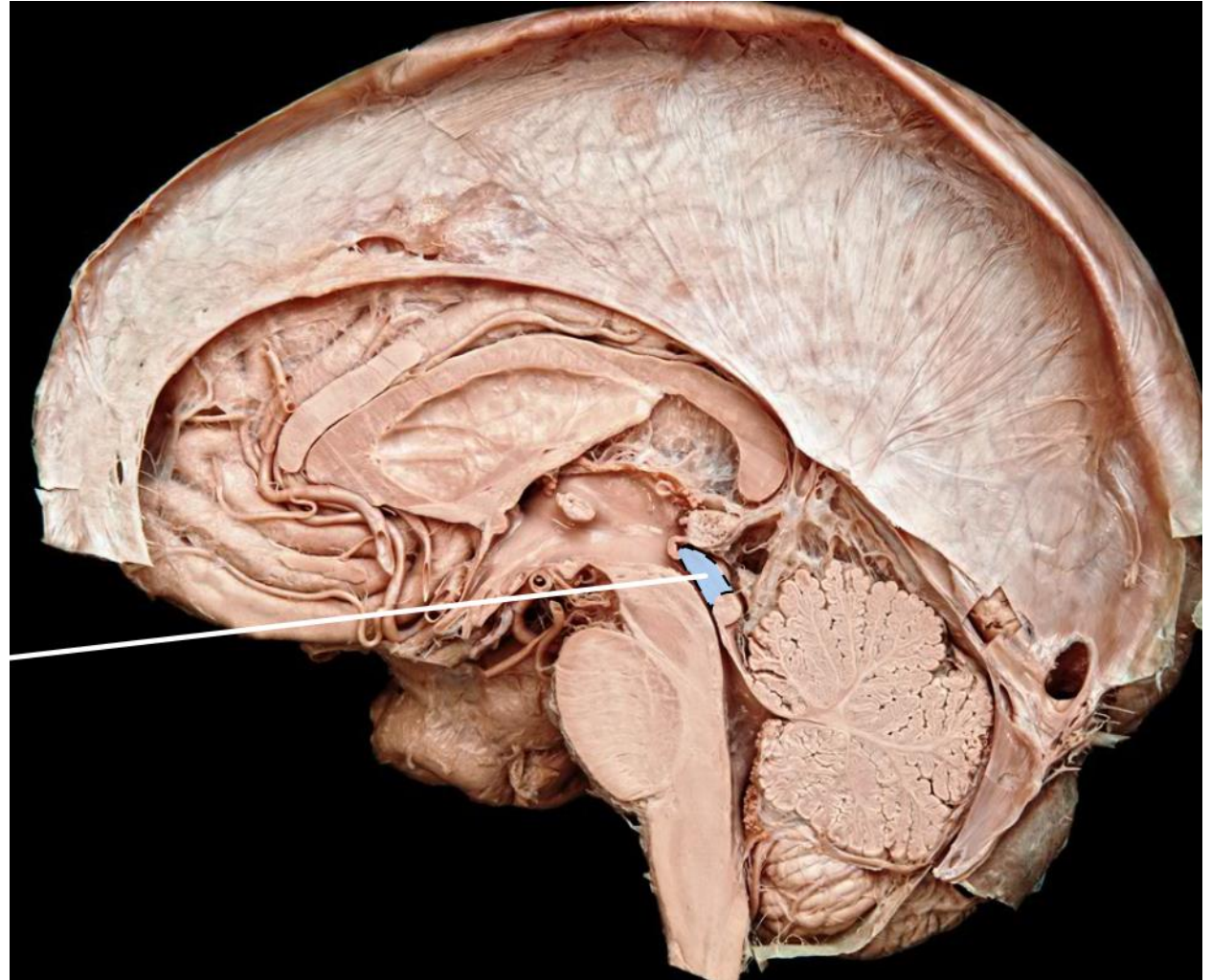
General Plan

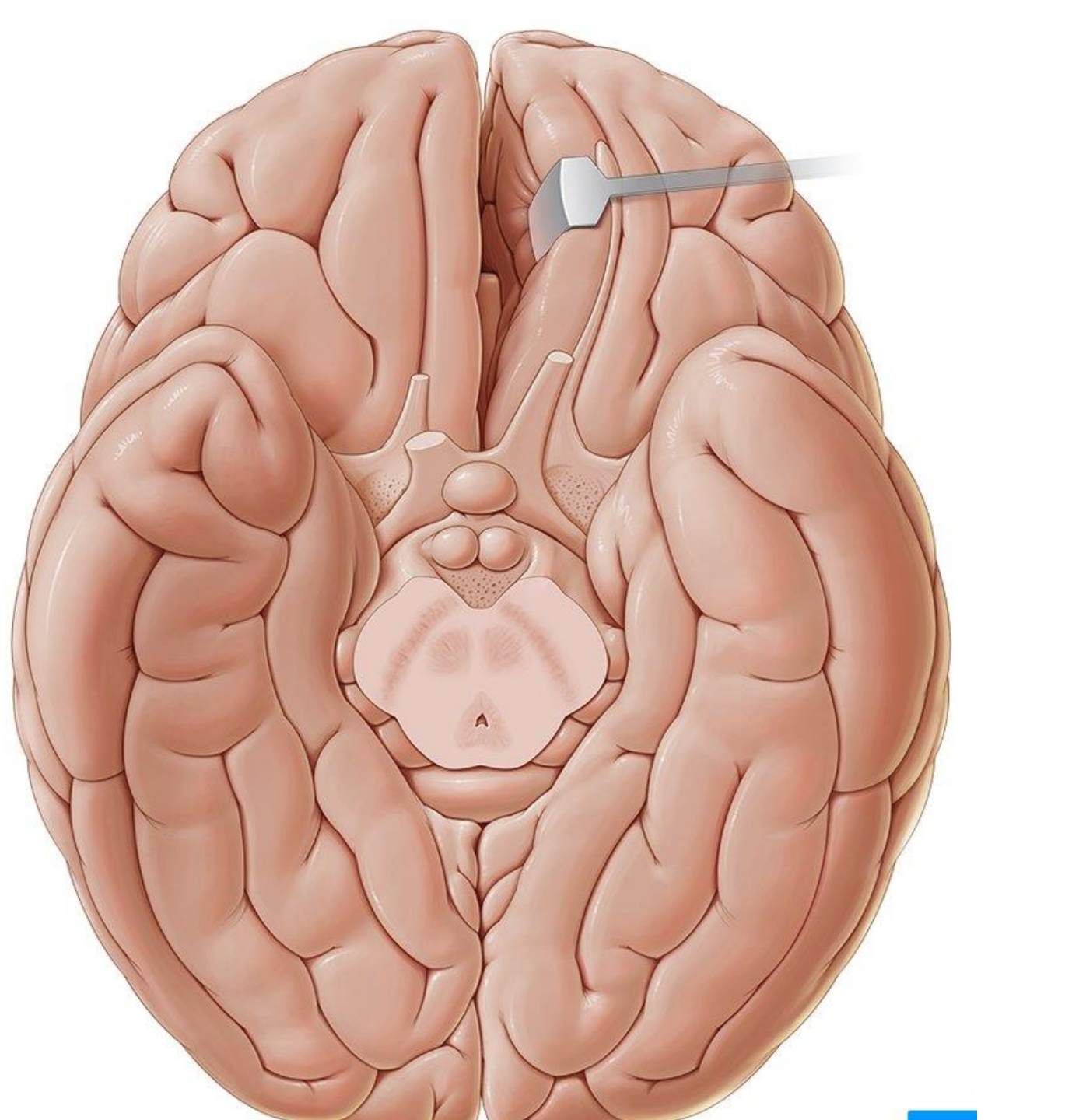
- Consists of brain and spinal cord
- Brain made up of
- **Forebrain**
 - Two cerebral hemispheres (telencephalon)
 - lateral ventricles
 - Diencephalon
 - Thalamus
 - Hypothalamus
 - Third ventricle
- **Midbrain**
 - Lies between forebrain to hindbrain
 - aqueduct
- **Hindbrain**
 - Pons
 - Medulla oblongata
 - cerebellum



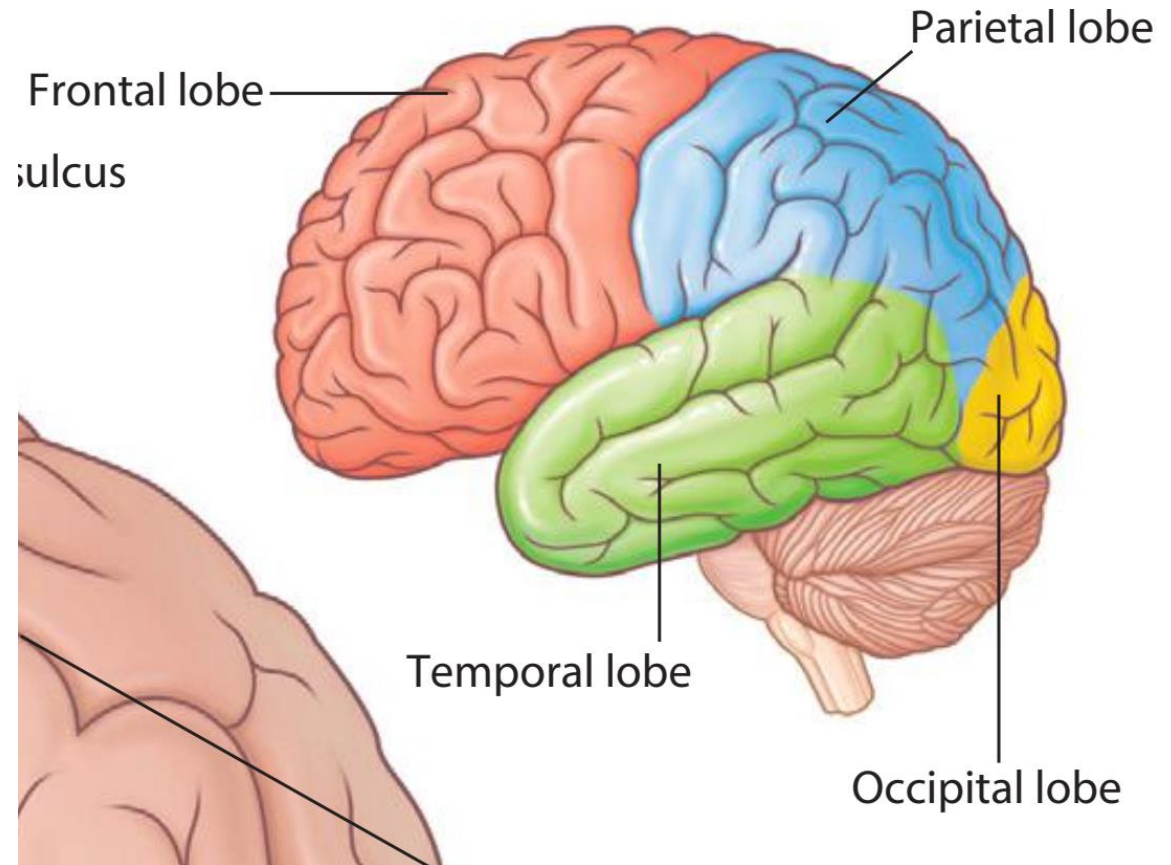
Forebrain (cerebrum)

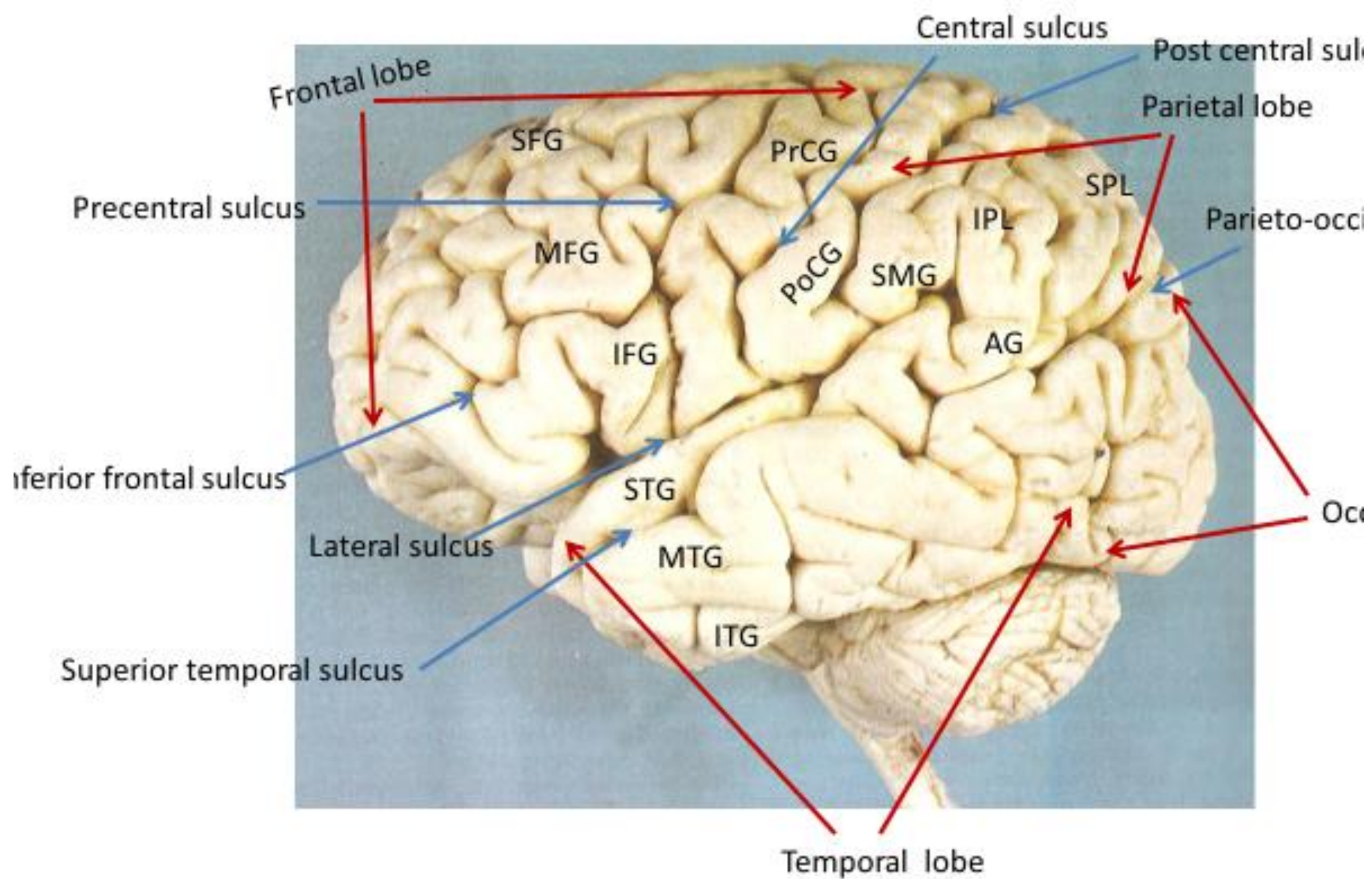
- Lies above
 - the floor of the cranial fossae
 - Tentorium cerebelli
- Medial surface (related to falx cerebri) is flat
- Two hemispheres joined by corpus callosum
- Superolateral surface: convex (skull shape)
- Inferior surface: irregular shaped
- All surfaces covered with cortex of grey matter
- Internally has basal nuclei and thalamus
- Cortex thrown into gyri and sulci





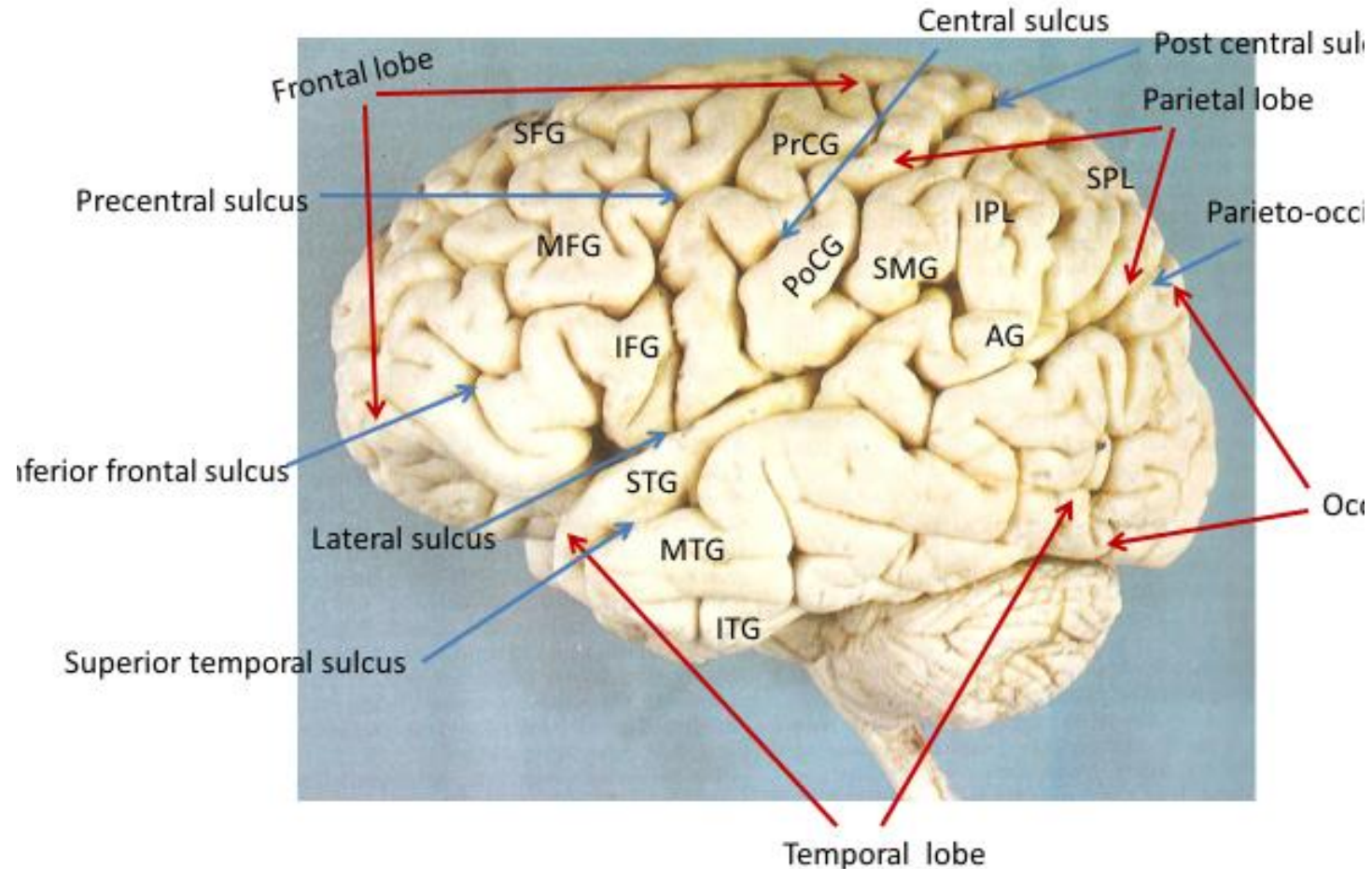
- Dominant hemisphere slightly larger than the other
- Surfaces divided into lobes by larger sulci
- **Lobes** named according to adjacent cranial bone
 1. Frontal
 2. Parietal
 3. Occipital
 4. Temporal
- **Sulci**
 1. Central
 2. Lateral
 3. Parieto-occipital

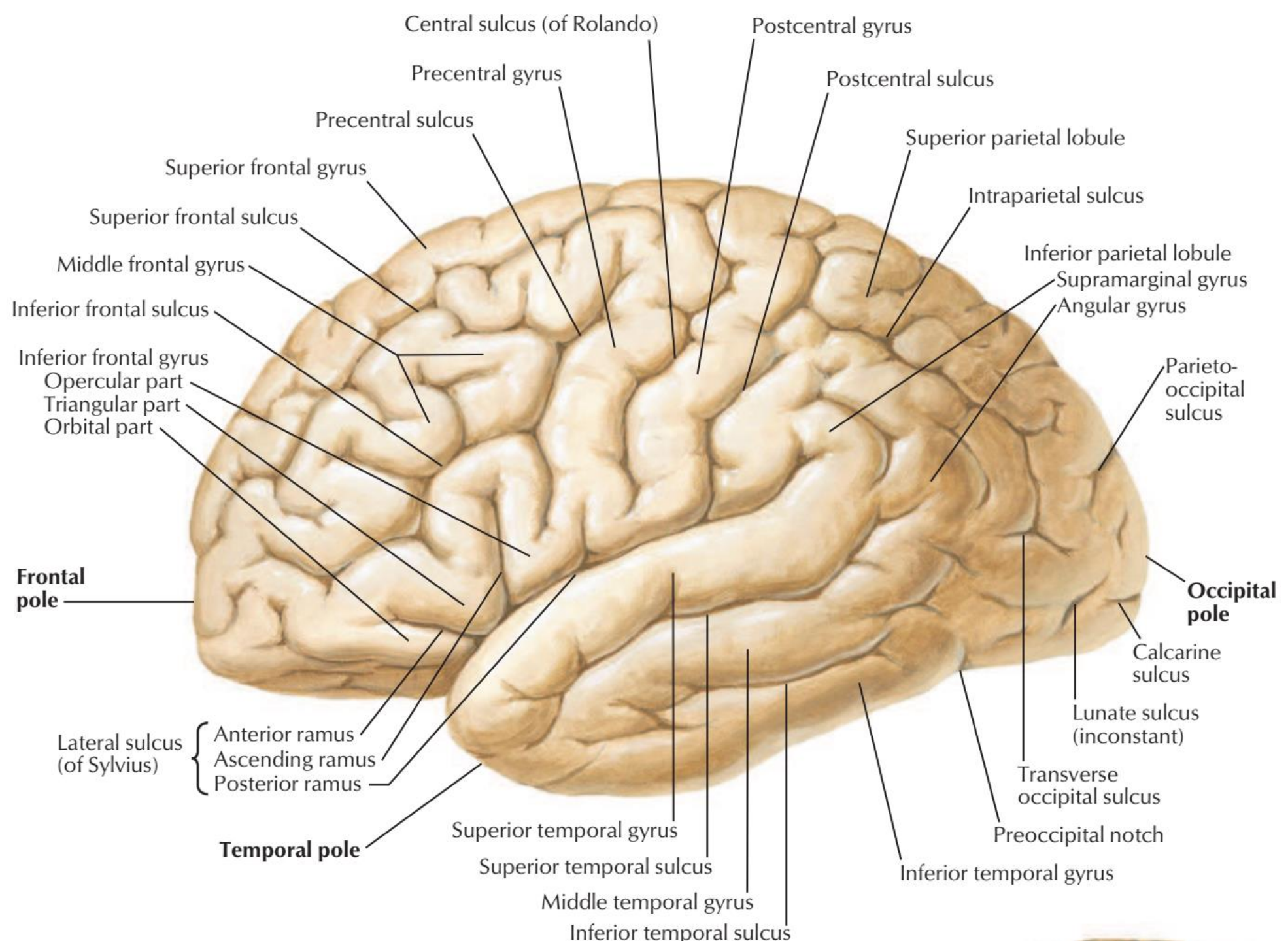




Main lobes and sulci

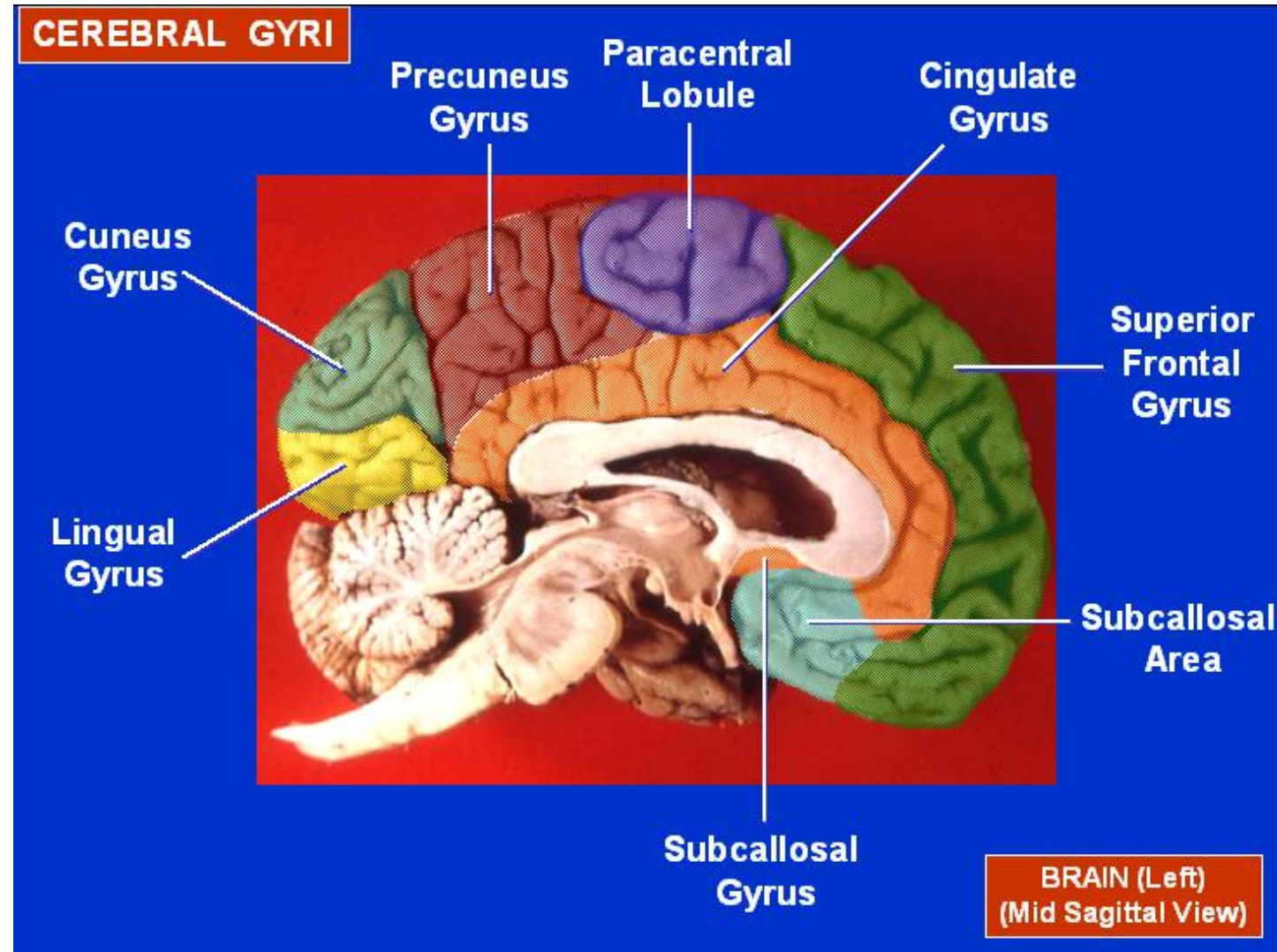
- The frontal lobe occupies the area anterior to the *central sulcus* (*fissure of Rolando*) and superior to the *lateral sulcus* (*fissure of Sylvius*)
- The parietal lobe occupies the area between the central sulcus and the *parieto-occipital sulcus* and superior to the lateral sulcus
- The temporal lobe occupies the area inferior to the lateral sulcus
- The occipital occupies the area posterior to the parieto-occipital sulcus



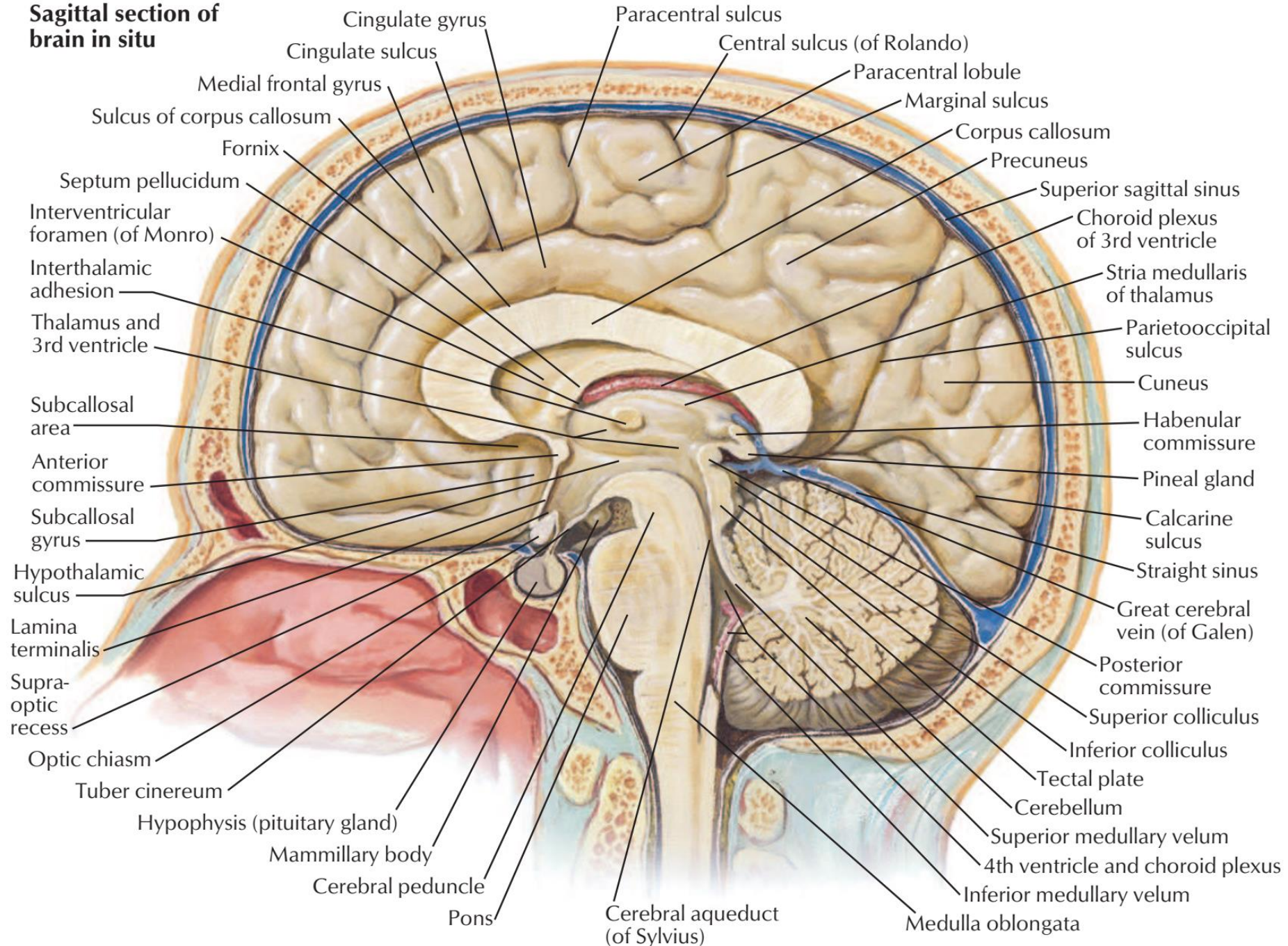


Medial surface

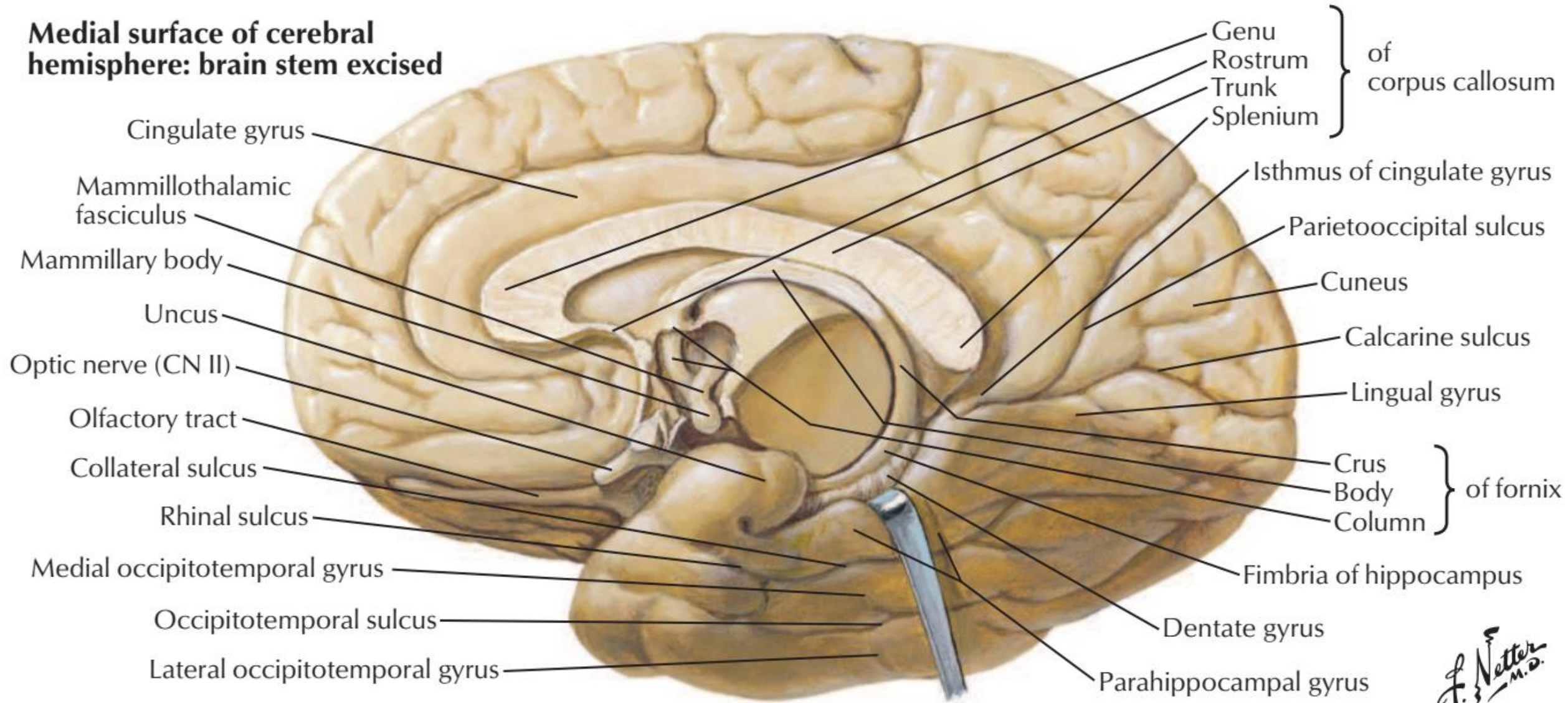
- lies against the falx cerebri in the longitudinal fissure
- Below the falx cerebri the 2 hemisphere are held together by corpus callosum
- Above corpus callosum: **note in front**
 - cingulate gyrus
 - cingulate sulcus
 - medial frontal gyrus
 - central sulcus
 - Paracentral lobule
 - Precuneus gyrus
- At the posterior part:
 - parieto-occipital sulcus
 - cuneus
 - calcarine sulcus
 - lingual gyrus



Sagittal section of brain in situ



Medial surface of cerebral hemisphere: brain stem excised



- Different areas of the cerebral cortex are functionally specialized
- Brodmann area divides cortex into areas of specializations

- Area 8: Frontal Eye Field**
- Contralateral horizontal gaze
 - Lesion: slow deviation of eyes towards the side of the lesion (bc of intact opposite cortex)
- Area 6: Premotor Cortex**
- Motor Planning
 - Lesion: **Apraxia** (unable to perform movements in the correct sequence)
- Area 4: Primary Motor Cortex**
- Precentral gyrus
 - Contains an orderly skeletal motor map of CL side of body
 - Lesion: contralateral (CL) spastic paresis
- Areas 3, 1, 2: Primary Somatosensory Cortex**
- Postcentral gyrus
 - Discriminative touch, vibration, position sense, pain & T₂
 - Lesion: impairment of all somatic sensations on the CL body
- Prefrontal Cortex**
- Areas 9 (Dorsolateral) & 10 (Ventral)**
- 1/4 cerebral cortex
 - Organizes & plans the intellectual & emotional aspects of behavior
 - Lesion: **Frontal Lobe Syndrome**
 - Lack of CT, easily distracted
 - Lack of initiative, foresight & perspective
 - Apathy + Abulia + slowing of intellectual activities + slow speech + ↓ social interaction
 - Involving group activities
- Area 43: Primary Gustatory Cortex**
- Areas 41 & 42: Primary Auditory Cortex**
- Area 39: Angular Gyrus**
- Area 40: Supramarginal gyrus**
- Area 22: Auditory Association Cortex**
- Area 37: Wernicke's area**
- Area 20: Inferior temporal gyrus**
- Area 21: Superior temporal gyrus**
- Area 18: Middle temporal gyrus**
- Area 17: Primary Visual Cortex**
- Area 19: Middle occipital gyrus**
- Area 18 & 19: Magnocellular system & alpha Rod Vision (Peripheral Retina)**
- Area 20 & 21: Parvocellular system & color cone vision (central Retina)**
- Area 22: Auditory Association Cortex**
- Area 39: Angular Gyrus**
- Area 40: Supramarginal gyrus**
- Area 43: Primary Gustatory Cortex**
- Area 41 & 42: Primary Auditory Cortex**
- Area 37: Wernicke's area**
- Area 20: Inferior temporal gyrus**
- Area 21: Superior temporal gyrus**
- Area 18: Middle temporal gyrus**
- Area 17: Primary Visual Cortex**
- Area 19: Middle occipital gyrus**
- Area 18 & 19: Magnocellular system & alpha Rod Vision (Peripheral Retina)**
- Area 20 & 21: Parvocellular system & color cone vision (central Retina)**
- Area 22: Auditory Association Cortex**
- Area 39: Angular Gyrus**
- Area 40: Supramarginal gyrus**
- Area 43: Primary Gustatory Cortex**
- Area 41 & 42: Primary Auditory Cortex**
- Area 37: Wernicke's area**
- Area 20: Inferior temporal gyrus**
- Area 21: Superior temporal gyrus**
- Area 18: Middle temporal gyrus**
- Area 17: Primary Visual Cortex**
- Area 19: Middle occipital gyrus**
- Area 18 & 19: Magnocellular system & alpha Rod Vision (Peripheral Retina)**
- Area 20 & 21: Parvocellular system & color cone vision (central Retina)**
- Area 22: Auditory Association Cortex**
- Area 39: Angular Gyrus**
- Area 40: Supramarginal gyrus**
- Area 43: Primary Gustatory Cortex**
- Area 41 & 42: Primary Auditory Cortex**
- Area 37: Wernicke's area**
- Area 20: Inferior temporal gyrus**
- Area 21: Superior temporal gyrus**
- Area 18: Middle temporal gyrus**
- Area 17: Primary Visual Cortex**
- Area 19: Middle occipital gyrus**
- Area 18 & 19: Magnocellular system & alpha Rod Vision (Peripheral Retina)**
- Area 20 & 21: Parvocellular system & color cone vision (central Retina)**
- Area 22: Auditory Association Cortex**
- Area 39: Angular Gyrus**
- Area 40: Supramarginal gyrus**
- Area 43: Primary Gustatory Cortex**
- Area 41 & 42: Primary Auditory Cortex**
- Area 37: Wernicke's area**
- Area 20: Inferior temporal gyrus**
- Area 21: Superior temporal gyrus**
- Area 18: Middle temporal gyrus**
- Area 17: Primary Visual Cortex**
- Area 19: Middle occipital gyrus**
- Area 18 & 19: Magnocellular system & alpha Rod Vision (Peripheral Retina)**
- Area 20 & 21: Parvocellular system & color cone vision (central Retina)**
- Area 22: Auditory Association Cortex**
- Area 39: Angular Gyrus**
- Area 40: Supramarginal gyrus**
- Area 43: Primary Gustatory Cortex**
- Area 41 & 42: Primary Auditory Cortex**
- Area 37: Wernicke's area**
- Area 20: Inferior temporal gyrus**
- Area 21: Superior temporal gyrus**
- Area 18: Middle temporal gyrus**
- Area 17: Primary Visual Cortex**
- Area 19: Middle occipital gyrus**
- Area 18 & 19: Magnocellular system & alpha Rod Vision (Peripheral Retina)**
- Area 20 & 21: Parvocellular system & color cone vision (central Retina)**
- Area 22: Auditory Association Cortex**
- Area 39: Angular Gyrus**
- Area 40: Supramarginal gyrus**
- Area 43: Primary Gustatory Cortex**
- Area 41 & 42: Primary Auditory Cortex**
- Area 37: Wernicke's area**
- Area 20: Inferior temporal gyrus**
- Area 21: Superior temporal gyrus**
- Area 18: Middle temporal gyrus**
- Area 17: Primary Visual Cortex**
- Area 19: Middle occipital gyrus**
- Area 18 & 19: Magnocellular system & alpha Rod Vision (Peripheral Retina)**
- Area 20 & 21: Parvocellular system & color cone vision (central Retina)**
- Area 22: Auditory Association Cortex**
- Area 39: Angular Gyrus**
- Area 40: Supramarginal gyrus**
- Area 43: Primary Gustatory Cortex**
- Area 41 & 42: Primary Auditory Cortex**
- Area 37: Wernicke's area**
- Area 20: Inferior temporal gyrus**
- Area 21: Superior temporal gyrus**
- Area 18: Middle temporal gyrus**
- Area 17: Primary Visual Cortex**
- Area 19: Middle occipital gyrus**
- Area 18 & 19: Magnocellular system & alpha Rod Vision (Peripheral Retina)**
- Area 20 & 21: Parvocellular system & color cone vision (central Retina)**
- Area 22: Auditory Association Cortex**
- Area 39: Angular Gyrus**
- Area 40: Supramarginal gyrus**
- Area 43: Primary Gustatory Cortex**
- Area 41 & 42: Primary Auditory Cortex**
- Area 37: Wernicke's area**
- Area 20: Inferior temporal gyrus**
- Area 21: Superior temporal gyrus**
- Area 18: Middle temporal gyrus**
- Area 17: Primary Visual Cortex**
- Area 19: Middle occipital gyrus**
- Area 18 & 19: Magnocellular system & alpha Rod Vision (Peripheral Retina)**
- Area 20 & 21: Parvocellular system & color cone vision (central Retina)**
- Area 22: Auditory Association Cortex**
- Area 39: Angular Gyrus**
- Area 40: Supramarginal gyrus**
- Area 43: Primary Gustatory Cortex**
- Area 41 & 42: Primary Auditory Cortex**
- Area 37: Wernicke's area**
- Area 20: Inferior temporal gyrus**
- Area 21: Superior temporal gyrus**
- Area 18: Middle temporal gyrus**
- Area 17: Primary Visual Cortex**
- Area 19: Middle occipital gyrus**
- Area 18 & 19: Magnocellular system & alpha Rod Vision (Peripheral Retina)**
- Area 20 & 21: Parvocellular system & color cone vision (central Retina)**
- Area 22: Auditory Association Cortex**
- Area 39: Angular Gyrus**
- Area 40: Supramarginal gyrus**
- Area 43: Primary Gustatory Cortex**
- Area 41 & 42: Primary Auditory Cortex**
- Area 37: Wernicke's area**
- Area 20: Inferior temporal gyrus**
- Area 21: Superior temporal gyrus**
- Area 18: Middle temporal gyrus**
- Area 17: Primary Visual Cortex**
- Area 19: Middle occipital gyrus**
- Area 18 & 19: Magnocellular system & alpha Rod Vision (Peripheral Retina)**
- Area 20 & 21: Parvocellular system & color cone vision (central Retina)**
- Area 22: Auditory Association Cortex**
- Area 39: Angular Gyrus**
- Area 40: Supramarginal gyrus**
- Area 43: Primary Gustatory Cortex**
- Area 41 & 42: Primary Auditory Cortex**
- Area 37: Wernicke's area**
- Area 20: Inferior temporal gyrus**
- Area 21: Superior temporal gyrus**
- Area 18: Middle temporal gyrus**
- Area 17: Primary Visual Cortex**
- Area 19: Middle occipital gyrus**
- Area 18 & 19: Magnocellular system & alpha Rod Vision (Peripheral Retina)**
- Area 20 & 21: Parvocellular system & color cone vision (central Retina)**
- Area 22: Auditory Association Cortex**
- Area 39: Angular Gyrus**
- Area 40: Supramarginal gyrus**
- Area 43: Primary Gustatory Cortex**
- Area 41 & 42: Primary Auditory Cortex**
- Area 37: Wernicke's area**
- Area 20: Inferior temporal gyrus**
- Area 21: Superior temporal gyrus**
- Area 18: Middle temporal gyrus**
- Area 17: Primary Visual Cortex**
- Area 19: Middle occipital gyrus**
- Area 18 & 19: Magnocellular system & alpha Rod Vision (Peripheral Retina)**
- Area 20 & 21: Parvocellular system & color cone vision (central Retina)**
- Area 22: Auditory Association Cortex**
- Area 39: Angular Gyrus**
- Area 40: Supramarginal gyrus**
- Area 43: Primary Gustatory Cortex**

Area 8: Frontal Eye Field

- Contralateral horizontal gaze
- Lesion: slow deviation of eyes towards the side of the lesion (bc of intact opposite cortex)

Area 6: Premotor Cortex

- Motor Planning
- Lesion: **Apraxia** (unable to perform movements in the correct sequence)

Area 4: Primary Motor Cortex

- Precentral gyrus
- Motor control
- contains an orderly skeletal motor map of CL side of body
- Postcentral gyrus
- Discriminative touch, vibration, pain & temp
- Lesion: contralateral (CL) spastic paresis
- Lesion: impairment of all somatic sensations on the CL body.

Areas 3, 1, 2: Primary Somatosensory Cortex

Prefrontal Cortex

Areas 9 (Dorsolateral) & 10 (Ventral)

- 1/4 cerebral cortex
- Organizes & plans the intellectual & emotional aspects of behavior
- Lesion: **Frontal Lobe Syndrome**
- Lack of [I] easily distracted
- Lack of initiative, foresight & perspective
- Apathy + Abulia + slowing of intellectual faculties + slow speech + ↓ social interaction
- Ruckelshausen's criteria

Areas 44 & 45:

BROCA'S AREA

- Dominant hemisphere only
- Center of motor speech
- Lesion: motor, nonfluent or expressive **aphasia** → understand written & spoken language but can't verbalize their thoughts
- **Agaphia**
- Frustrated
- DT!

Area 43: Primary Gustatory Cortex

Areas 41 & 42: Primary Auditory Cortex

- Transverse gyrus of Heschl
- Reception & recognition of auditory stimuli
- Projects to area 22
- Lesion: unilateral slight hearing loss & difficulty understanding speech in noisy environment

Area 22: Auditory Association Cortex

- Language comprehension center
- Lesion: sensory, fluent or receptive **aphasia** → can't comprehend spoken language, sometimes w/ **alexia** (left hemisphere) but lacks meaning "word salad"
- PT's unaware

Areas 22, 39, 40: Wernicke's area

Area 39: Angular Gyrus

Area 40: Supramarginal gyrus

- Regions of convergence of visual, auditory & somatosensory information
- Lesion of Area 40: **Gerstmann Syndrome**
- Apraxia
- Acalculia
- Agnosia
- Impaired left hand/foot

Areas 5 & 7: Somatosensory Association Cortex

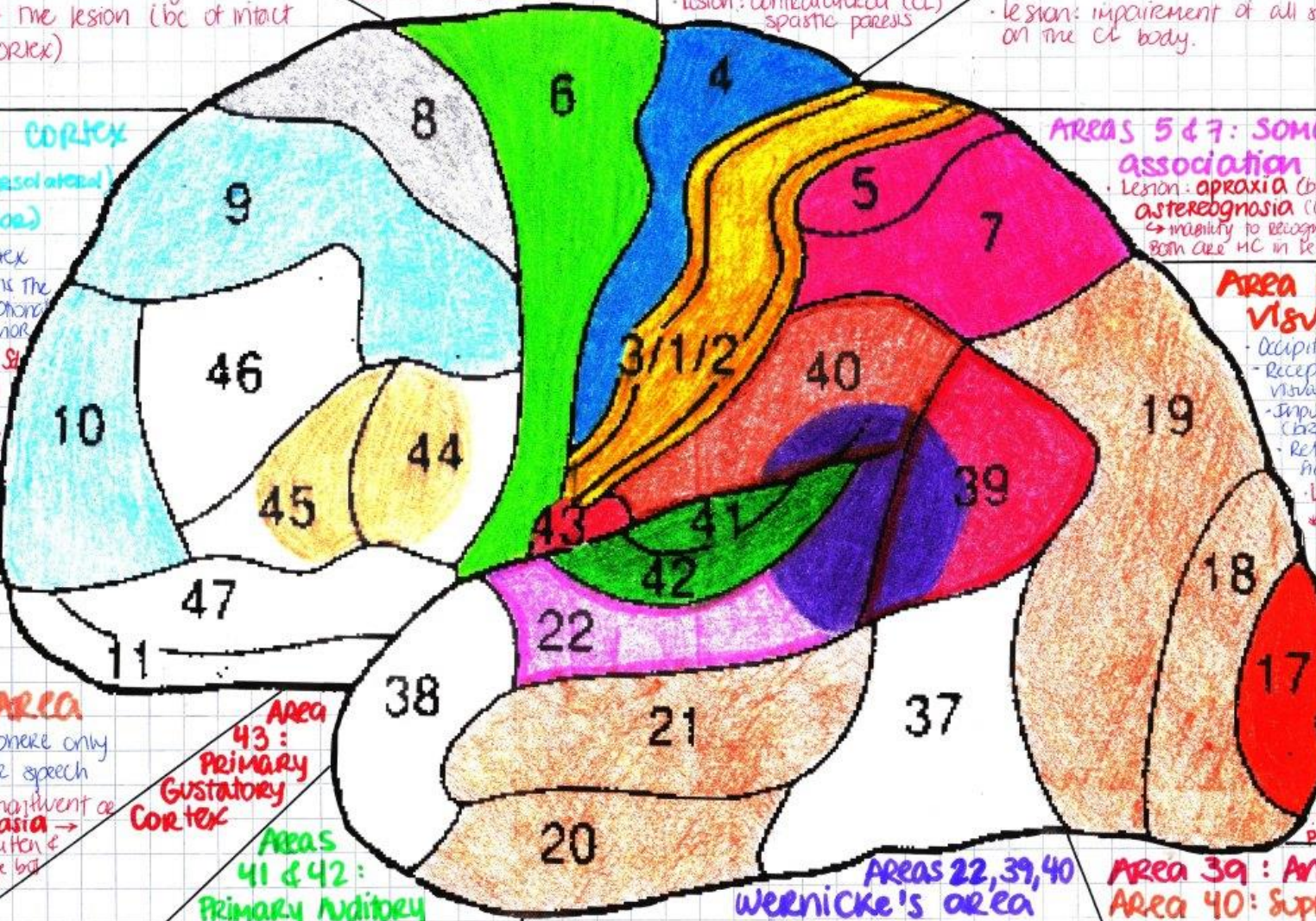
- Lesion: **apraxia** (bilateral) & **astereognosis** (contralateral)
- → inability to recognize objects by touch
- Both are MC in left hemisphere damage

Area 17: Primary Visual Cortex

- Occipital lobe → medial portion
- Reception & recognition of visual stimuli
- Input fibres: Line of Gennari (brain surface)
- Retinal surface (visual field) is represented here
- Lesion: **scotomas**
- Unilateral lesion (PCA): homonymous hemianopia w/ macular sparing

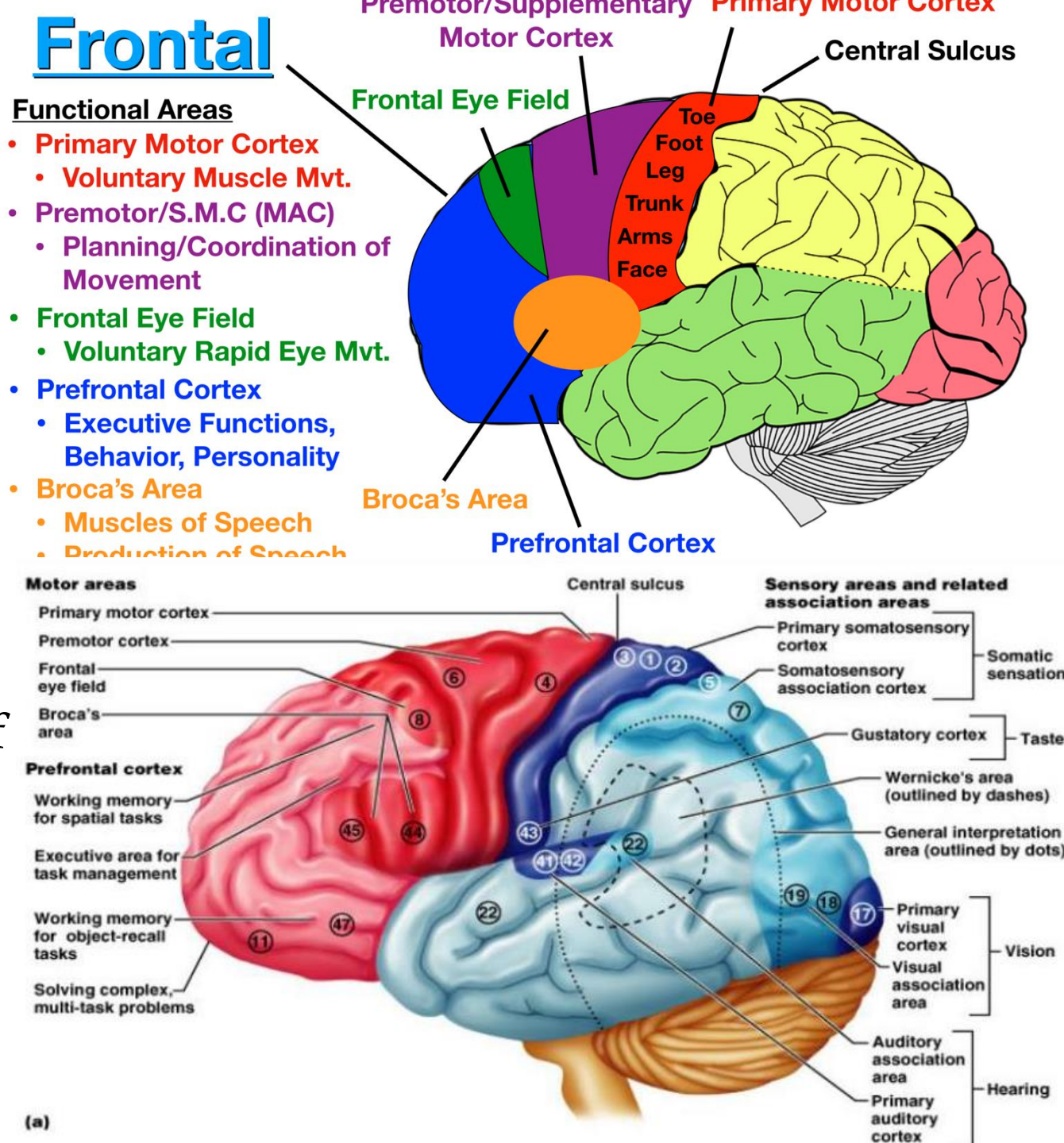
Areas 18, 19, 20 & 21: Visual Association C.

- 18 & 19: Magnocellular system & depth; Rod stream (peripheral retina)
- Lesion: deficit in perceiving visual motion
- 20 & 21: Parvocellular system & color; cone stream (central retina) vis agnosia
- Lesion: CL achromatopsia & prosopagnosia (→ faces)



Frontal lobe

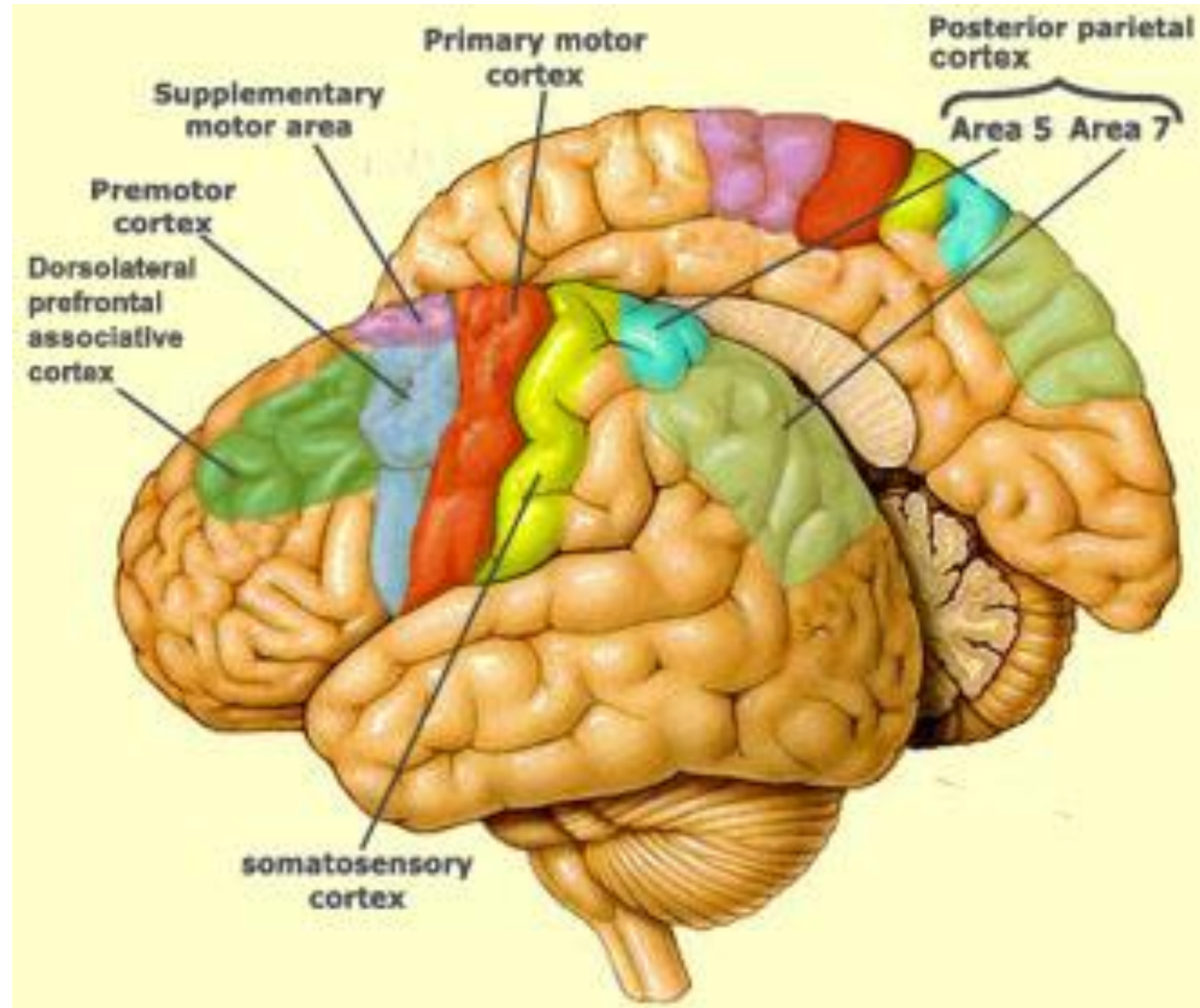
- Precentral area: precentral gyrus + posterior parts of superior, middle and inferior frontal gyrus.
 - Divided into
 - Posterior: primary motor area (4)
 - Voluntary movement of the contralateral side
 - Bilateral movements of extraocular muscles
 - Anterior: premotor area/secondary motor area (6, 8, 44, 45)
- Frontal eye field: voluntary movement of the eye
- Prefrontal area: personality
- Motor speech area of Broca (44 & 45):
 - Loc: Inferior frontal gyrus
 - Fxn: Formation of words



Parietal Lobe

Composed of

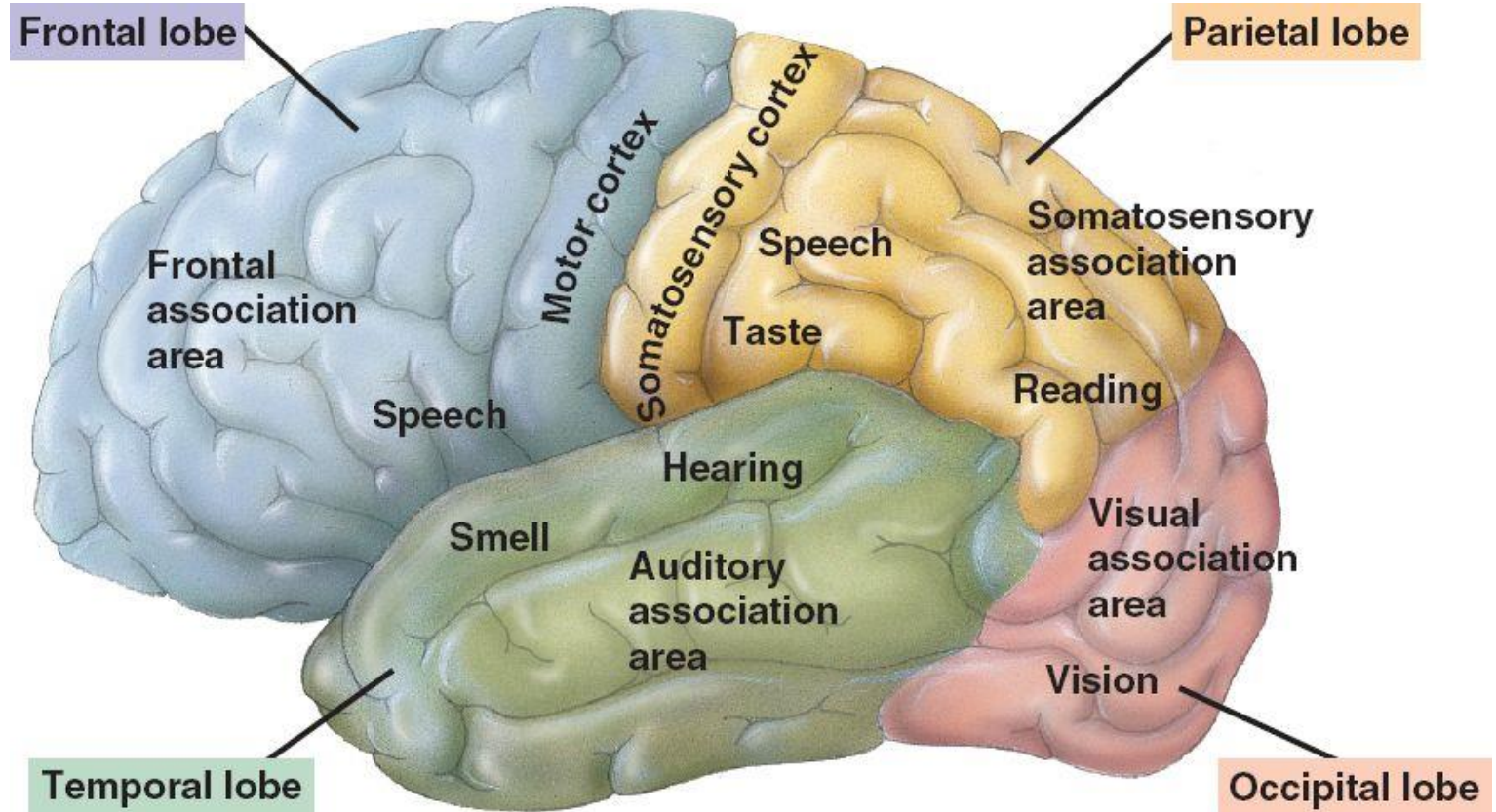
- Post central gyrus (sensory cortex)
 - primary somesthetic area
 - Secondary somesthetic area
 - Somesthetic associated area
- Allow to distinguish between the various modality sensation
- e.g. stereognosis, vibration sense



Occipital Lobe

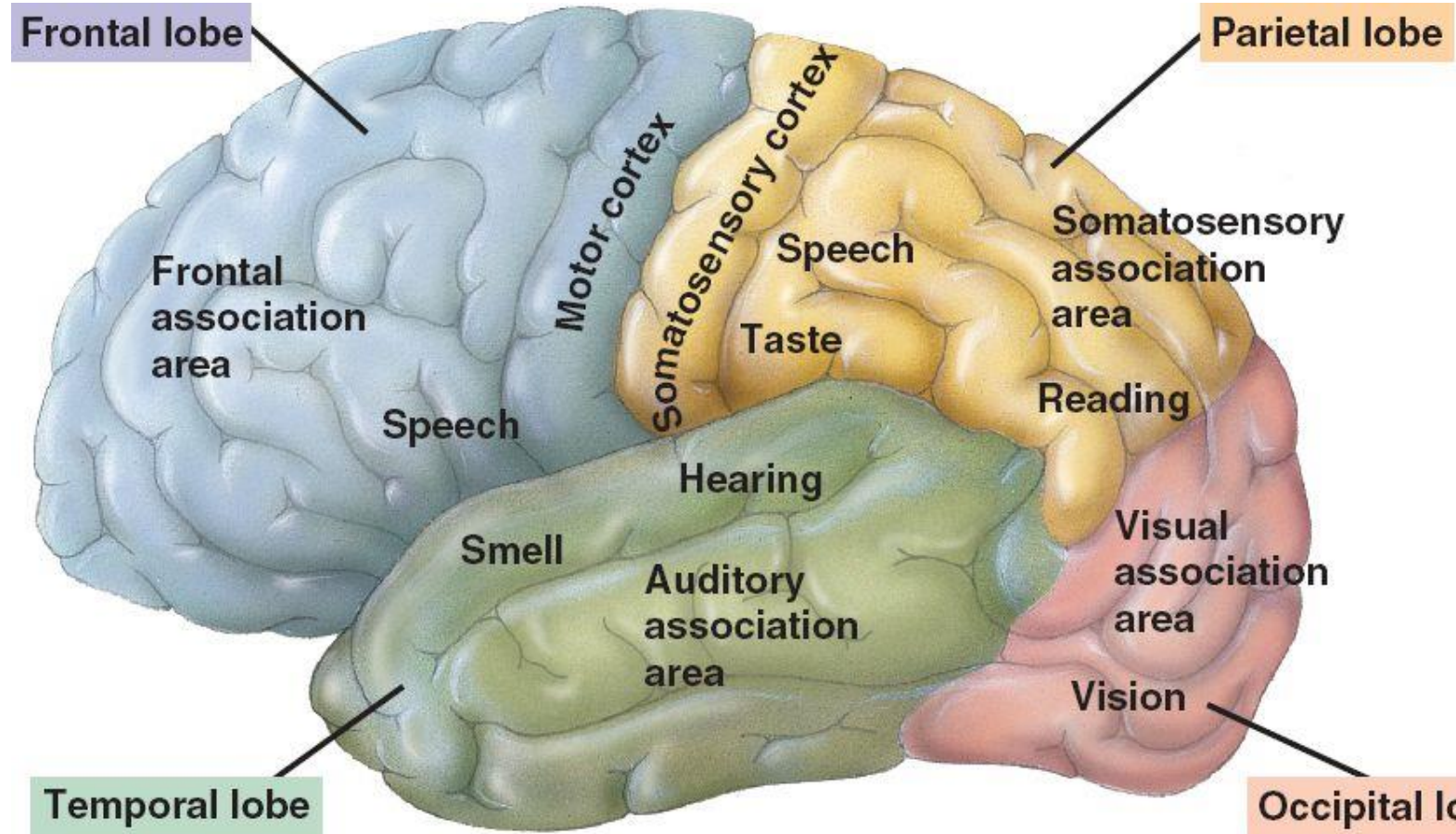
Composed of

- Primary visual cortex (Brodmann area 17 macula lutea is presented)
- Secondary visual cortex (allow individual to appreciate what is seeing)
- Occipital eye field (reflexes associated with movement of the eye)



Temporal lobe

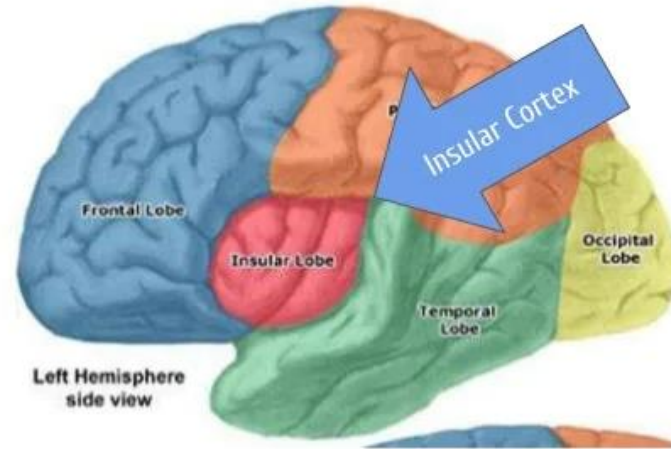
- Compose of:
- Primary auditory area (Brodmann's areas 41,42)
- Secondary auditory area (concern in the interpretation of sounds)
- Sensory speech area of Wernicke (concern in the understanding of reading and writing of sentences).



Other cortical areas

- Taste area
- Vestibular area
- Insular

The Insular Cortex



Lesion of cerebral cortex

- Primary motor cortex – paralysis of the contralateral side, Jacksonian epileptic seizure and Muscle spasticity
- Frontal eye field – the two eye deviating to the side of the lesion
- Motor speech area of Broca – expressive aphasia
- Motor speech area of Wernicke – receptive aphasia
- Both speech areas – global aphasia
- Dominant angular gyrus – alexia or agraphia

Lesion of cerebral cortex (cont.)

Prefrontal cortex –

- lost of initiative and judgment,
- emotional changes occur particularly euphoria.
- Personality change occurs and patient no longer conforms to the acceptable social behavior – become careless of dressing and appearance
- Frontal leukotomy and frontal lobectomy are used to reduce emotional respond in patient with obsessive emotional state and terminal patient with intractable pain

Lesion of cerebral cortex (cont.)

Somesthetic cortex –

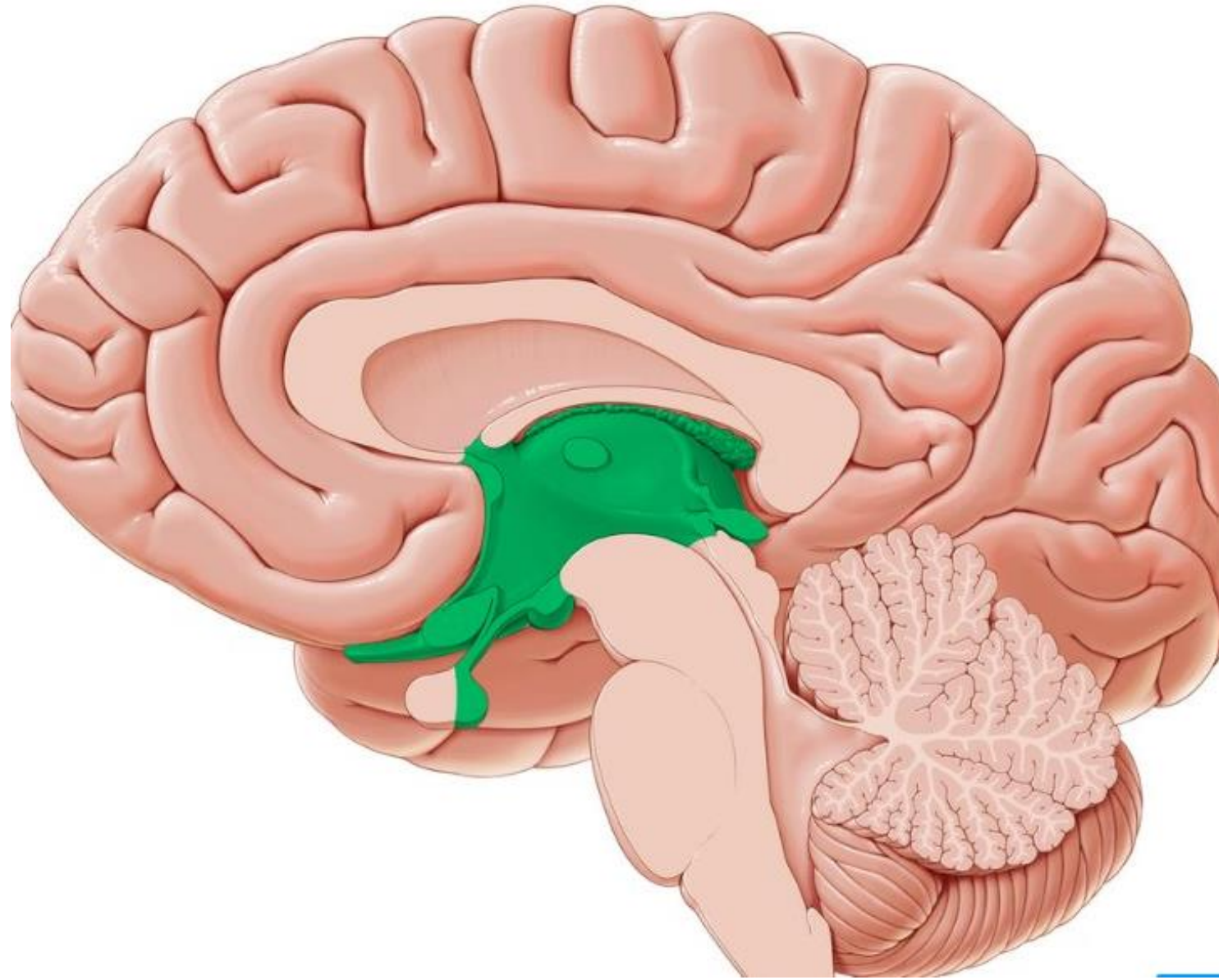
- can not judge the degree of sensory modality e.g. temperature, weight of object, can not localize tactile stimuli accurately

Visual cortex –

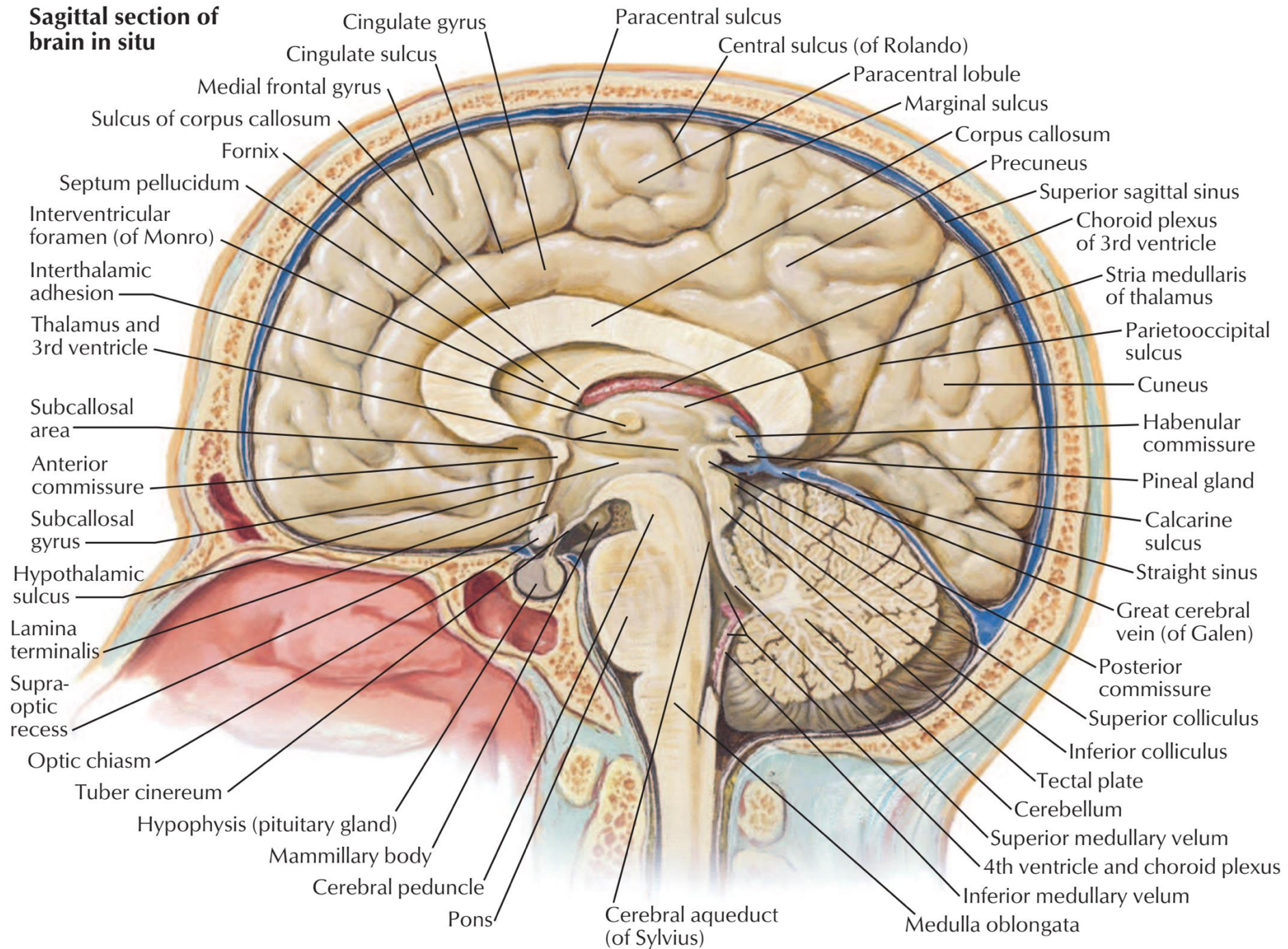
- contralateral loss of sight in the contralateral visual field (hemianopia)
- What is agnosia

Diencephalon

- The diencephalon consists of the
 - 3rd ventricle
 - Thalamus
 - Subthalamus
 - Epithalamus
 - metathalamus
 - hypothalamus



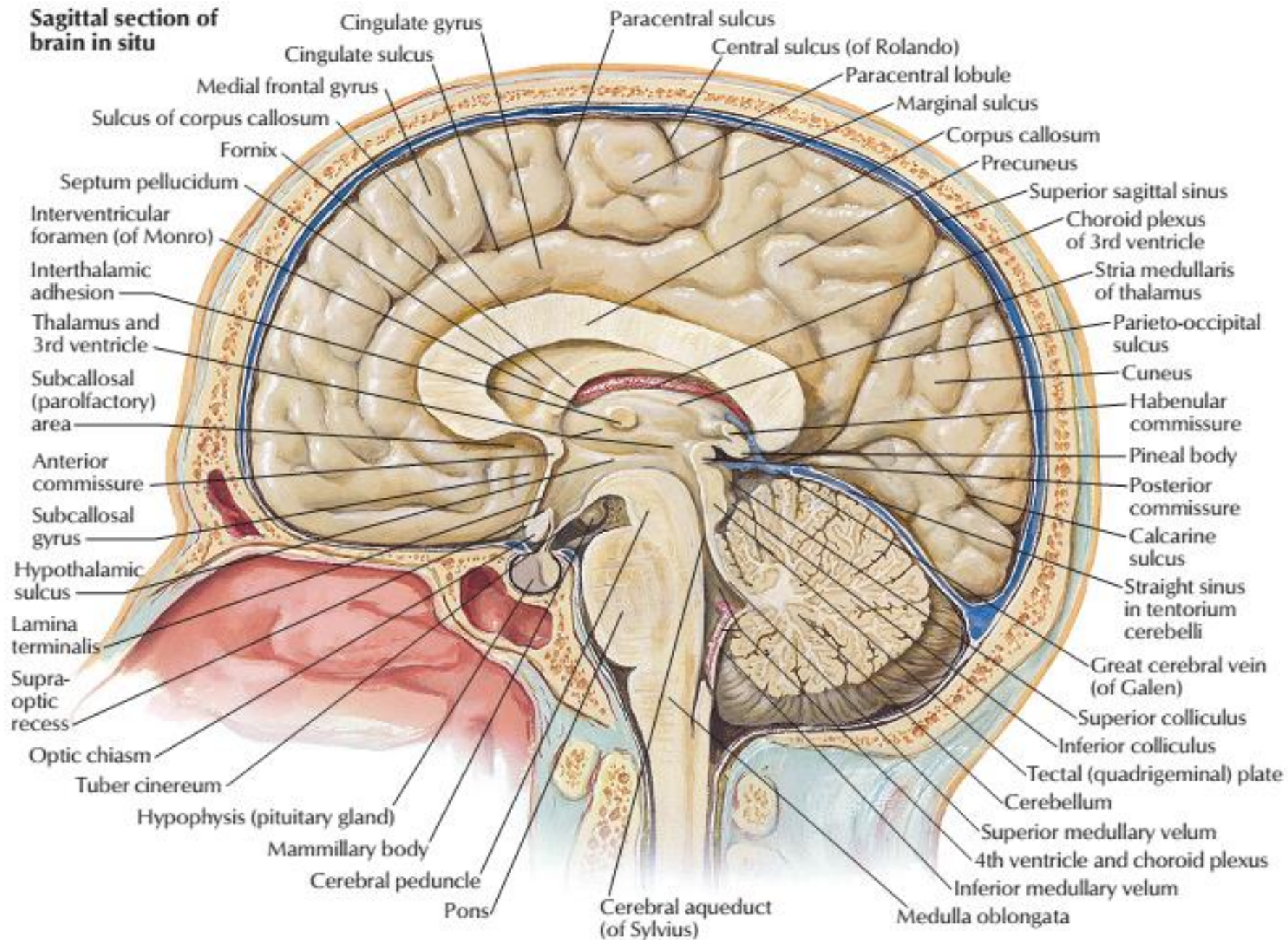
Sagittal section of brain in situ

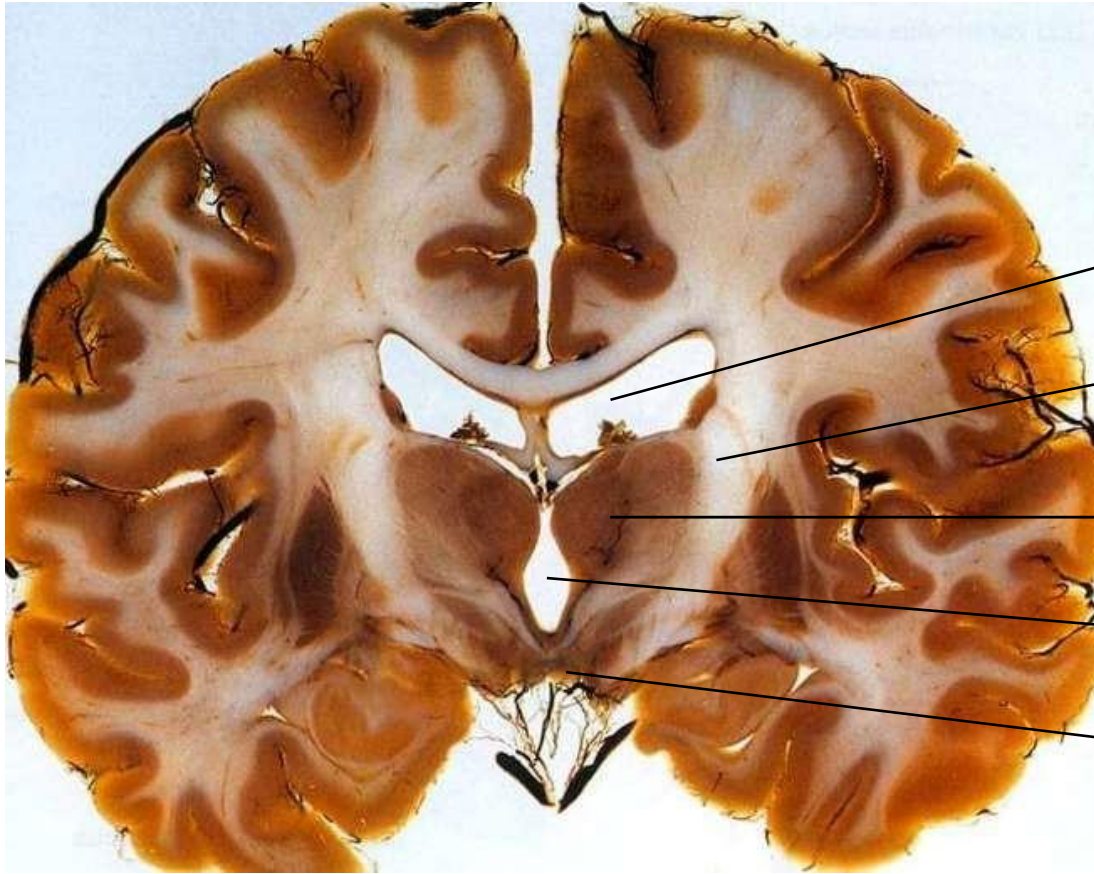


Thalamus- “inner chamber” or “meeting place”.

- large, egg-shaped mass of gray matter, located on each side of the superior aspect of the 3rd ventricle
- largest part of the diencephalon, about 4 cm long and 1.5 cm broad.
- Is an important relay and integrative centre for sensory and motor information passing to all areas of the cerebral cortex, the basal ganglia, the hypothalamus, and the brainstem.
 - There is reciprocal innervation between the thalamus and cerebral cortex, via corticothalamic and thalamocortical projections

Thalamus





→ **Dorsal** – Lateral Ventricle

→ **Lateral** – Internal capsule

→ **Thalamus**

→ **Medial** - 3rd Ventricle

→ **Ventral** – Sub thalamus & hypothalamus



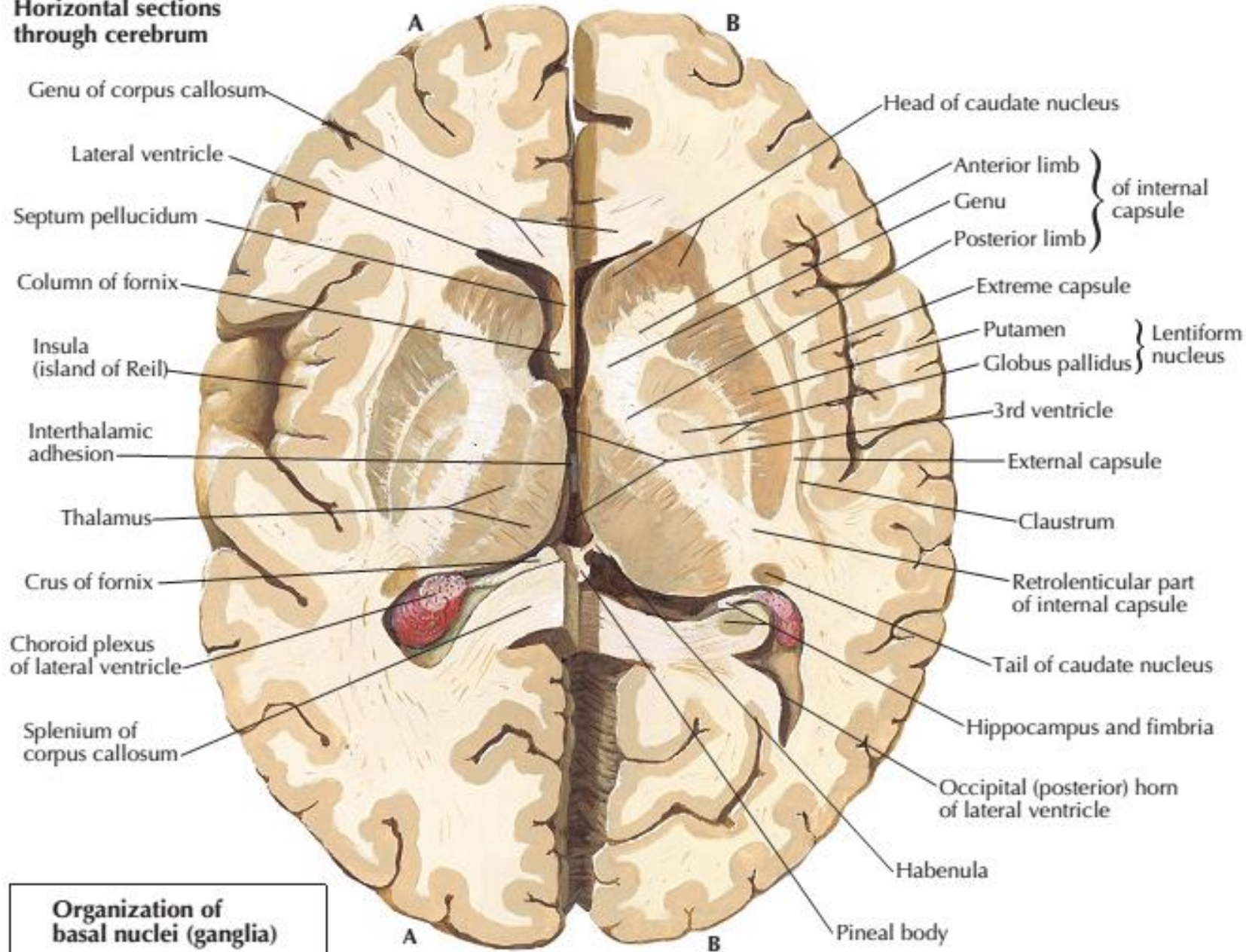
→ Massa intermedia

Thalamus- “inner chamber” or “meeting place”.

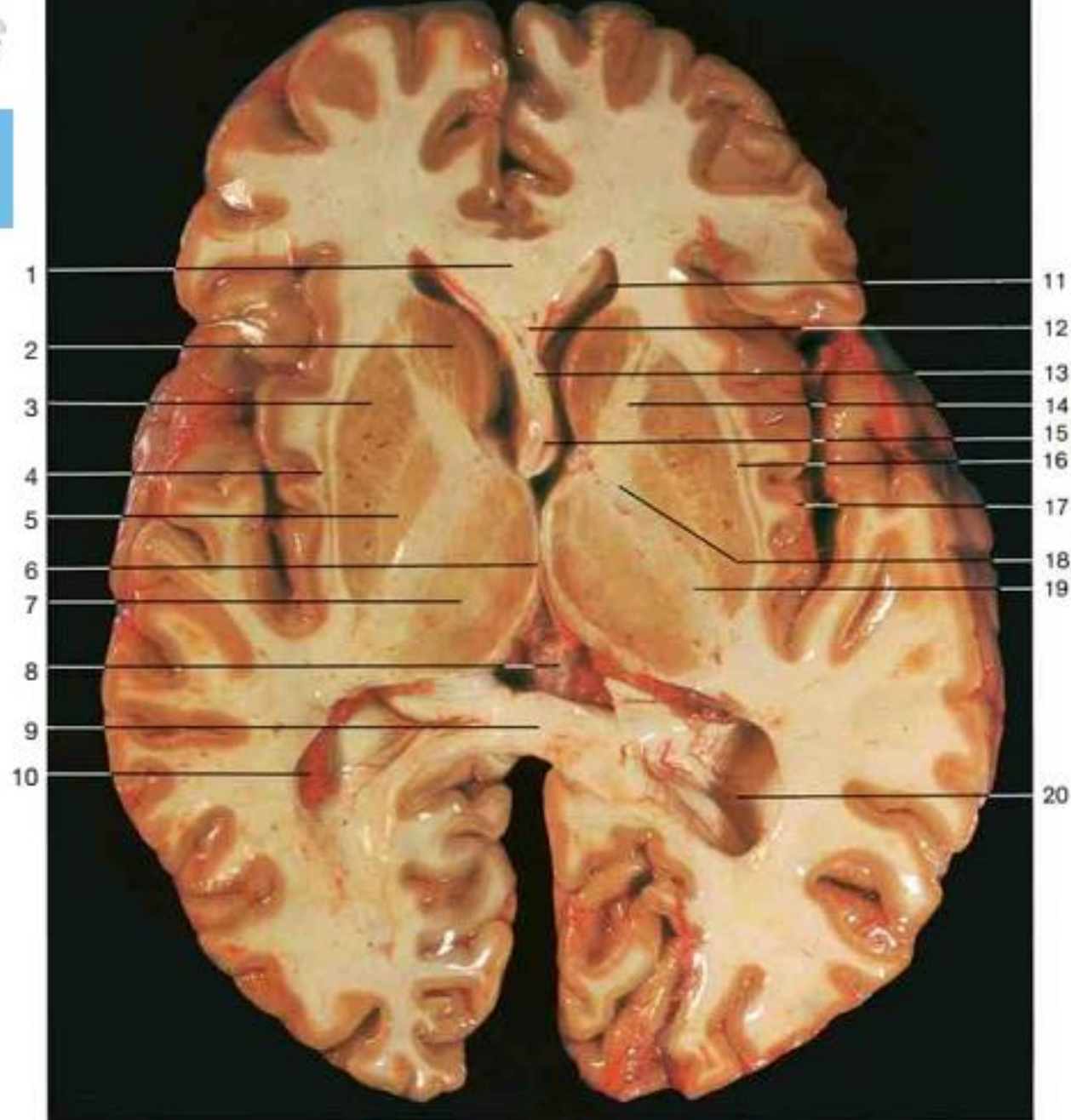
- All types of sensory information, **with the exception of olfaction**, reach the thalamic nuclei via afferent fibers.
- Other functions include roles in consciousness (connection with RAS), the sleep/wake cycle, emotion and memory

Thalamus

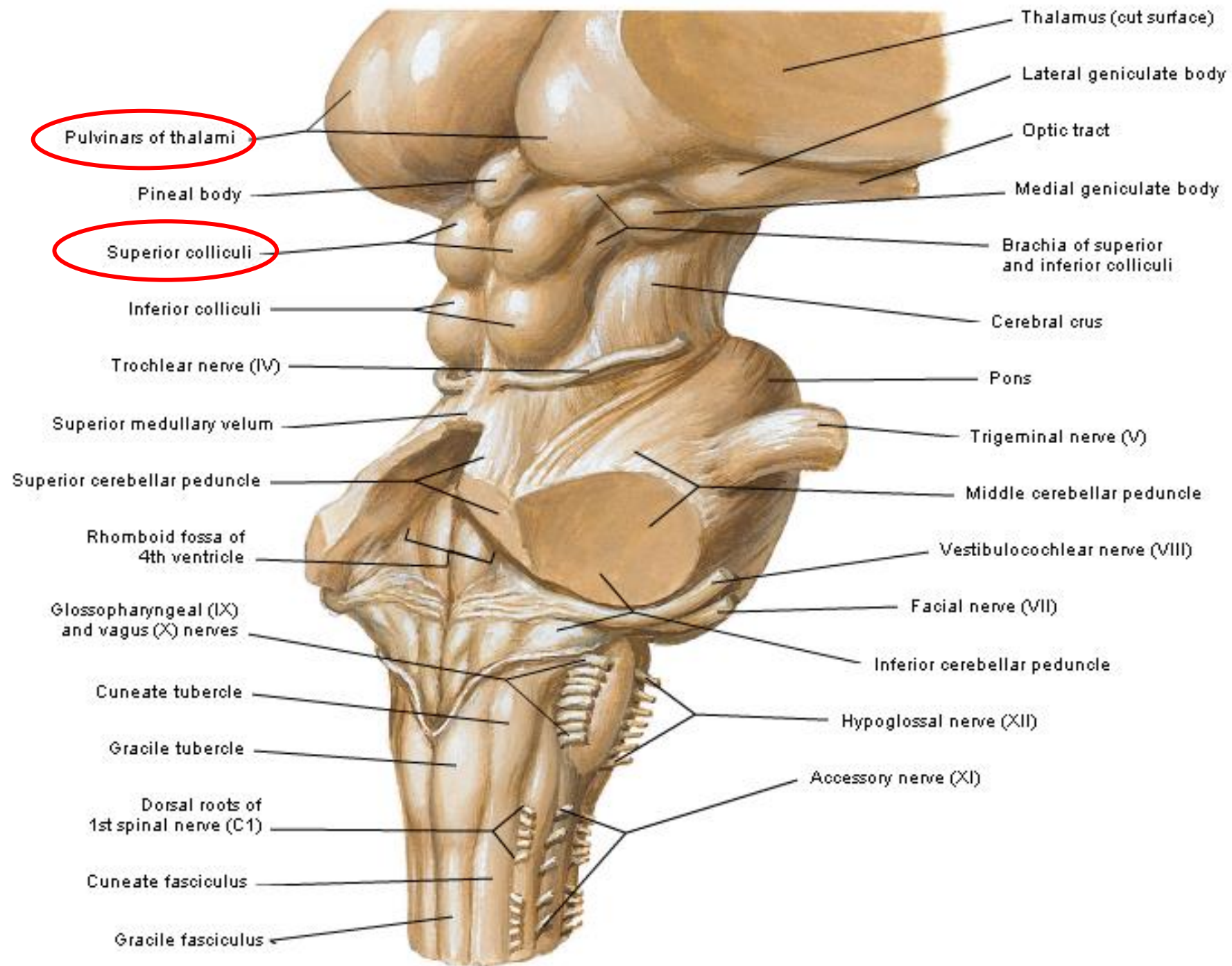
Horizontal sections through cerebrum



Thalam



Horizontal section through the brain, showing the subcortical nuclei and internal capsule. Section 1.



Thalamic nuclei

- **Relay nuclei**

- receive information (e.g. visual, acoustic, primary somatosensory, motor input) that is very specific and well-defined, and project this information to very specific, functionally discrete cortical areas.

- **Association nuclei**

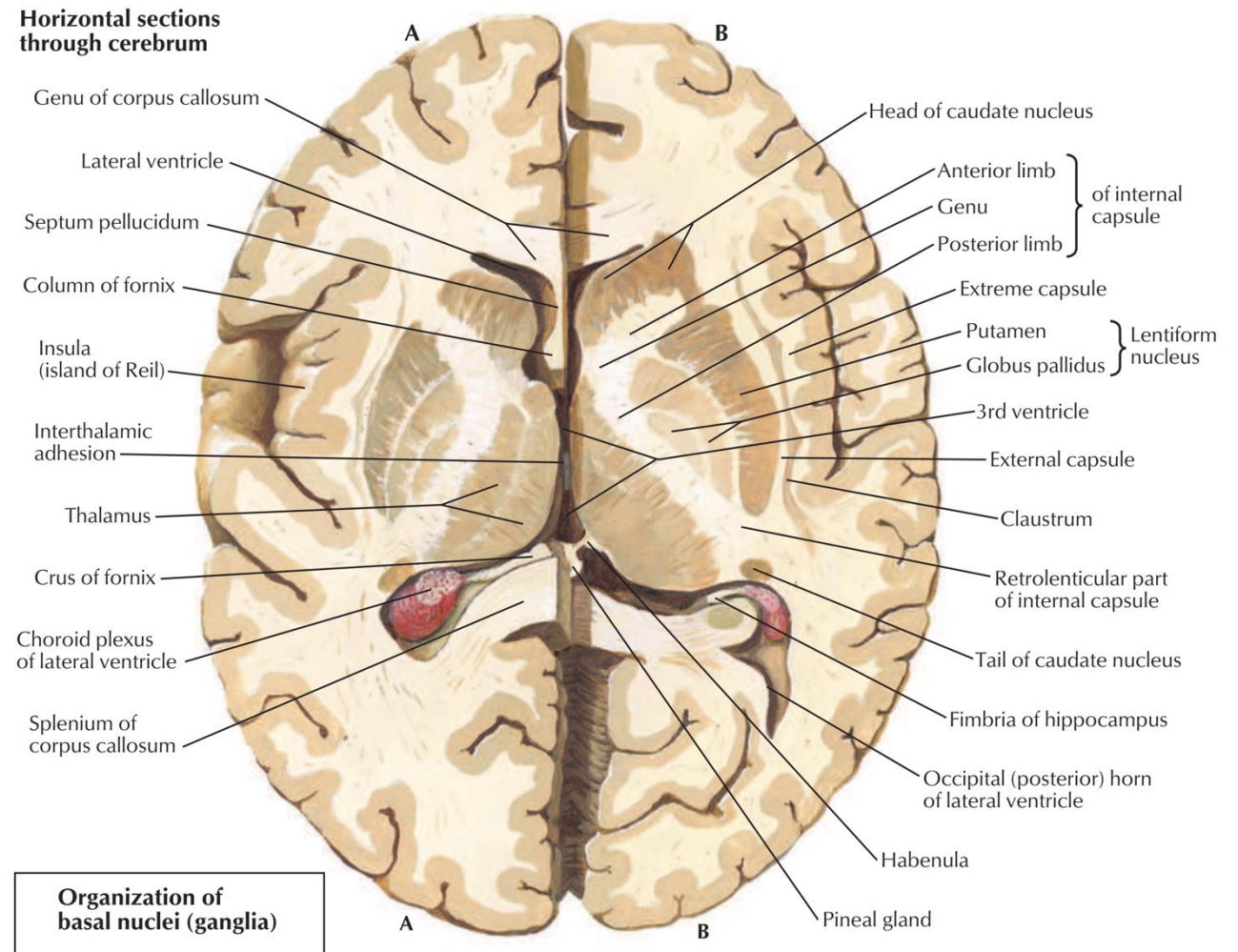
- receive most of their information from the cerebral cortex itself, and then project back to the “association areas” in the cortex to regulate the integration and interpretation of the sensory information.

- **Nonspecific nuclei**

- show broad and diffuse projections through the cerebral cortex
- may be involved in general functions such as consciousness and attention.

Basal nuclei (basal ganglia)

- The term basal nuclei referred to collection of masses of grey matter cells situated within the cerebral hemisphere
- The basal nuclei consists of
 - caudate nucleus,
 - lentiform nucleus,
 - amygdala
 - claustrum



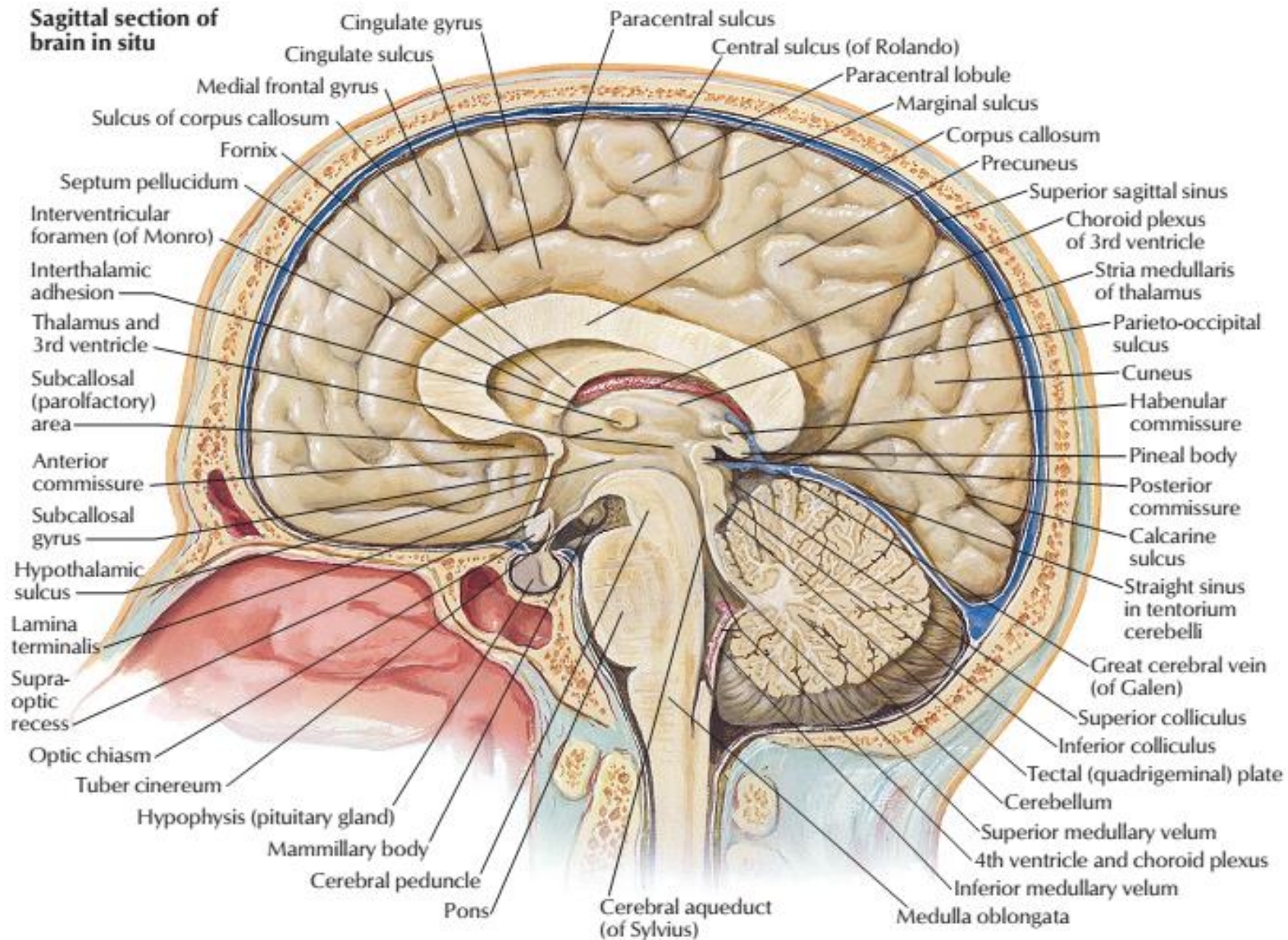
Brainstem

- Makes up about 2.5% of the total brain mass
- Includes the midbrain, the pons and the medulla oblongata, and the and occupies the posterior cranial fossa of the skull
- Lying rostrally is the diencephalon and caudally is the spinal cord
- Connected to the cerebellum by three pairs of cerebellar peduncles

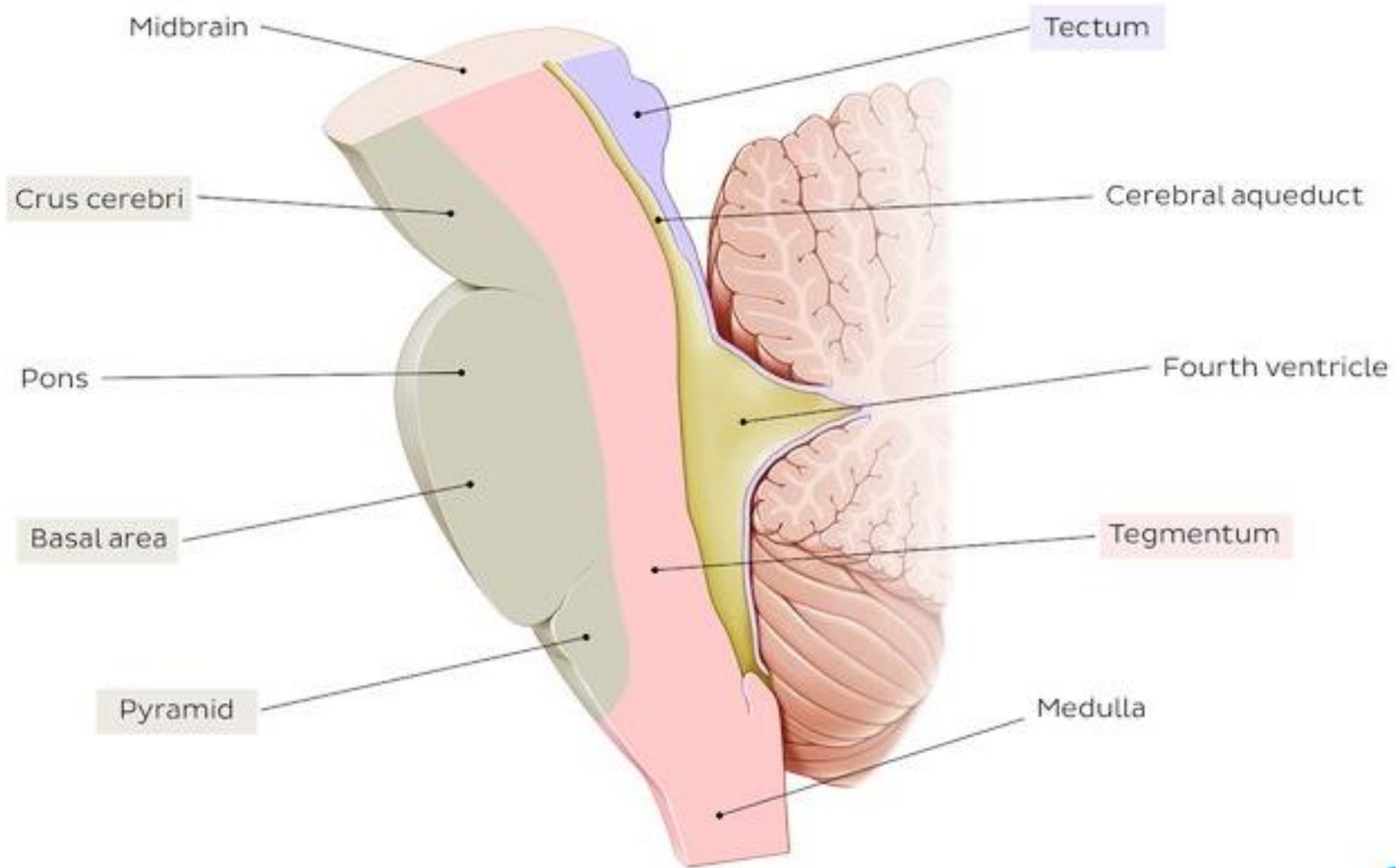
Functions.

- It serves as a conduit for ascending and descending tracts between the spinal cord, cerebellum and cerebrum
- It contains the reticular formation.
- It contains CN III – XII nuclei, red nucleus, substantia nigra, pontine nuclei and olivary nucleus

Brainstem



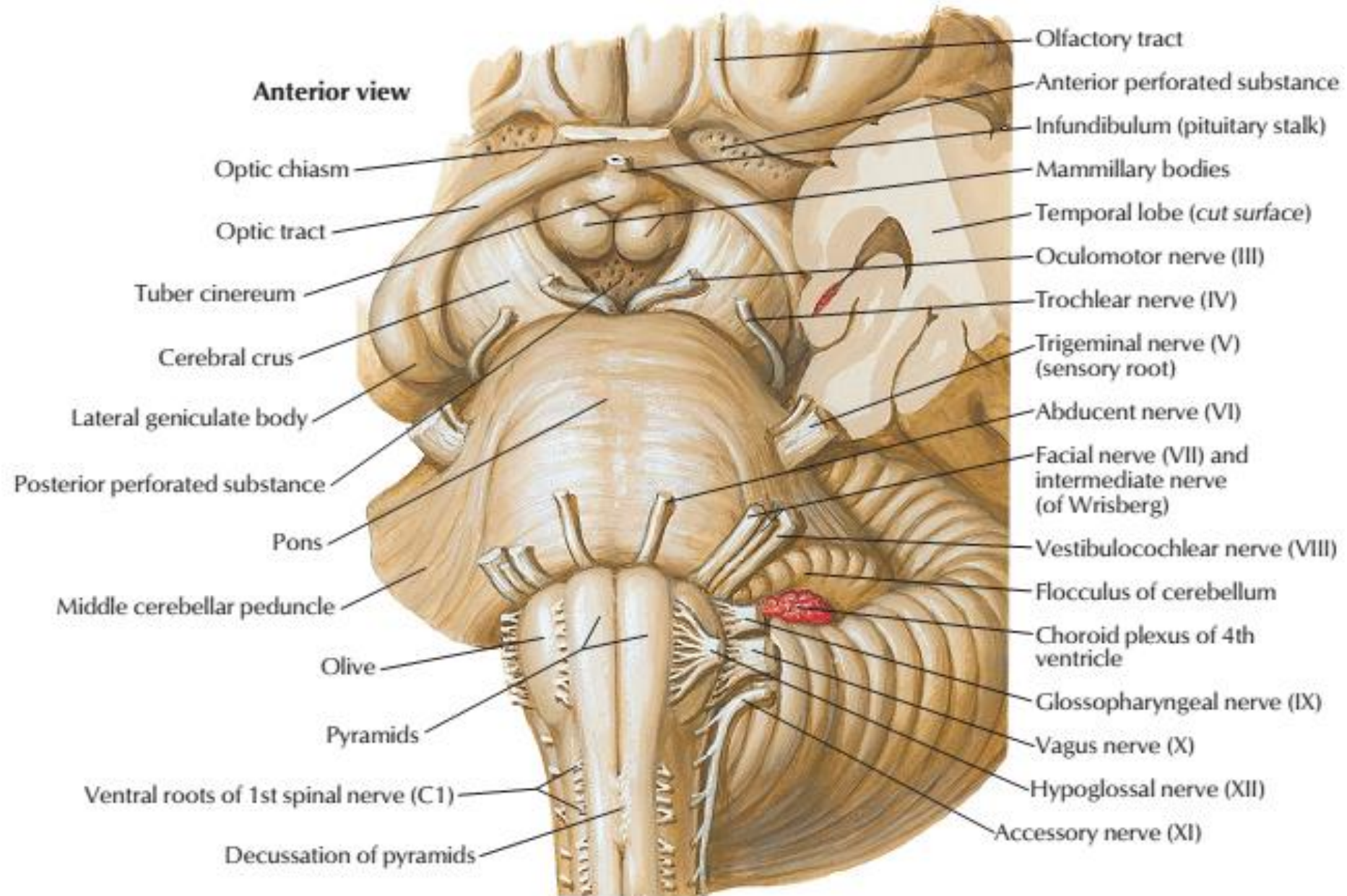
Brainstem



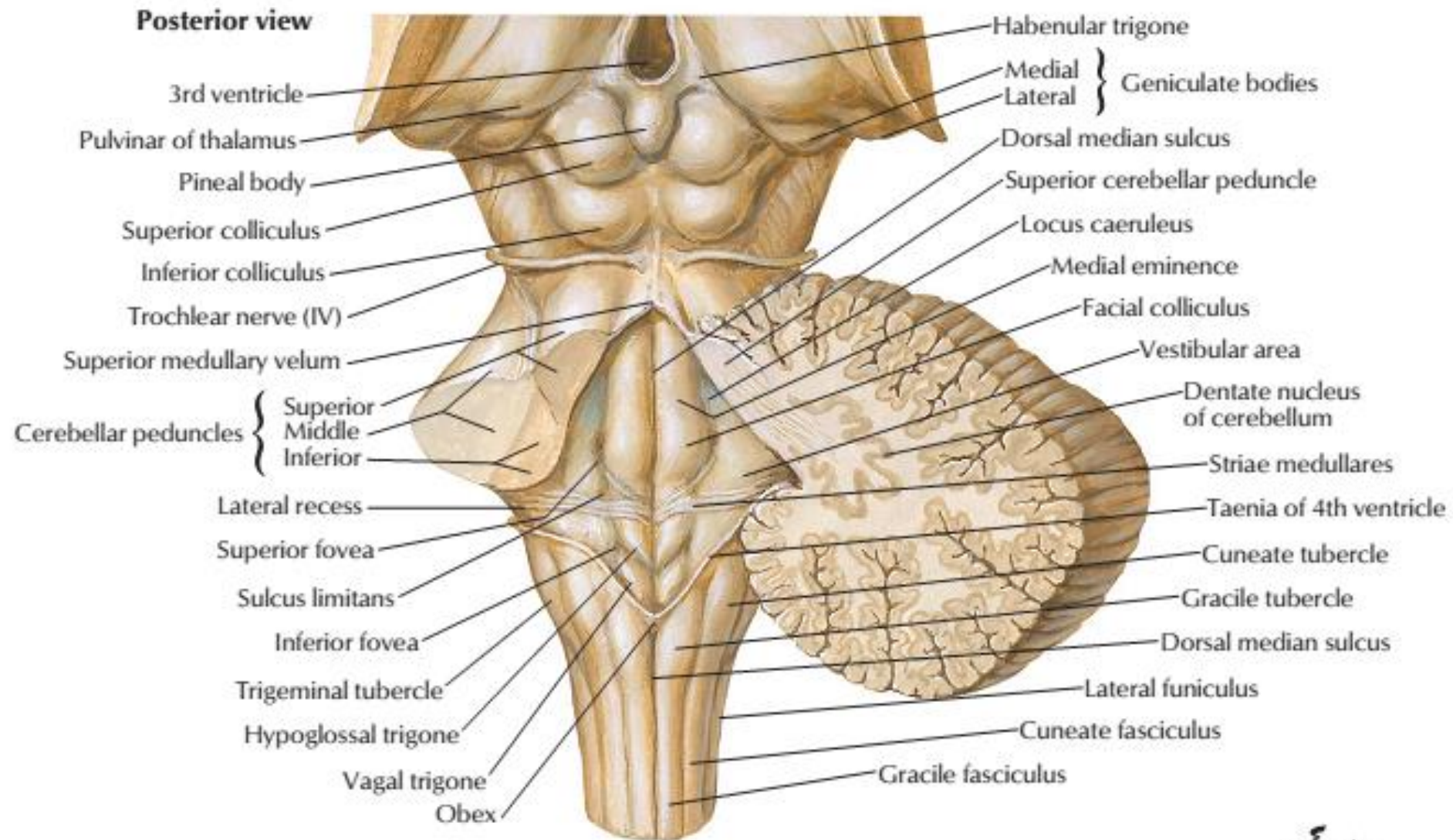
Midbrain (Mesencephalon)

- most rostral part of the brainstem that connects the pons and cerebellum with the forebrain
- about 2 cm in length. Its long axis ascends through the hiatus in the tentorium cerebelli
- traversed by the **cerebral aqueduct** (of Sylvius)
- connected to cerebellum by superior cerebellar peduncle
- **Functions**
 - serves as a conduit for ascending and descending tracts
 - Houses CN III (oculomotor nucleus, Edinger–Westphal nucleus or accessory oculomotor nucleus) and CN IV (trochlear nucleus) **nuclei** and, **reflex centers** (the colliculi), **red nucleus** and **substantia nigra**,
 - houses the mesencephalic nucleus of CN V

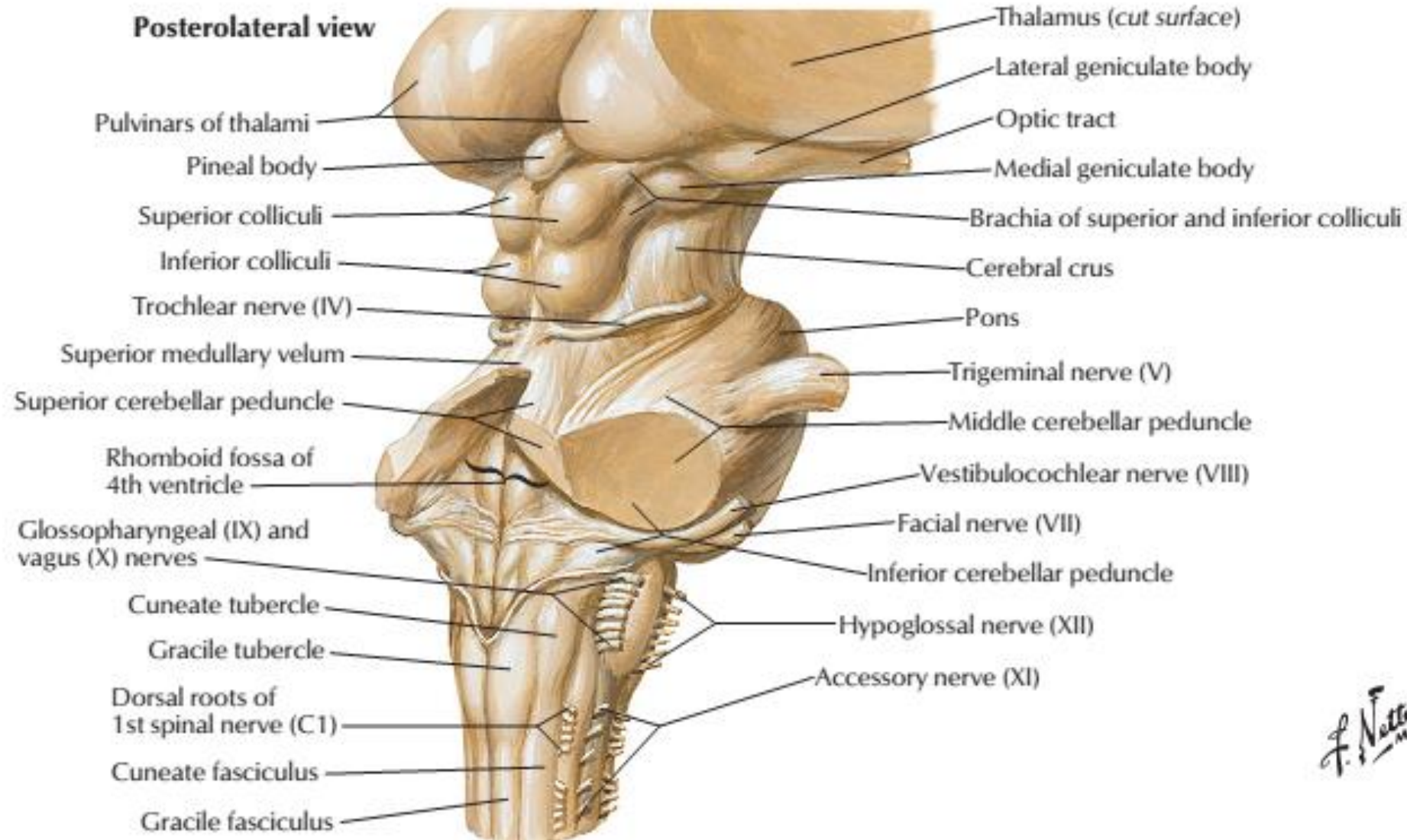
Midbrain



Midbrain

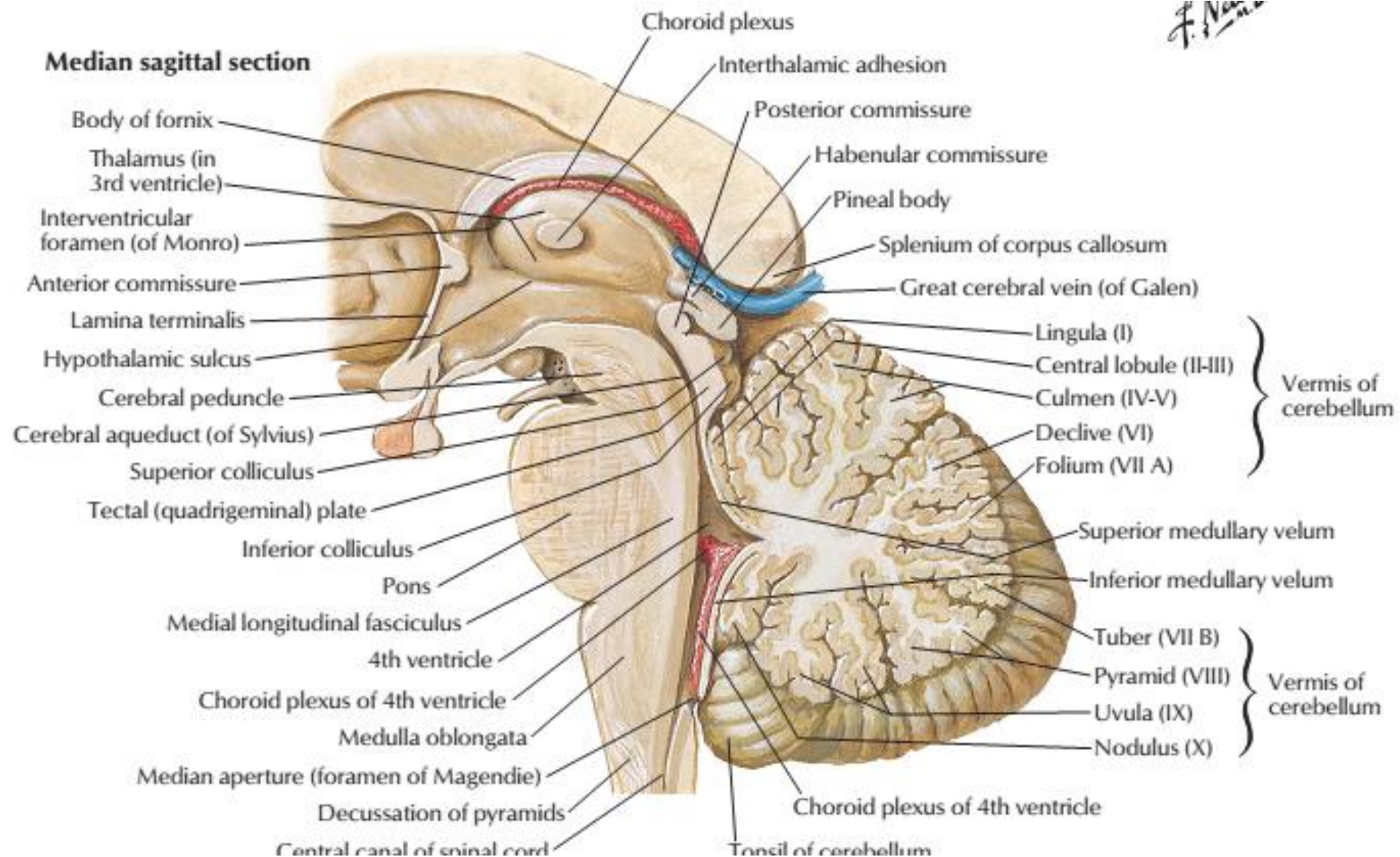


Midbrain



F. Netter M.D.

Midbrain



Pons - “bridge”

- Situated anterior to the cerebellum, and between the midbrain and medulla oblongata, about 1 inch (2.5 cm) long
- Connected to the cerebellum by the **middle cerebellar peduncle**
- Dorsal surface forms the upper part of the floor of the 4th ventricle.
- Ventral surface shows a shallow midline groove (occupied by the basilar artery) with a bulge (containing pontine nuclei, corticospinal and corticonuclear tracts) on either side
- Possesses CN nuclei (V, VI, VII and VIII)
 - **CN V** emerges from the lateral mid-pons at the junction of the pons and middle cerebellar peduncle.
 - **Cranial nerves** emerge, from medial to lateral, at the pontomedullary junction (CN VI) and cerebellopontine angle (CN VII and VIII).

Pons

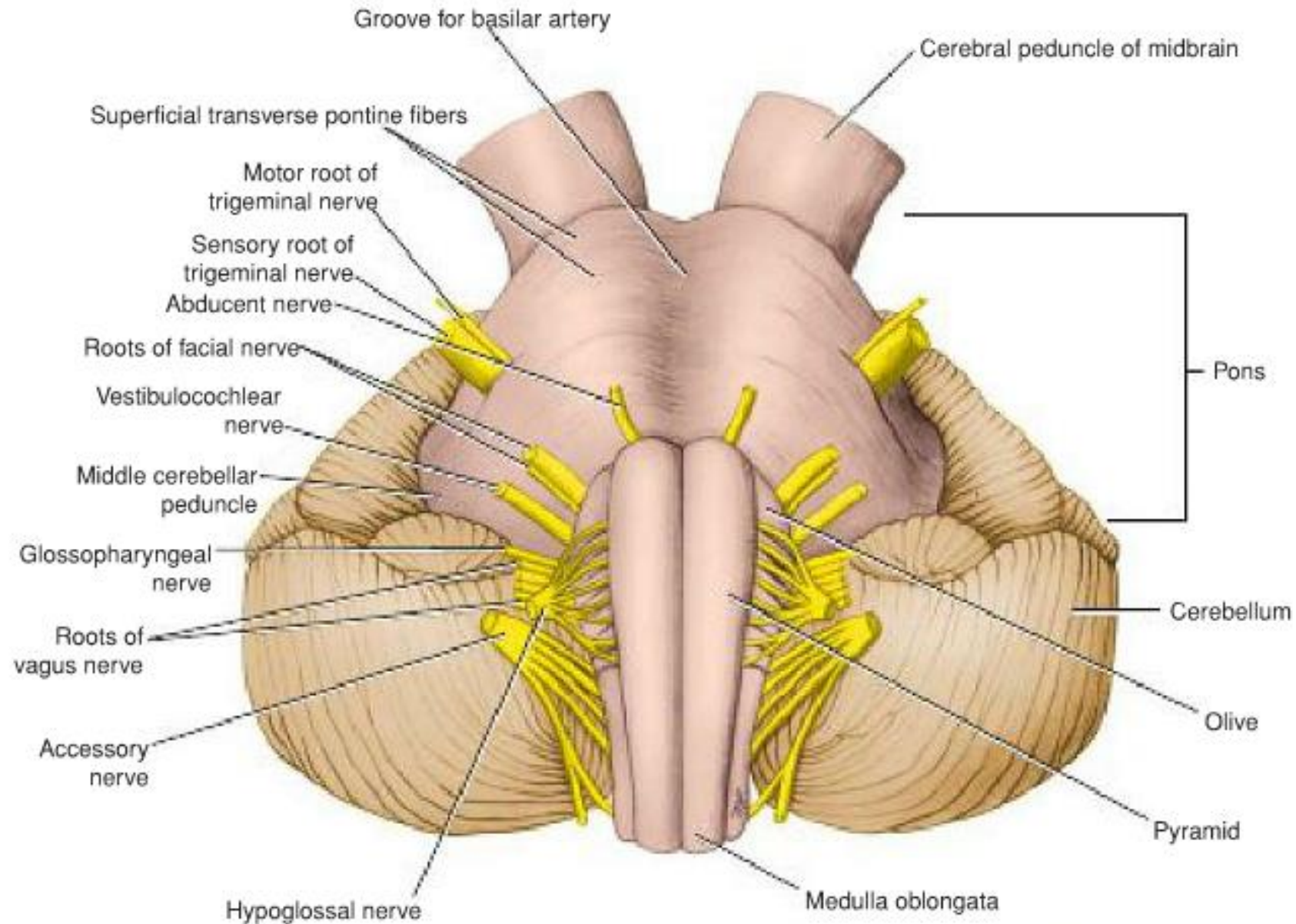


Figure 5-17 Anterior surface of the brainstem showing the pons.

Medulla Oblongata

- Most caudal part of the brain stem between the pons and the spinal cord
- Lies in the posterior cranial fossa, below the tentorium cerebelli.
- Upper part, the **open medulla**, forms the lower half of the floor of the 4th ventricle. The lower part, the **closed medulla**, has a narrow central canal and resembles the structure of the spinal cord
- Connected to the cerebellum by the **inferior cerebellar peduncle**

Medulla Oblongata

- Houses the centers for **vital functions** of the body, such as those for the heart rate, blood pressure, and breathing (dorsal and ventral respiratory group).
 - **Alcohol overdose also suppresses the medullary rhythmicity center and may result in death**
- Contain autonomic nuclei that control reflexes for vomiting, swallowing, sneezing, coughing, and hiccupping.
- Also contains nuclei that are components of sensory pathways for gustation (taste), audition (hearing), and equilibrium (balance)
- Contains the nuclei of CN IX, X, XI and XII

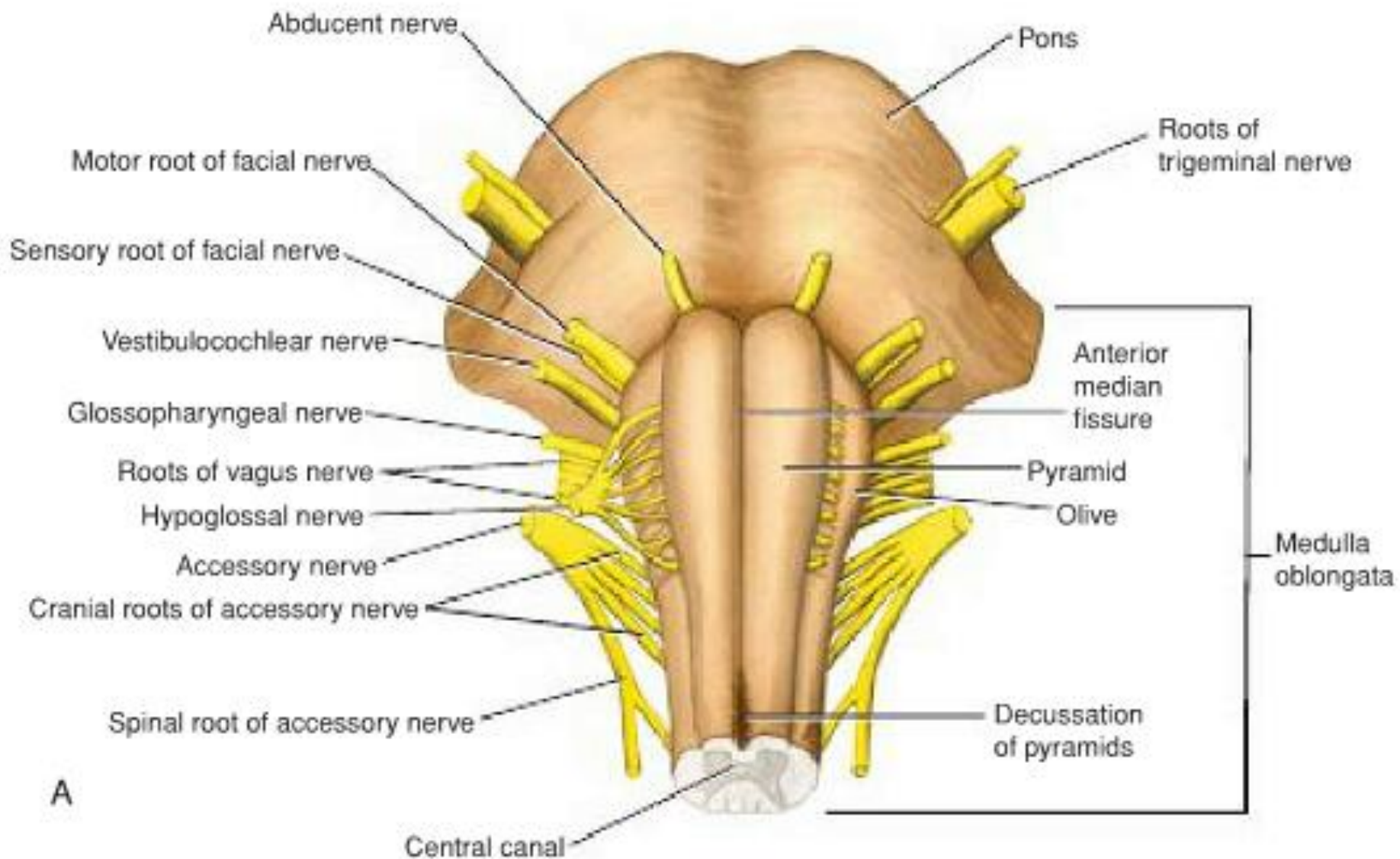


Illustration of the **ventral surface** of the medulla oblongata

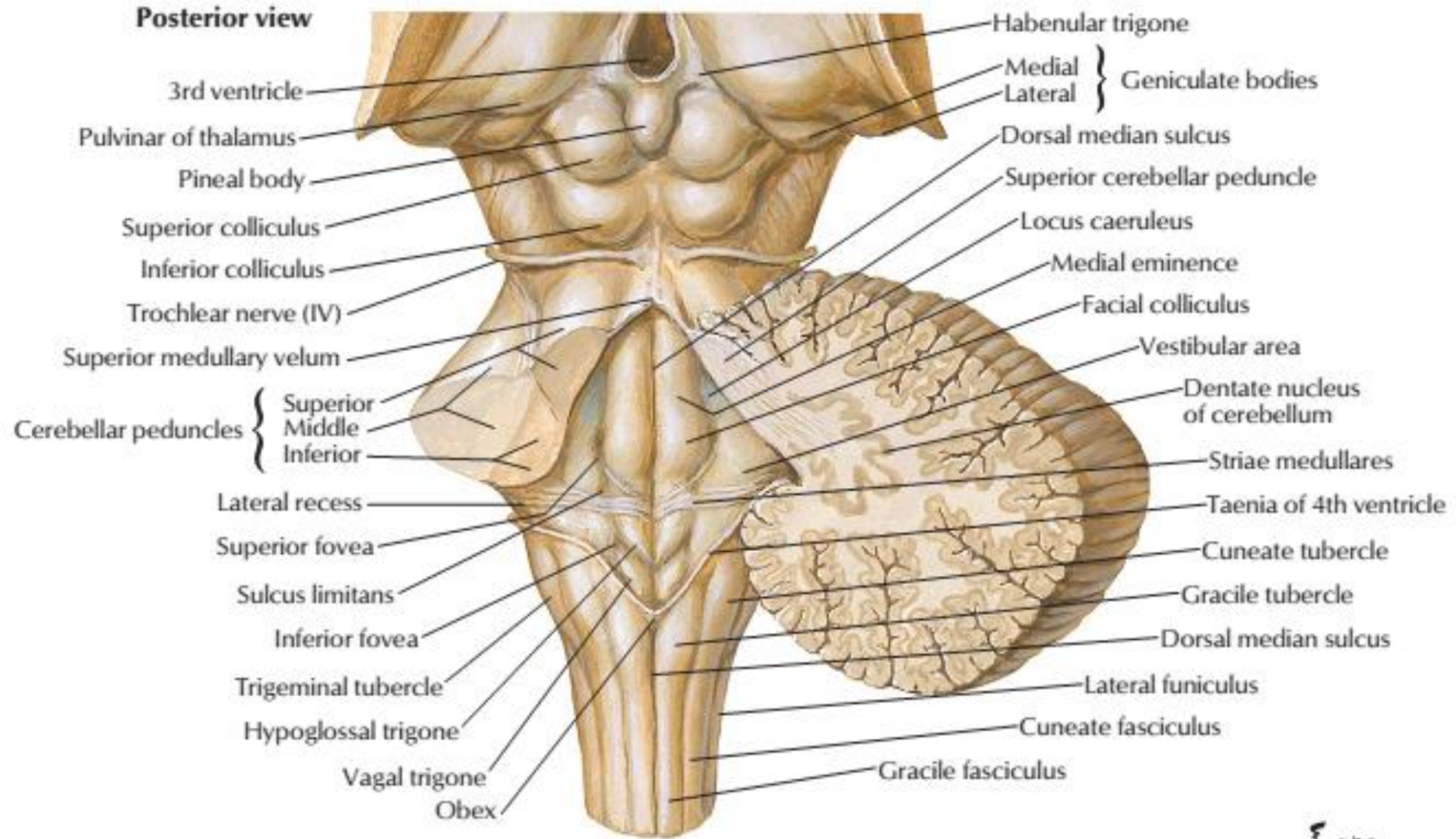
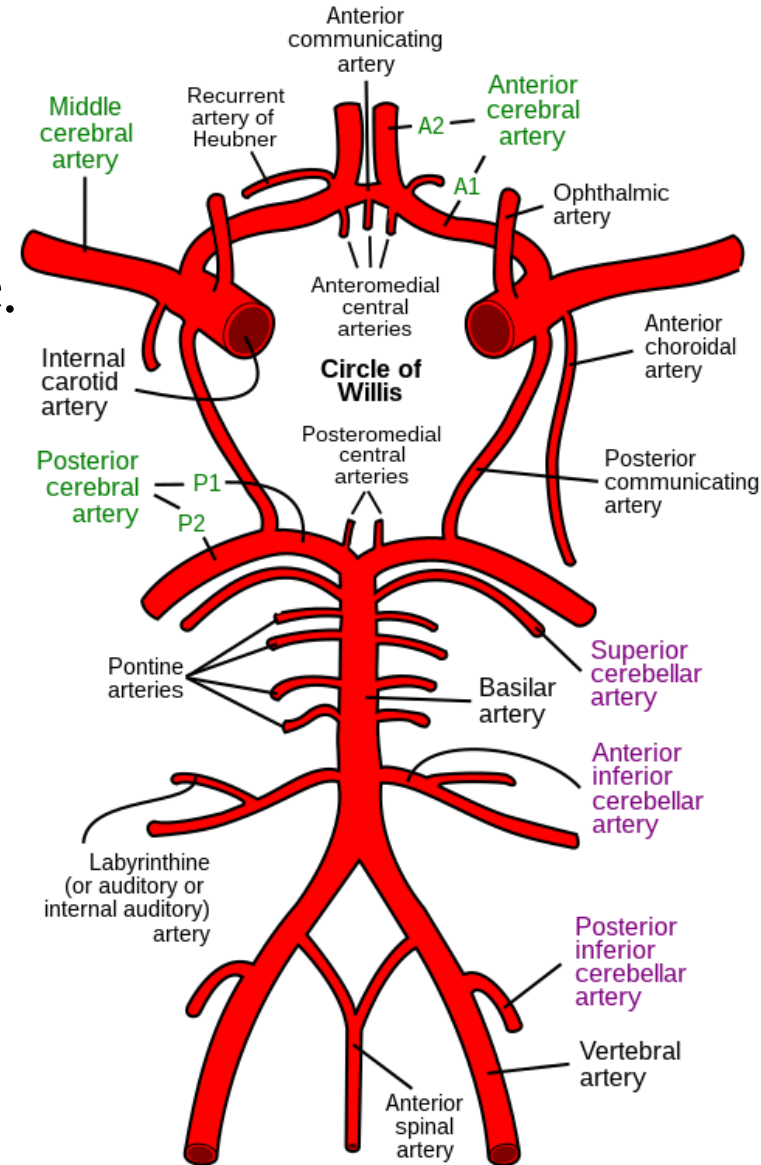


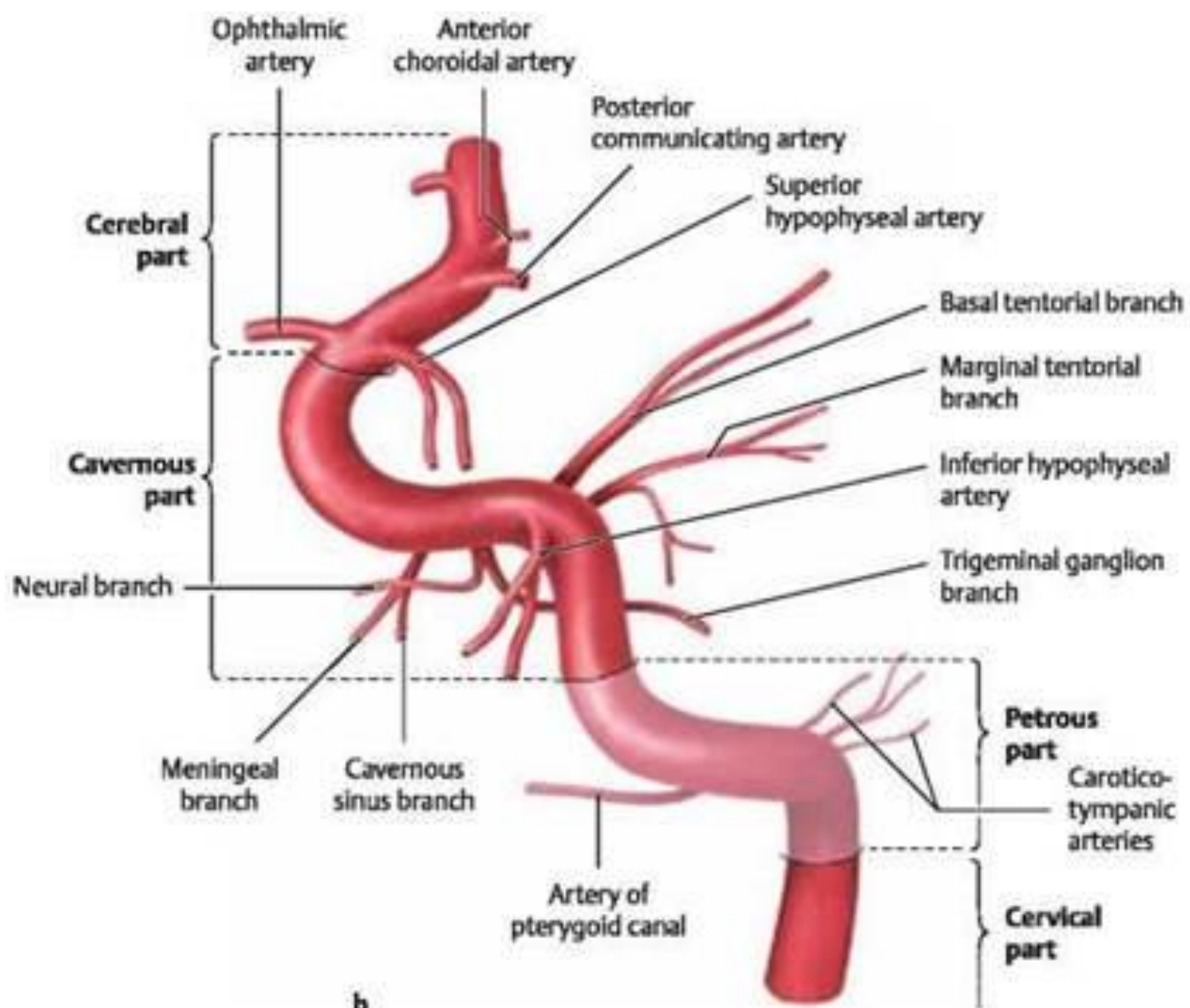
Illustration of the **dorsal surface** of the medulla oblongata

The blood supply of the brain

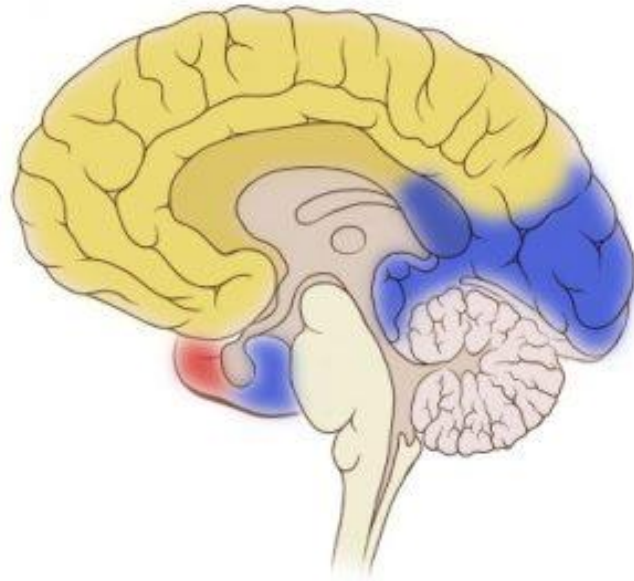
Supplied by

- two **internal carotid**
- two **vertebral** arteries.
 - lie in the subarachnoid space.
- Their branches anastomose at the base of the brain to form circulus arteriosus (circle of willis)
- **Intracranial braches of ICA:**
 - ophthalmic artery
 - Choroidal artery
 - Anterior cerebral artery
 - Middle cerebral artery
 - Posterior communicating artery





Medial view



Lateral view



- Posterior cerebral artery
- Middle cerebral artery
- Anterior cerebral artery

The Spinal Cord

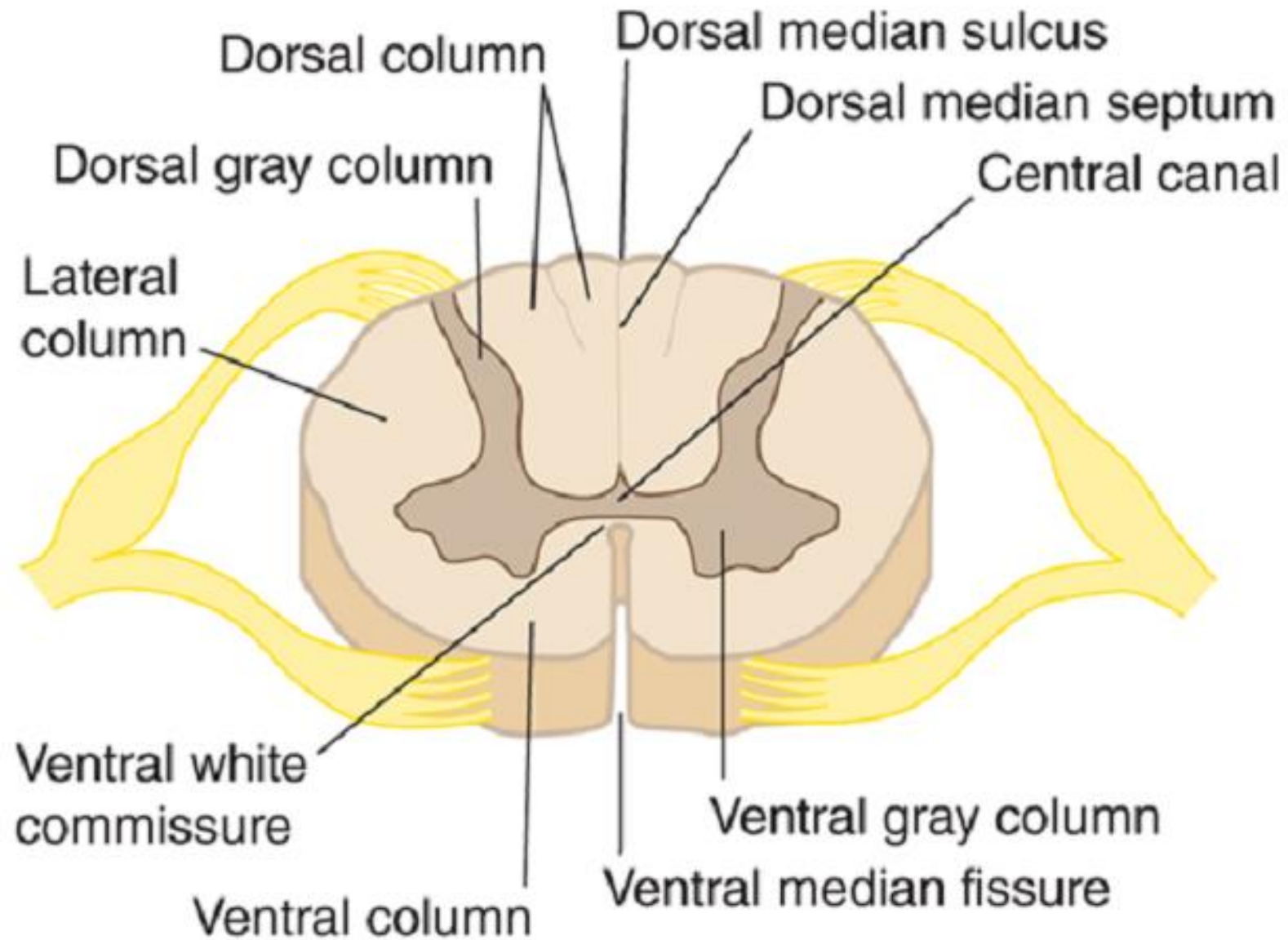
- **Functions**

- Spinal nerves attach to it
 - Provides two-way communication to and from the brain
 - Contains spinal reflex centers
-
- Is a cylindrical caudal continuation of the medulla at the foramen magnum within the vertebral cavity
 - Terminates as the **conus medullaris** at the inferior border of L1/2 or (L3 in newborn)
 - About 42-45 cm long, 1/2 inch (14 mm) wide
 - Is enlarged in the **cervical (C3-T2)** and **lumbar regions (L1-S3)** due to the increased number of sensory and motor neurons of the upper and lower limbs.
 - Nerves of the brachial plexus originate at the cervical enlargement; the nerves of the lumbosacral plexus arise from the lumbar enlargement.

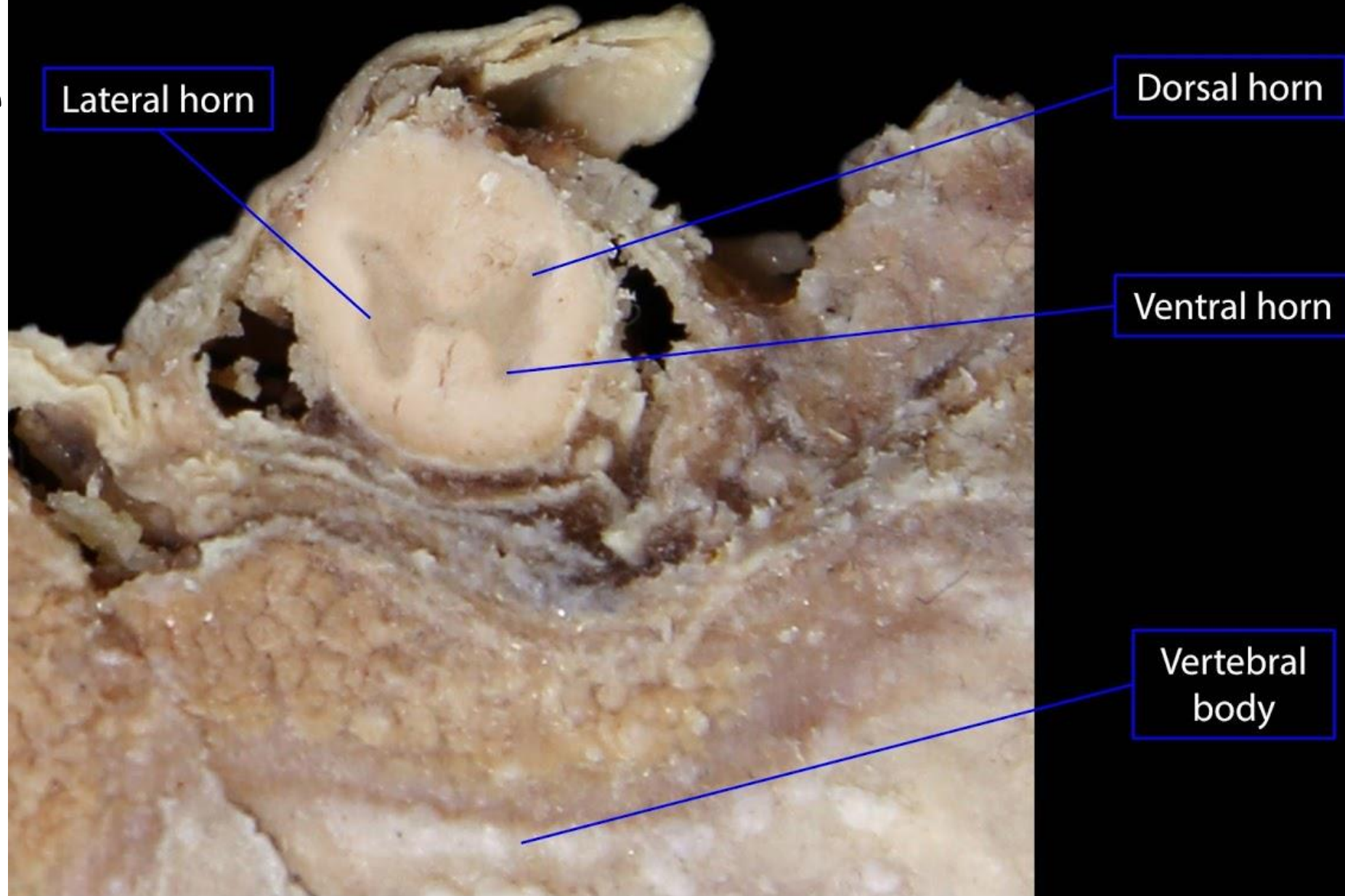
Internal Structure of Spinal Cord-Transverse Section

- Divided partially into right and left sides by the **ventral median fissure** and the **dorsal median sulcus**
- composed of an inner core of **gray matter** surrounded by an outer covering of **white matter**
- gray matter is H-shaped with **ventral and dorsal gray horns**, united by a thin gray commissure containing the small **central canal**.
 - A small **lateral gray horn (intermediolateral columns)** is present in the thoracic and upper lumbar segments of the cord.
- central canal is closed inferiorly but opens superiorly into the fourth ventricle
- White matter is divided into **ventral, lateral, and dorsal white columns or funiculi**

Trans



Ce



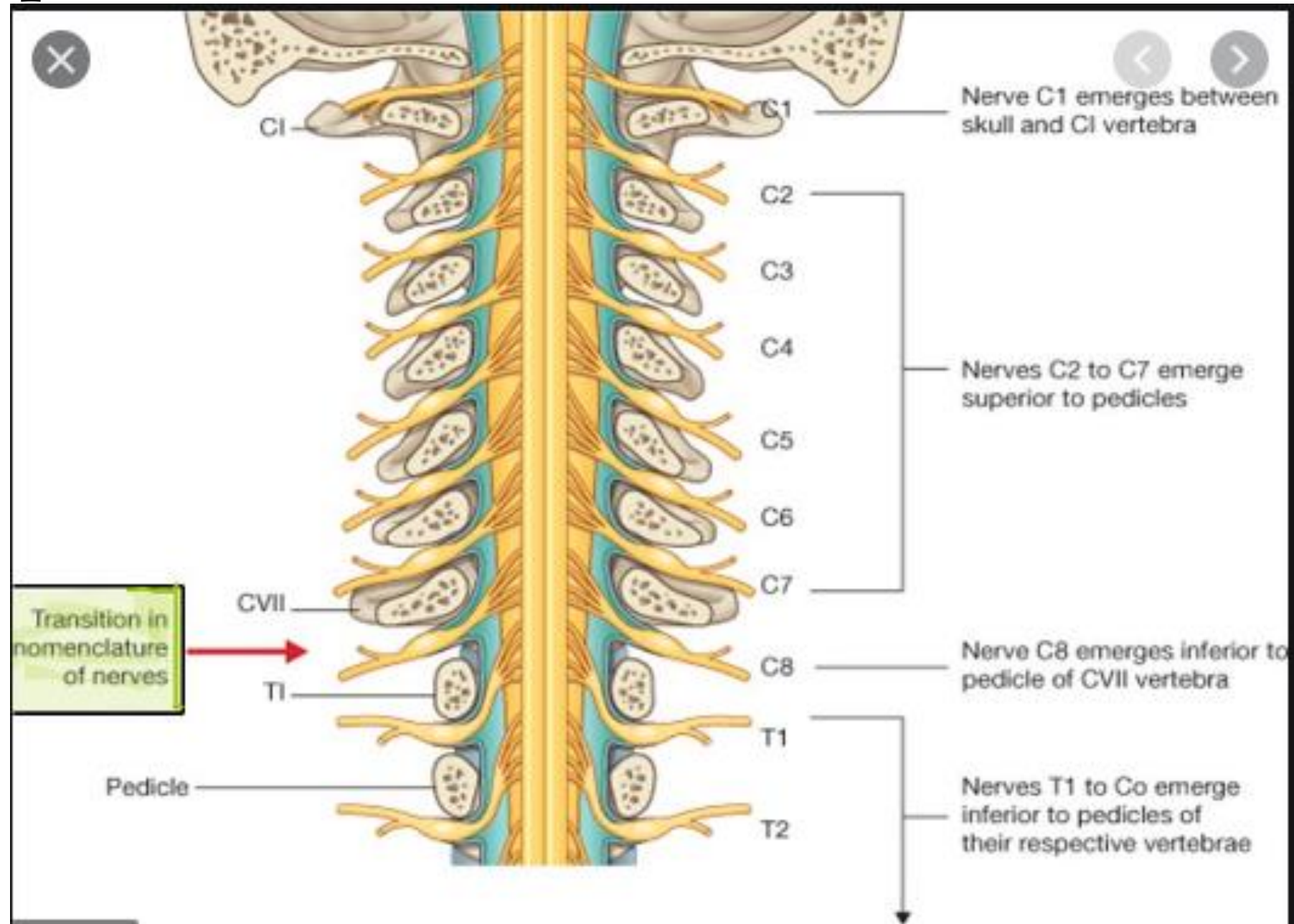
Gray Horns

- **Ventral horn**
 - contains the large **motor neurons** that supply voluntary striated muscles.
- **Dorsal horn**
 - contains the terminal axons of first-order **sensory neurons** and the cell bodies of the second-order sensory neurons that project cranially.
- **Intermediolateral group (Lateral horn)**
 - Small lateral gray horn from T1 – L2/3
 - Neurons are relatively small and give rise to **preganglionic sympathetic fibers**.
 - Group of cells present at S2 – S4 give rise to **preganglionic parasympathetic fibers (nervus erigentes)**

Spinal Nerves

- 31 pairs—all are mixed nerves
- Named for point of issue from the spinal cord
 - 8 pairs of cervical nerves (C_1 – C_8)
 - 12 pairs of thoracic nerves (T_1 – T_{12})
 - 5 pairs of lumbar nerves (L_1 – L_5)
 - 5 pairs of sacral nerves (S_1 – S_5)
 - 1 pair of coccygeal nerves (Co_1)
- Spinal nerves C_1 – C_7 exit the vertebral canal *above* their corresponding vertebrae.
- Spinal nerve C_8 exits the vertebral canal between vertebrae C_7 and T_1 .
- Spinal nerves T_1 – L_5 exit the vertebral canal *below* their corresponding vertebrae.

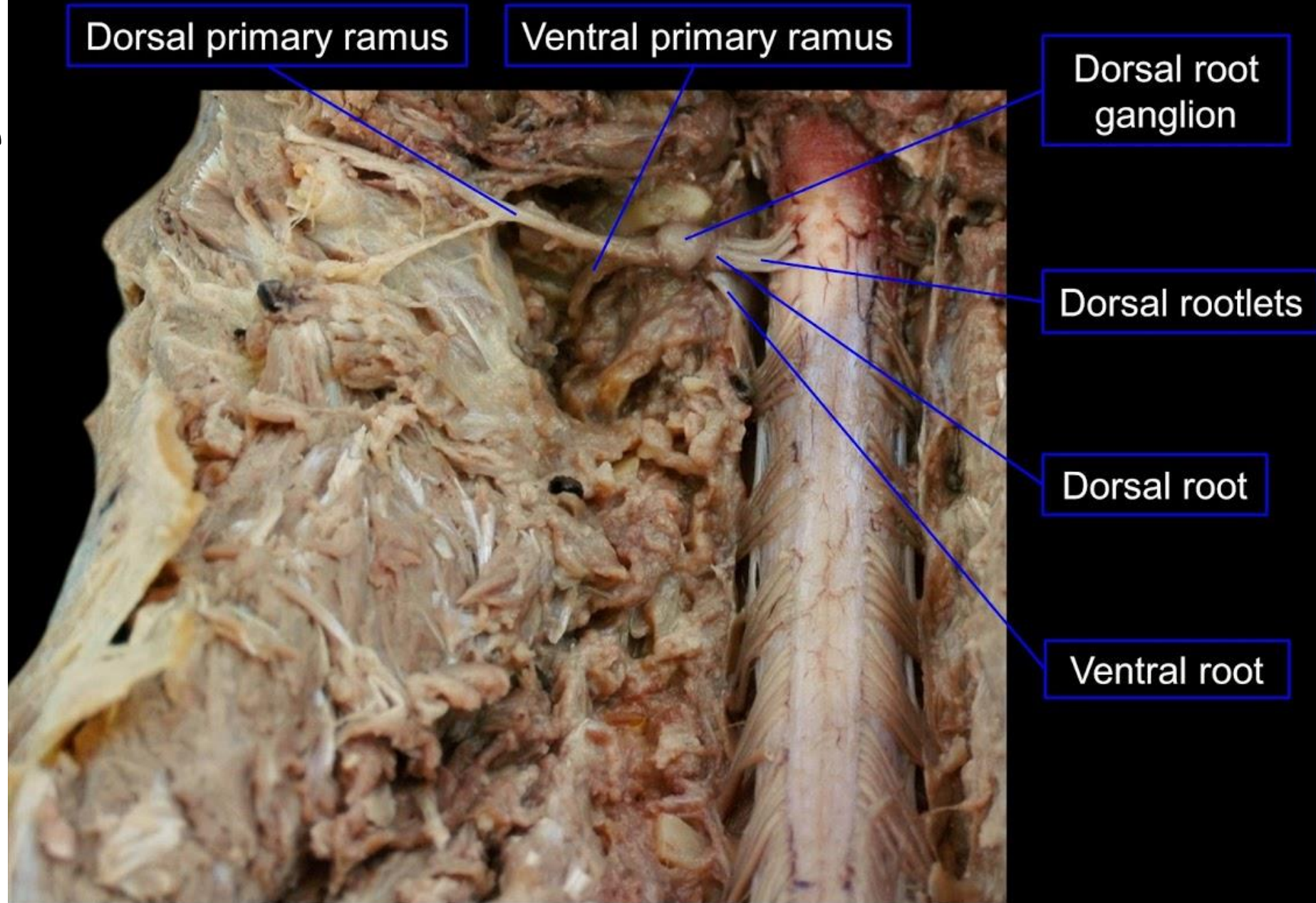
Spinal Nerves



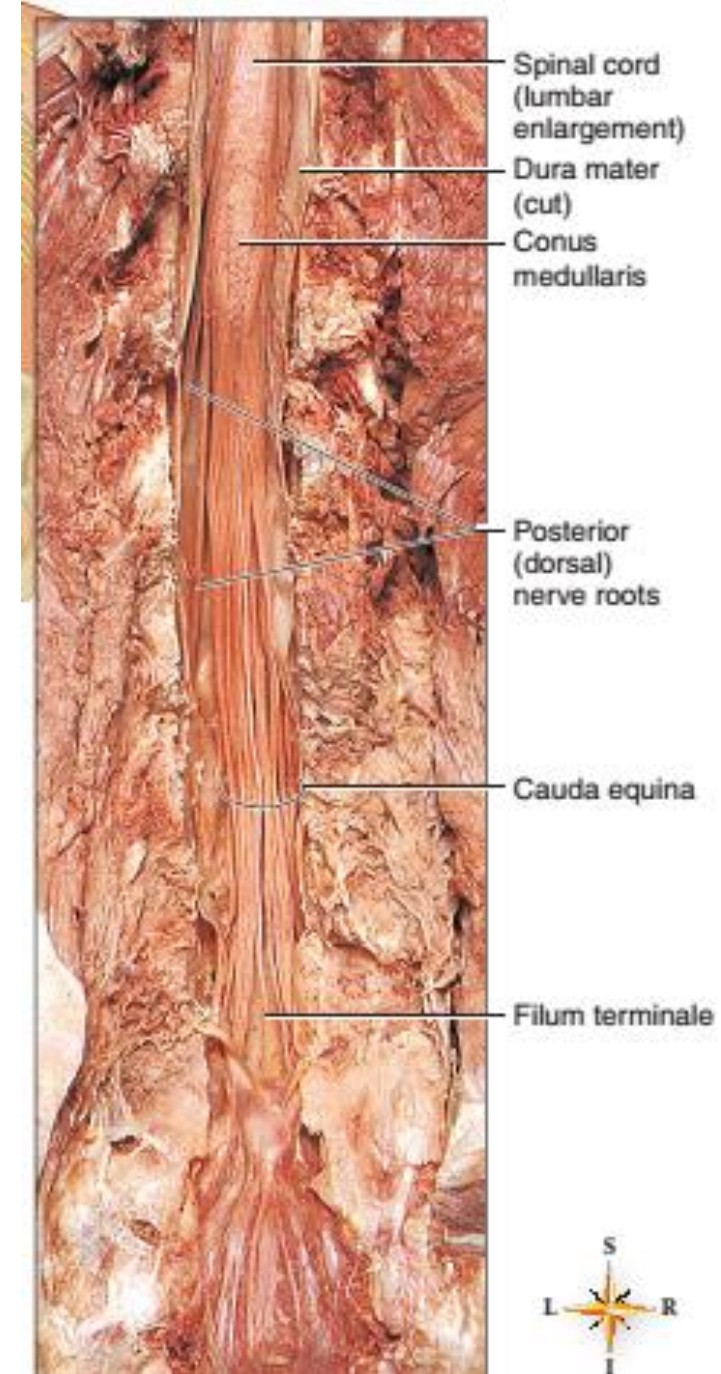
Spinal Nerves

- Formed by the union of ventral and dorsal roots, distal to the dorsal root ganglia, **within the intervertebral foramen**. Hence, all spinal nerves are mixed nerves.
- Each ventral or dorsal root is formed by the union of 1-8 rootlets.
- Sensory information enters the spinal cord through the **dorsal roots**. *Cell bodies of the first-order sensory neurons located in the dorsal root ganglion*
- Motor information leaves the spinal cord through the **ventral roots**, and LMNs are located in the anterior horn at each spinal level.

Ce



- Because the spinal cord is shorter (stops growing by age 4 years) than the vertebral column, the spinal cord segments do not correspond numerically with the vertebrae that lie at the same level.
- The filum terminale and the long ventral and dorsal roots of the lumbar, sacral and coccygeal nerves form the cauda equina (horse's tail) below the conus medullaris

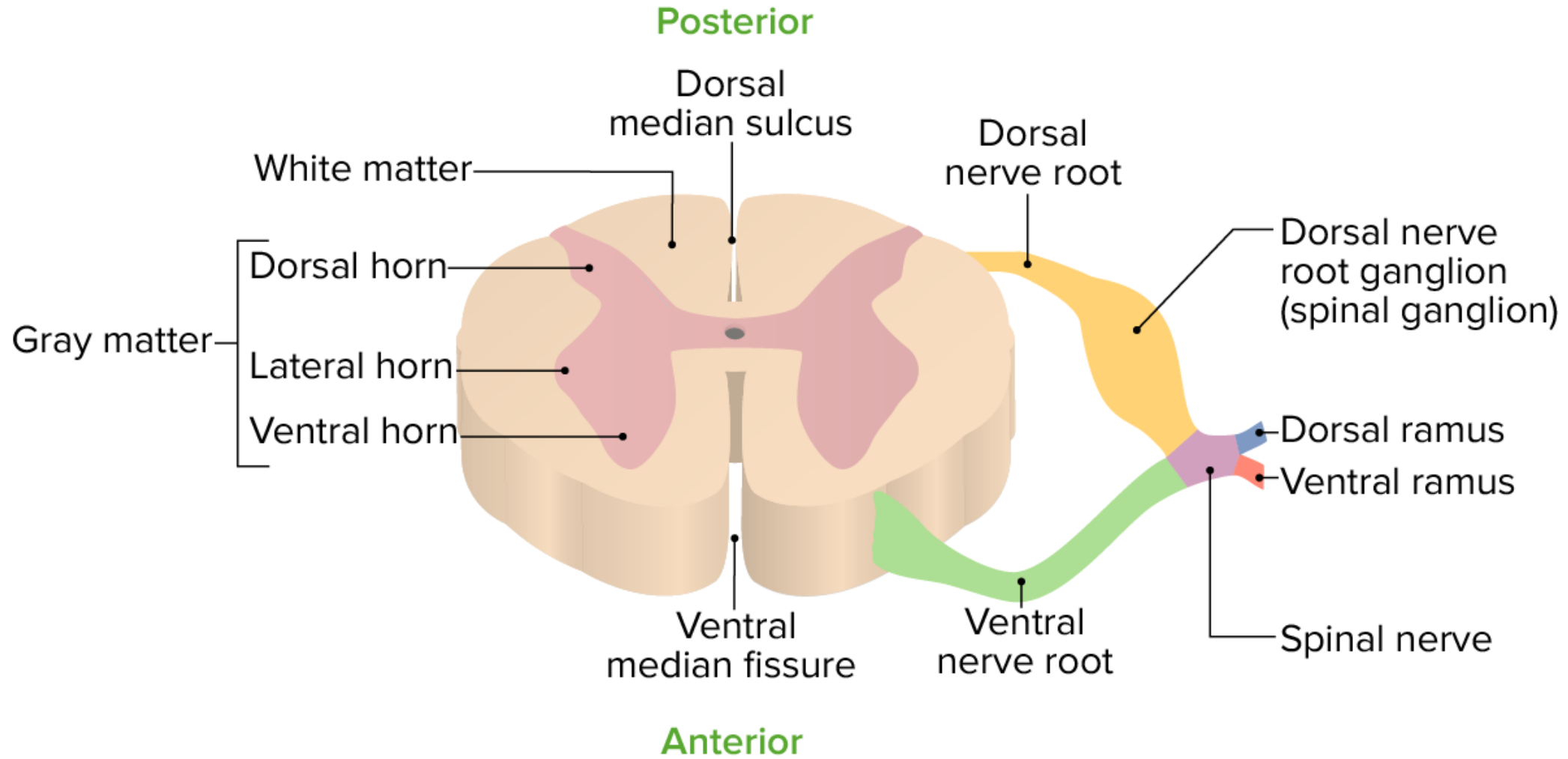


Spinal Nerves

- Each spinal nerve (1-2mm long) divides into **dorsal ramus** and **ventral ramus**
 - **Dorsal and ventral rami** contain both sensory *and* motor fibers.
 - Meningeal branch
 - **Smaller dorsal rami**
 - medial branch, largely sensory, and a lateral branch, mainly motor
 - receives sensory innervation from, and sends motor innervation to, the skeletal muscles of the back
 - **Larger ventral rami**
 - innervates the muscles and skin of the lateral & anterior trunk, and limbs
 - Forms network of nerves called **plexuses** (except for T2–T12).

NB. C1 has no cutaneous branch

Transverse section of spinal cord



Transverse section of spinal cord

